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SECTION 22 05 00 - COMMON WORK RESULTS FOR PLUMBING

PART 1 GENERAL

1.01 PREFACE

- A. The Work shall be performed using the General Contracting (GC) / Single-Prime Contractor project delivery method and the Design-Bid-Build process. References to Work to be performed by different Trades Contractors or Subcontractors are all subject to the Scope of Work by the Prime Contractor. Any conflicts regarding Scope of Work shall be directed to the Prime Contractor for resolution. THE PRIME CONTRACTOR FOR THIS PROJECT SHALL ALSO BE KNOWN AS THE LEAD CONTRACTOR.
- B. Terminology: The terms below will be used throughout Division 22, in accordance with Division 00 and Division 01.
1. Owner: Eureka Apostolic Christian Home.
 - a. Contracting Authority: [REDACTED]
 - b. Using Agency (Facility): [REDACTED]
 - c. Owner's Agent: [REDACTED]
 2. Architect/Engineer (A/E): pH7 / Roger D. Fields Associates, Inc.
 3. Prime Contractor / Lead Contractor: [REDACTED]
 4. Contractors:
 - a. All Trade Contractors listed below, other than the Lead Contractor, are Subcontractors.
 - b. General Trades: Refer to Divisions 02 through 14.
 - c. Fire Suppression: Refer to Division 21.
 - d. Plumbing: Refer to Division 22.
 - e. HVAC: Refer to Division 23.
 - f. Electrical: Refer to Division 26.
 - g. Communications: Refer to Division 27.
 - h. Electronic Safety and Security: Refer to Division 28.
 - i. Site: Refer to Divisions 31 through 35.

5. Authority Having Jurisdiction (AHJ) mentioned in Division 22:
 - a. Eureka-Goodfield Fire Department.
 - b. City of Eureka Building Department.
 - c. Woodford County Health Department.
 - d. State of Illinois Department of Commerce.
- C. Scope of Work Summary for the Contractor Performing Plumbing Work on this Project:
 1. Provide the domestic water and sanitary systems shown in the project documents.
 2. Provide restroom fixtures and sinks shown in the project documents.
 3. Provide connections for the food service kitchen equipment.

1.02 RELATED DOCUMENTS

- A. Refer to the additional Divisions below for specific requirements, responsibilities, and methods relating to Division 22 Work. If there is a conflict, comply with the more restrictive requirements.
 1. Division 00, "Procurement and Contracting Requirements": All sections.
 - a. The "General Conditions" will provide additional information on the Contractor's Scope of Work responsibilities.
 2. Division 01, "General Requirements": All sections.
- B. Read the entire Specification covering other branches of the Work. Contractor shall be held responsible for the coordination of its work with the work performed by other Trades.

1.03 DESCRIPTION

- A. Plumbing Work includes the Work described in Division 22, "Plumbing," Sections 22 00 00 through 22 99 99 (as included), unless otherwise noted.
- B. Plumbing Work also includes the Work described in the following Sections:
 1. Division 11, "Equipment":
 - a. Section 11 46 83, "Ice Machines."
 2. Division 21, "Fire Suppression," Section 21 13 13, "Wet-Pipe Sprinkler Systems."
 3. Division 23, "Heating, Ventilating, and Air Conditioning":
 - a. Section 23 11 23, "Facility Natural-Gas Piping."

4. Division 33, "Utilities":
 - a. Section 33 11 00, "Groundwater Sources."
 - b. Section 33 14 00, "Water Utility Transmission and Distribution."
 - c. Section 33 30 00, "Sanitary Sewerage."
 - d. Section 33 32 00, "Sanitary Sewerage Equipment."
 - e. Section 33 40 00, "Stormwater Utilities."
- C. In the Sections listed above, the term "Contractor" shall mean the Contractor performing Plumbing Work on this Project, unless otherwise noted.
- D. Plumbing Work includes, but is not limited to, the following:
 1. Unit Prices, Allowances, or Alternates, as shown, if any.
 2. Ice machines. Refer to Section 11 46 83.
 3. Wet-pipe sprinkler systems. Refer to Section 21 13 13.
 4. Concrete work.
 5. Underground plumbing systems.
 6. Plumbing insulation.
 7. Commissioning.
 8. Plumbing piping:
 - a. Interior domestic water system.
 - b. Domestic water pumps.
 - c. Interior sanitary system. Connect at 5 ft.-0 in. outside building, except as shown otherwise.
 - d. Sump pumps.
 9. Plumbing equipment:
 - a. Water heaters.
 10. Plumbing fixtures.
 11. Interior natural gas piping system. Connect at 5 ft.-0 in. outside building shown otherwise. Refer to Section 23 11 23.
 12. Exterior natural gas piping system. Refer to Section 33 52 16.
 13. Site plumbing work: Refer to Division 33.

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E. Work Not Included:

1. Site work: Other Contractor(s) shall be responsible for all site work not listed above, including the following work (includes all work from point of connection to existing utilities):
 - a. Water service: Entire system, to inside building, 18 in. above floor.
 - b. Sanitary system: Entire system.
 - c. Storm system: Entire system.
 - d. Gas service: Entire system.

F. Contractor is responsible for the more restrictive requirement between these general provisions and other requirements contained elsewhere in the Contract Documents.

G. Furnish material, labor, tools, accessories, and equipment to complete and leave ready for operation all plumbing systems of this Project as described in these Specifications and as shown on the Drawings.

H. Use sufficient skilled workers and competent supervisors in the execution of this portion of the Work to ensure proper and adequate installation throughout.

I. Refer to Section 22 05 00.31, "Project Coordination for Plumbing," regarding final electrical connections.

J. Work Sequence:

1. The Work shall be conducted in one phase.
2. Complete work in one Area before proceeding to the next Area. Coordinate with all Trades, the Owner, and the A/E.

K. All plumbing systems shall be provided as complete and operational, and shall not be considered complete unless, and until, at a minimum, all Code and Governing Agency requirements are satisfied.

1.04 COORDINATION MEETINGS AND COORDINATION DRAWINGS

A. Terminology:

1. Coordination Meetings: Coordination Meetings shall be held to identify Coordination Areas and the need for a Coordination Drawing for any Coordination Area. The Lead Contractor shall schedule and conduct the Coordination Meetings.
2. Coordination Areas: Areas of the project that require special attention, i.e., where there are potential conflicts between Trades.

3. Coordination Drawings: Drawings and electronic files prepared by the Coordination Participants to demonstrate how multiple-system and interdisciplinary work shall be coordinated.
4. Coordination Participants: All Contractors shall participate in the process of developing the Coordination drawings, including, at a minimum, the following Trades:
 - a. General Trades.
 - b. Fire Suppression.
 - c. Plumbing.
 - d. HVAC.
 - e. Electrical.
 - f. Communications.
 - g. Electronic Safety and Security.

B. Refer to Section 23 05 00, "Common Work Results for HVAC," for additional requirements.

1.05 QUALITY ASSURANCE

- A. Standards: Refer to individual Specification Sections for additional referenced standards. If a standard is undated, the referenced edition or most recent version of the standard applies.
- B. Standards: The following is a list of referenced standards. Not all of these standards necessarily apply to this Project. Refer to individual Specification Sections. If a standard is undated, the referenced edition or most recent version of the standard applies.
 1. American Institute of Architects Committee on Architecture for Health (AIA/CAH), Health Guidelines Revision Committee (HGRC): 2014 Guidelines for Design and Construction of Hospitals and Outpatient Facilities.
 2. American Society of Heating, Refrigerating and Air Conditioning Engineers (ASHRAE):
 - a. Standard 62.1, Current Enforced Edition, "Ventilation for Acceptable Indoor Air Quality."
 - b. Standard 90.1, Current Enforced Edition, "Energy Standard for Buildings Except Low-Rise Residential Buildings."
 3. American Society of Mechanical Engineers (ASME).
 - a. Boiler and Pressure Vessel Code (BPVC), Section IX, "Welding and Brazing Qualifications."

4. Americans with Disabilities Act (ADA), 2010, Accessibility Guidelines, Code of Federal Regulations (CFR), 36 CFR 1191.
5. National Fire Protection Association (NFPA):
 - a. 2023 NFPA 70, "National Electrical Code® (NEC)."
 - b. 2021 NFPA 99, "Health Care Facilities Code."
6. The Joint Commission.
7. United States Department of Housing and Urban Development.
8. United States Department of Labor, 29 CFR 1910.7, Occupational Safety and Health Administration (OSHA), Part 1926, "Safety and Health Regulations for Construction."
9. United States Environmental Protection Agency (EPA).
10. United States Green Building Council (USGBC).
11. State of Illinois:
 - a. Illinois Administrative Code (IAC):
 - 1) IAC Title 35: Environmental Protection.
 - 2) IAC Title 41: Fire Protection.
 - 3) IAC Title 71: Public Buildings, Facilities, and Real Properties.
 - 4) IAC Title 77: Public Health.
 - 5) IAC Title 83: Public Utilities.

END OF SECTION 22 05 00

SECTION 22 05 00.31 - PROJECT COORDINATION FOR PLUMBING

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Division 26, "Electrical."
- B. Section 26 05 00.31, "Project Coordination for Electrical."
- C. Section 26 29 13, "Enclosed Controllers."

1.02 DESCRIPTION

- A. Provide plumbing work associated with equipment being provided by other Trades.
- B. Electrical Contractor shall provide all line-voltage wiring, and all control, interlock, and power wiring and conduit, regardless of voltage, required to power, control, monitor, or interlock plumbing equipment, unless noted otherwise.
 - 1. Exception: Provide faucet sensor and solenoid valve low-voltage plug-in connections to control module.
- C. Coordinate all wiring requirements for Plumbing Work with the Electrical Contractor.
- D. Electrical Contractor shall provide power wiring, rough-in, and final connection, for each piece of plumbing equipment requiring power or other wiring.
- E. Alternative Manufacturer Electrical Requirements: Contractor furnishing the equipment shall compensate the Electrical Contractor for the cost of any additional labor and materials required to wire and hook up equipment other than the Design Base Model and Manufacturer.
- F. All electrical equipment shall be UL-listed and furnished in accordance with Division 26 requirements.

1.03 QUALITY ASSURANCE

- A. Standards:
 - 1. National Electrical Manufacturers Association (NEMA).
 - 2. National Fire Protection Association (NFPA): 2023 NFPA 70, "National Electrical Code® (NEC)."
 - 3. Underwriters Laboratories, Inc. (UL).

PROJECT COORDINATION FOR PLUMBING

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PART 2 PRODUCTS

2.01 DISCONNECT SWITCHES

- A. Factory-Installed Disconnect Switches: NEMA Type 3R, heavy-duty, disconnect switch within the housing of the equipment, for exterior equipment or other NEMA Type required for wet and dusty environments.

2.02 MOTOR STARTERS

- A. Factory-Installed Starters: NEMA Size 0 minimum, manual or magnetic as shown on the Drawings, with selector switch or pushbutton as noted, pilot lights where noted, overloads, auxiliary contacts, fuses, and control power transformer, unless unit being controlled includes integral 24 volt control system.

PART 3 EXECUTION

3.01 PREPARATION

- A. Consult Drawings that may affect location of equipment, conduit, or wiring, and make minor adjustments in locations to secure coordination.

3.02 INSTALLATION

- A. Installation shall comply with NEC and shall be inspected and approved by the AHJ.
- B. Motor Starters:
 - 1. Provide factory-installed starters only where specifically listed in Division 22 Specification Sections.
 - 2. Electrical Contractor shall provide all other starters.
 - 3. On all starters provided by the Plumbing Contractor, if motor starter heater element's designated amperage is more than 110% of the full load current after balancing, the Plumbing Contractor shall provide new starter heater elements.
- C. Assist in the check-out and start-up operations for plumbing equipment.
- D. Plumbing Contractor is responsible for the final operation of all equipment furnished under its Work.
- E. Electrical Contractor shall complete the applicable sections of start-up checklists for all Plumbing equipment.

- F. Provide the Electrical Contractor with a full detailed set of all required wiring diagrams for equipment furnished under the Plumbing Work.

END OF SECTION 22 05 00.31

SECTION 22 05 00.33 - SHOP DRAWING SUBMITTALS FOR PLUMBING

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Division 01, "General Requirements."
- B. Section 22 05 00, "Common Work Results for Plumbing."

1.02 DESCRIPTION

- A. Submittal dates:
 - 1. Within (14) calendar days after the Award of Contract, submit complete materials list of all items proposed to be provided on this Project.
 - 2. Submit shop drawings within (28) days after the Award of Contract.
- B. Materials and equipment installed under the Plumbing Work shall meet all the requirements of the Contract Documents. Do not order or install materials or equipment until submittals are reviewed by the A/E.
- C. Submit shop drawings for the installation of the Plumbing Work for review by the A/E before any Plumbing Work is installed, enlarged, or extended. Under NO circumstances shall any Plumbing Work be performed prior to receiving shop drawings reviewed by the A/E.
- D. The A/E's review is ONLY for conformance with the design concept of the Project and with the Contract Documents.
- E. Catalog data for equipment reviewed by the A/E shall not take precedence over the requirements of the Contract Documents. Review by the A/E shall not relieve the Contractor from the responsibility for deviations from Drawings or Specifications, nor from the responsibility for providing proper clearance and coordination with other Contractors.
- F. Color Selections: The A/E shall approve all color selections for plumbing equipment, including fixtures, faucets, and trim.

1.03 SHOP DRAWINGS

- A. Submit shop drawings for each manufactured item of equipment and all components to be used in the Work. Refer to the Shop Drawing Log attached to this Section and to the Sections listed in Section 22 05 00, Article 1.03, for items.

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- B. Submit shop drawings electronically in Portable Document Format (PDF). Submit number of copies indicated in Division 01. Contractor shall be responsible for making all copies for distribution and file.
- C. Include the following information:
 - 1. Complete original PDF or first-generation scanned PDF copies of the Manufacturer's catalog sheets, brochures, diagrams, schedules, performance charts, illustrations, and other standard descriptive product data.
 - 2. Brand name.
 - 3. Catalog number.
 - 4. Specific performance data.
 - 5. Material description.
 - 6. Characteristics.
 - 7. Ratings.
 - 8. Capacities.
 - 9. Working pressure.
 - 10. Dimensional data.
 - 11. Material gauge or thickness.
 - 12. Wiring diagrams and control diagrams.
 - 13. Arrangement of component parts, physical dimensions, clearances, materials, service requirements, colors (where required), controls, and accessories.
- D. Clearly mark or highlight each shop drawing copy to identify pertinent materials, products, models, and colors (where appropriate).
- E. Coordinate each submittal with requirements of the Work and the Contract Documents.
- F. When (2) or more items of the same material or equipment are required, they shall be of the same Manufacturer.
- G. Completely review shop drawings, product data, and samples prior to submission.
- H. When submitted for review, all shop drawings shall bear the Contractor's stamped and signed certification of the following:
 - 1. Contractor has reviewed, checked, and approved the shop drawings.
 - 2. Contractor has coordinated shop drawings with the requirements of the Project and with the provisions of the Contract Documents.

**SHOP DRAWING SUBMITTALS
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3. Contractor has ensured that the equipment shown on the shop drawings meets all the requirements of the Contract Documents.
4. Contractor has determined and verified all of the following:
 - a. Materials.
 - b. Field measurements.
 - c. Field construction criteria.
 - d. Catalog numbers and similar data.
 - e. Conformance with Specifications.
 - f. Hands of equipment.
 - g. Quantities and sizes.
- I. UNSIGNED COPIES WILL NOT BE REVIEWED.
- J. Notify the A/E, in writing, at the time of submission, of any deviations in the submittals from the requirements of the Contract Documents.
- K. Provide the following certifications information:
 1. Test Agency results verifying capacities, operating conditions, and power requirements at design conditions.
 2. Manufacturer's statement of compliance with Standards discussed in individual Specification Sections.
 3. Equipment labels indicating Certification requirements.
 4. Quality standard designations on each unit piece, e.g., each pipe length, pressure vessel, or valve.
 5. Typed verification that noted mixes, chemical compositions, and testing procedures were complied with.
 6. Other Certifications listed in other Sections of the Specifications.
- L. Contractor shall make submittals promptly in accordance with the approved schedule, and in such sequence as to cause no delay either in its Work or in the work of any other Contractor.
- M. If a submittal is rejected, immediately make any corrections or changes required by the A/E. Resubmit until accepted.
- N. If the Contractor orders equipment or materials, or begins installation, fabrication, or work prior to the return of approved submittals, it shall be "at the Contractor's own risk."

**SHOP DRAWING SUBMITTALS
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- O. Incorporate Shop Drawings into the Operating and Maintenance Manuals.

1.04 REQUIRED SUBMITTAL INFORMATION

- A. The form attached to this section contains a list of required submittals. This list may not be a complete list of all required submittals. Submit for approval all items to be provided, whether listed or not.

END OF SECTION 22 05 00.33

SHOP DRAWING LOG FOR PLUMBING

PROJECT NAME: EUREKA APOSTOLIC CHRISTIAN HOME

PROJECT NUMBER: RDFA 25059, PH7 #2025-11

Roger D. Fields Associates, Inc.
580 N. 4th ST., Suite 240
Columbus, Ohio 43215
614-451-2248

SUBMITTAL KEY

A	Catalog Cuts/Shop Drawings	G	Completed start-up test forms
B	Installation instructions	H	Coordination Drawings / Studies
C	Color samples	I	Calculations
D	Product samples	J	Schedule
E	Typed statement of material to be furnished	K	Sample of finish
F	Typed verification of compliance with certification requirements		

STATUS KEY

APP	Approved (conforms)
CAN	Corrections Are Noted (conforms as noted)
ICN	Incorporate Corrections Noted
R&R	Revise & Resubmit
R	Rejected

Plumbing Submittals Required	Specification Section	Submittal Key	Contractor	Original Submittal			Re-submittal			Remarks / Reviewed by
				Date A/E Received	Date A/E Returned	Status Key	Date A/E Received	Date A/E Returned	Status Key	
Division 11										
Ice Machine Filters	11 47 00	A								
Ice Machines	11 47 00	A								
Division 21										
Backflow Preventers, Sprinkler System	21 10 00	A								
Hydrants, Fire	21 10 00	A								
Water Flow Switch	21 10 00	A								
Division 22										
Access Panels	22 05 00.73	A								
Health Department Certificate (Commercial Kitchens)	22 05 00.78	F								
Gauge Cocks	22 05 19	A								
Pressure Gauges	22 05 19	A								
Thermometers	22 05 19	A								
Regulators, Water-Pressure	22 05 23	A								
Valves	22 05 23	A								
Valves, Water-Pressure Reducing	22 05 23	A								
Concrete Tests	22 05 29	F								
Hangers, Equipment	22 05 29	A								
Hangers, Piping	22 05 29	A								
Vibration Isolators	22 05 48	A								
Pipe-Coding Materials	22 05 53	D								
Valve-Tagging Chart	22 05 53	D								
Valve-Tagging Materials	22 05 53	D								
Cleanouts	22 05 76	A								

Plumbing Submittals Required	Specification Section	Submittal Key	Contractor	Original Submittal			Re-submittal			Remarks / Reviewed by
				Date A/E Received	Date A/E Returned	Status Key	Date A/E Received	Date A/E Returned	Status Key	
Cleanouts, Exterior	22 05 76	A								
Insulation, Equipment	22 07 00	A								
Insulation, Lavatory Trap/Supply	22 07 00	A								
Insulation, Pipe and Fittings	22 07 00	A								
Pipe and Fittings	22 10 00	E								
Pipe Boots	22 10 00									
Certificate of Completion of Water Line Chlorination	22 11 16	F								
Lead-Free Solder Certificate	22 11 16	F								
Pipe and Fittings, Domestic Water	22 11 16	E								
Water Meter	22 11 16	A								
Air Gap Fittings, Backflow Preventers	22 11 19	A								
Backflow Preventer Test Certification	22 11 19	F								
Backflow Preventer, Main Line	22 11 19	A								
Backflow Preventers, Branch Lines	22 11 19	A								
Backflow Preventers, Equipment	22 11 19	A								
Backflow Preventers, Vending Equipment	22 11 19	A								
Connection Boxes, Ice Makers	22 11 19	A								
Connection Boxes, Washer/Dryer	22 11 19	A								
Connection Boxes, Whirlpool	22 11 19	A								
Connectors, Piping Hose	22 11 19	A								
Connectors, Pumps	22 11 19	A								
Drain Pans, Water Heater	22 11 19	A								
Hose Bibbs	22 11 19	A								
Hydrants, Roof	22 11 19	A								
Hydrants, Wall	22 11 19	A								
Pressure Gauges, TMV	22 11 19	A								
Quick Couplers	22 11 19	A								
Shock Absorbers	22 11 19	A								
Strainers	22 11 19	A								
Tanks, Expansion	22 11 19	A								
Tanks, Well Water Hydropneumatic	22 11 19	A								
Thermometers, Thermostatic Mixing Valves	22 11 19	A								
Thermostatic Mixing Valves	22 11 19	A								
Thermostatic Mixing Valves, with Recirculating Pump and Aquastat/Timer	22 11 19	A								
Trap Primers	22 11 19	A								
Vacuum Breakers	22 11 19	A								

Plumbing Submittals Required	Specification Section	Submittal Key	Contractor	Original Submittal			Re-submittal			Remarks / Reviewed by
				Date A/E Received	Date A/E Returned	Status Key	Date A/E Received	Date A/E Returned	Status Key	
Valves, Washer Shut-Off	22 11 19	A								
Pumps, In-Line Recirculating	22 11 23.23	A								
Pumps, In-Line Recirculating — Aquastat/Timer	22 11 23.23	A								
Floor Drains	22 13 00	A								
Floor Sinks and Accessories	22 13 00	A								
Pipe and Fittings, Sanitary Waste	22 13 00	E								
Shower Drains	22 13 00	A								
Valves, Air Admittance	22 13 00	A								
Vent Caps	22 13 00	A								
Grease Interceptors	22 13 23	A								
Hair Interceptors	22 13 23	A								
Lint Traps	22 13 23	A								
Solids Interceptors	22 13 23	A								
Oil/Water Sanitary Waste Separators	22 13 26									
Sewage Ejectors, Basin, Controls	22 13 29	A								
Valve Box Basin	22 13 29	A								
Downspout Nozzles / Boots	22 14 00	A								
Emergency Roof Drains	22 14 00	A								
Pipe and Fittings, Storm	22 14 00	E								
Roof Drains	22 14 00	A								
Sump Pumps, Basin, Controls	22 14 29	A								
Pressure-Relief Valves	22 34 36	A								
Storage Tanks, Hot Water	22 34 36.13	A								
Water Heaters, Gas-Fired	22 34 36.23	A								
Bathtubs	22 40 00	A								
Clinic Sink	22 40 00	A								
Disposers, Garbage (Food Waste Grinders)	22 40 00	A								
Flush Valves	22 40 00	A								
Laundry Trays (Laundry Tubs)	22 40 00	A								
Lavatories	22 40 00	A								
Lavatory Carriers	22 40 00	A								
Lavatory Faucets	22 40 00	A								
Mop Service Basin and Accessories	22 40 00	A								
Shampoo Bowls	22 40 00	A								
Shower Basins	22 40 00	A								
Shower Faucets	22 40 00	A								
Sink Faucets and Bubblers	22 40 00	A								

Plumbing Submittals Required	Specification Section	Submittal Key	Contractor	Original Submittal			Re-submittal			Remarks / Reviewed by
				Date A/E Received	Date A/E Returned	Status Key	Date A/E Received	Date A/E Returned	Status Key	
Sinks	22 40 00	A								
Sinks, Service / Utility / Janitor's	22 40 00	A								
Sinks, Stainless Steel	22 40 00	A								
Transformers	22 40 00	A								
Traps, Plumbing Fixtures	22 40 00	A								
Wash Fountains	22 40 00	A, C								
Water Closet Carriers	22 40 00	A								
Water Closet Flush Valves	22 40 00	A								
Water Closet Seats	22 40 00	A								
Water Closets	22 40 00	A								
Water Cooler Carriers	22 40 00	A								
Water Coolers	22 40 00	A, C								
<i>Division 23</i>										
Pipe and Fittings, Natural-Gas	23 11 23	E								
Sediment Traps, Natural-Gas	23 11 23	A								
Valves, Natural-Gas	23 11 23	A								
Division 33										
Area Drains	33  	A								
Manholes	33  	A								

SECTION 22 05 00.35 - SPECIAL PROCEDURES FOR PLUMBING

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 22 05 43, "Underground Plumbing Systems."

1.02 QUALITY ASSURANCE

- A. Standard: Occupational Safety and Health Administration (OSHA), including 29 CFR 1910.147.

1.03 SAFETY

- A. Contractor shall be solely responsible for safety precautions and programs in connection with the Work. The Owner or A/E will not be responsible for the Contractor's failure to employ proper safety procedures.
- B. Maintain the Project in accordance with Federal, State, and Local Safety and Insurance Standards. All procedures shall comply with the latest OSHA regulations.
- C. Contractor shall provide head, eye, and ear protection for each of its employees in accordance with OSHA regulations.
- D. Exercise precaution for the protection of persons and property.
 - 1. Provide guard rails, barricades, enclosures, canopies, passageways, lanterns, warning lights, and other protective safety devices required by the AHJ, and required to protect persons and property against accidentally dropped materials or other construction hazards.
 - 2. Provide guard lights at all barricades, railings, obstructions in the streets or sidewalks, trenches, or pits.
- E. Provide protection as may be required to prevent glass breakage. Replace broken glass at no cost to the Owner.
- F. DO NOT USE POWDER-ACTUATED ANCHORS OR FASTENERS.
- G. Excavation Safety: Refer to Section 22 05 43.

SPECIAL PROCEDURES FOR PLUMBING

1.04 ENVIRONMENTAL PROCEDURES

A. Hazards Control:

1. Do not dispose of volatile wastes, such as mineral spirits, oil, or paint thinner, in storm or sanitary drains. Dispose of such wastes properly.
2. Store volatile wastes in covered metal containers, and remove from premises daily.
3. Prevent accumulation of waste that creates hazardous conditions.
4. Provide adequate ventilation during use of volatile or noxious substances.

END OF SECTION 22 05 00.35

SECTION 22 05 00.40 - QUALITY REQUIREMENTS FOR PLUMBING

PART 1 GENERAL

1.01 DESCRIPTION

- A. Provide Work of the highest quality, complete in every respect and conforming to the accepted practices and standards of the Trades involved.
- B. Further definition of quality is given by reference to various Laws, Codes, Standards, and Regulations.

1.02 REGULATORY REQUIREMENTS

- A. All work on this Project shall conform to the currently enforced Edition of all applicable National, Federal, State, and Local Laws, Rules, Regulations, Standards, and Building Codes, and any local ordinances and requirements pertaining to this work, unless otherwise noted.
- B. Any Law, Regulation, Standard, or Code referred to in other Sections of Division 22 is included in its entirety as a part of these Specifications.
- C. Give notice and comply with all Laws, Rules, Regulations, Ordinances, and lawful orders of the AHJ bearing on the performance of the Plumbing Work.
 - 1. If the Contractor observes that any of the Contract Documents are at variance therewith in any respect, it shall promptly notify the A/E in writing, and any necessary changes will be arranged by the A/E.
 - 2. If the Contractor performs any Work knowing it to be contrary to such Laws, Rules, Regulations, and Ordinances and fails to give prior notice to the A/E, the Contractor shall assume full responsibility for, and shall bear all costs associated with, correcting the Work.
- D. Only a Contractor and craftsmen licensed by the AHJ shall provide this Plumbing Work.
- E. Methods and materials shall be certified where noted in the individual Specification Sections.
- F. All equipment and appliances installed on this Project shall bear the label of an Approved Testing Agency, and shall be installed in accordance with the Manufacturer's instructions for the labeled equipment and appliances.

fixed

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revised to Illinois
Administrative Code.
Don't know a specific
reference, but this
shouldn't be on the
p.c. anyways

- G. All structural steel used on this ~~Project~~ shall be manufactured in the United States, in accordance with Ohio Revised Code 153.011, unless specifically approved otherwise by the Owner.

1.03 APPLICABLE PERMITS, FEES, BONDS, LICENSES, AND CERTIFICATIONS

- A. The A/E will submit all Contract Drawings and Specifications and the plan approval application to the AHJ, pay the application fees, and obtain plan approval.
- B. Contractor shall secure and pay for any ADDITIONAL permits, governmental fees, bonds, and licenses required for the proper execution and completion of the Plumbing Work.
- C. Contractor shall pay for all other fees and other charges related to Plumbing Work and payable to Utility Companies or Code Enforcement Agencies.
- D. The Owner will pay for all water and sewer tapping fees, capacity fees, front footage fees, and water meter costs. The Owner will apply for service.
- E. Contractor shall pay for the gas line tap charges, house line test and inspection, and all other fees assessed by the Gas Utility Company.

1.04 REFERENCES

- A. Titles, Abbreviations, and Symbols: Titles and abbreviations may be used in these Specifications. Abbreviations may be shown on the Plumbing Drawings. Refer to the list of abbreviations attached to this Section. Refer also to the symbols lists shown on the Drawings for further abbreviations. All titles and abbreviations may not necessarily apply to this Work.
- B. Definitions:
1. Provide: To furnish, install, and connect to make completely ready for regular operation.
 2. Furnish: To supply or deliver to site complete with all required accessories and installation instructions.
 3. Install: To mount, erect, hang, or fasten in place, and connect to make ready for regular operation.
 4. Concealed: Either embedded in masonry or other construction, or installed below floor slab, behind wall furring, within chases or soffits, within walls, within double partitions, above ceilings, in trenches, in tunnels, or within crawl spaces.
 5. Exposed: In full or partial view; not "Concealed" as defined above.
 6. Accessible Ceiling: Lay-in ceiling with removable ceiling tiles.

7. Piping: Pipe, fitting, flanges, valves, controls, specialties, hangers, bracing, insulation, and other items required or necessary.
8. Refer to additional Definitions in Division 01 and in the AHJ's Building and Mechanical Codes.

TITLES, ABBREVIATIONS, AND SYMBOLS

#	Number	ANSI	American National Standards Institute
&	And		
∠	Angle	APPROX.	Approximately
@	At	ARCH.	Architect
Ø	Diameter	ARR'T.	Arrangement
Ø	Round OR Phase	ASA	Acoustical Society of America
A.	Compressed air	ASHRAE	American Society of Heating, Refrigerating, and Air Conditioning Engineers
A.D.	Access door OR Area drain		
A.F.D.	Acid floor drain	ASME	American Society of Mechanical Engineers
A.F.F.	Above finished floor	ASPE	American Society of Plumbing Engineers
A.P.	Access panel		
A.W.C.O.	Acid wall cleanout	ASSE	American Society of Sanitary Engineering
A/E	Architect/Engineer	ASTM	American Society for Testing & Materials
AABC	Associated Air Balance Council		
AB.	Above	AUTO.	Automatic
ACI	American Concrete Institute	AV	Acid vent
ACPA	American Concrete Pipe Association	AW	Acid waste
ADA-AG	Americans with Disabilities Act — Accessibility Guidelines	AWG	American wire gauge
ADC	Air Diffusion Council	AWS	American Welding Society
AFCI	Arc-fault circuit interrupter	AWWA	American Water Works Association
AH	Air handler	B.D.D.	Backdraft damper
AHJ	Authority Having Jurisdiction	B.T.	Bathtub
AHRI	Air-Conditioning, Heating, and Refrigeration Institute	B.V.	Backwater valve
ALT.	Alternate	BAL.	Balancing
ALUM.	Aluminum	BFP	Backflow preventer
AMCA	Air Movement and Control Association	BHP	Brake horsepower
amp	ampere	BICSI	Building Industry Consulting Services International
		BLDG.	Building

BSBD.RAD.	Baseboard radiation	COORD.	Coordinate
BSMT.	Basement	CSA	CSA Group (formerly the Canadian Standards Association).
BTM.	Bottom	CSI	Construction Specifications Institute
BTU	British thermal unit	CT	Current transformer
c.	Conduit	CU IN.	Cubic inch
C.B.	Catch basin	CU. FT.	Cubic feet
C.I.	Cast iron	CUH	Cabinet unit heater
C.O.	Clean out	Cx	Commissioning Agent
C/C	Center to center	D.	Deep
C/L	Center line	D.F.	Drinking fountain
CAB.	Cabinet	D.L.	Door louver
CAP.	Capacity	D.M.	Damper motor
CEIL./CLG.	Ceiling	D.S.	Downspout
CFC	Chlorofluorocarbon	D-B	Design-Build OR Design-Builder
CFH	Cubic feet per hour	DB	Dry bulb
CFM	Cubic feet per minute	DBL.	Double
CFR	Code of Federal Regulations	DCW	Domestic cold water
CGA	Compressed Gas Association	DET.	Detail
CHEM.	Chemical	DHW	Domestic hot water
CISPI	Cast Iron Soil Pipe Institute	DHWR	Domestic hot water return
CMA	Construction Manager Agent/Advisor	DIA. / Ø	Diameter
CMR	Construction Manager at Risk	DIFF.	Diffuser
COL.	Column	DIM.	Dimension
COMB.	Combination	DISCH.	Discharge
CONC.	Concrete	DN.	Down
COND.	Condensate OR Condenser	DOE	United States Department of Energy
CONN.	Connection OR Connect	DPR.	Damper
CONST.	Construction	DR.	Door
CONTR.	Contractor	DW	Distilled water
CONV.	Convactor OR Converter	DWG.	Drawing

DWH	Domestic water heater	F.H.C.	Fire hose cabinet
DWV	Drain, waste and vent	F.H.E.C.	Fire hose/extinguisher cabinet
E.A.T.	Entering air temperature	F.V.	Flush valve
E.C.	Electrical Contractor	FCC	Federal Communications Commission
E.S.P.	External static pressure	FD	Floor drain
E.T.	Expansion tank	FDA	Food and Drug Administration
E.W.T.	Entering water temperature	FHA	Federal Housing Administration
EA.	Each	FIG.	Figure
EFF.	Efficiency	FIN.	Finish
EIA	Electronic Industries Alliance	FIN.RAD.	Finned radiation
ELEC.	Electric OR Electrical	FLEX.	Flexible
ELEM.	Element	FLR. / FL.	Floor
ELEV.	Elevation	FMG	Factory Mutual Global
ENGR.	Engineer	FOR	Fuel oil return
ENT.	Entering	FOS	Fuel oil supply
EPA	Environmental Protection Agency	FPC / FSC	Fire Protection / Fire Suppression Contractor
EQUIP.	Equipment	FPM	Feet per minute
ETL	Electrical Testing Laboratories	FT.	Feet
EW	Eye wash	FT.HD.	Feet of head
EWC	Electric water cooler	FURN.	Furnish(ed)
EX.	Existing	G	Gas
EXH.	Exhaust	G.C.	General Contracting OR General Contractor
EXIST.	Existing	G.I.	Grease interceptor
EXT.	Exterior	G.T.C.	General Trades Contractor
F & T	Float and thermostatic	gal.	Gallon
F.	Fan OR Fire	GALV.	Galvanized
F.D.	Fire damper	GEN.	General
F.E.	Fire extinguisher	GFI / GFCI	Ground-fault circuit interrupter
F.E.C.	Fire extinguisher cabinet OR Food Service Equipment Contractor	GPM	Gallons per minute
F.H.	Fire hydrant		

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GR	Grille	IEC	International Electrotechnical Commission
GRAV.	Gravity	IEEE	Institute of Electrical and Electronics Engineers, Inc.
Grd.	Ground	IESNA	Illuminating Engineering Society of North America
H./HT.	Height	IN.	Inside OR Inches
H.P.	High pressure	IND.	Indirect
H.S.	Hair strainer	IND. U.	Induction unit
H'STAT	Humidistat	INSUL.	Insulation
HAC	Heating and air conditioning	INT.	Interior
HB	Hose bibb	INV.	Invert
HCFC	Hydrochlorofluorocarbon	INV. ELEV.	Invert elevation
HHS	United States Department of Health and Human Services	IPS	Iron pipe size
HORIZ.	Horizontal	ISO	International Organization for Standardization
HP	Horsepower	J.R.	Janitor's receptor
HPC	High pressure steam condensate	JCAH	Joint Commission for Accreditation of Hospitals
HPS	High pressure steam	kVA	Kilovolt-ampere
hr.	Hour	L.	Length
HTG.	Heating	L.A.T.	Leaving air temperature
HTR.	Heater	L.W.T.	Leaving water temperature
HUD	United States Department of Housing and Urban Development	LAB	Laboratory
HV	Heating and ventilating OR High velocity	LAV	Lavatory
HVAC	Heating, ventilating, and air conditioning	lb.	pound
I.D.	Inside diameter	LEED	Leadership in Energy and Environmental Design
IAQ	Indoor air quality	LH	Left hand
IBC	International Building Code	LP	Liquefied petroleum
ICC	International Code Council	LPC	Low pressure steam condensate
ICEA	Insulated Cable Engineers Association, Inc.	LPS	Low pressure steam
IDF	Intermediate distribution frame	LV'G.	Leaving

M.A.	Mixed air	NEBB	National Environmental Balancing Bureau
M.O.	Motor operated	NEC	National Electrical Code
MAN. DPR.	Manual damper	NECA	National Electrical Contractors Association
MAT.	Material	NEMA	National Electrical Manufacturers Association
MAX.	Maximum	NESHAPs	National Emissions Standards for Hazardous Air Pollutants
MBH	1,000 British thermal units/hour	NFPA	National Fire Protection Association
MECH.	Mechanical	NIOSH	National Institute for Occupational Safety and Health
MERV	Minimum efficiency reporting value	NIST	National Institute of Standards and Technology
MET./MTL.	Metal	NO. OR #	Number
MEZZ.	Mezzanine	NOM.	Nominal
MFR.	Manufacturer	NP	Not permitted
MH	Manhole	NPT	National Pipe Thread Tapered
MIN.	Minimum	NSF	National Sanitation Foundation
MISC.	Miscellaneous	NSPE	National Society of Professional Engineers
MP	Multiple Prime	O.	Oxygen
MPS	Medium pressure steam	O.A.	Outside air
MS	Motor starter	O.D.	Outside diameter
MSS	Manufacturers Standardization Society of the Valve and Fittings Industry Inc.	O.I.	Oil interceptor
MTD./MT.	Mounted OR Mount	O.V.	Outlet velocity
N.I.C.	Not in contract	O/C	On center
N.T.S.	Not to scale	OAC	Ohio Administrative Code
N2	Nitrogen	ODH	Ohio Department of Health
N2O	Nitrous oxide	ODOE	Ohio Department of Energy
NADCA	National Air Duct Cleaners Association	ODOT	Ohio Department of Transportation
NAECA	National Appliance Energy Conservation Act	OPG.	Opening
NB	National Board of Boiler and Pressure Vessel Inspectors	OPP.	Opposite
NCPI	National Clay Pipe Institute	ORC	Ohio Revised Code

OSHA	Occupational Safety and Health Administration	RPM	Revolutions per minute
OU./OZ.	Ounce	RTA/C	Rooftop air-conditioning unit
P.D.	Pressure drop	RW	Raw water
P.I.V.	Post indicator valve	RWC	Rainwater conductor
P/L	Property line	S	Sink
PDI	Plumbing & Drainage Institute	S & R	Supply and return
PEI	Petroleum Equipment Institute	S. DPR.	Smoke damper
PLBG.	Plumbing	S.A.	Shock absorber OR Supply air
PNEU.	Pneumatic	S.F.	Square feet
PRESS.	Pressure	S.M., S/M	Sheet metal
PROP.	Propeller	S.P.	Static pressure
PRV	Pressure-reducing valve	S.S.	Service sink OR Storm sewer
PSF	Pounds per square foot	S.S., S/S	Stainless steel
PSI	Pound-force per square inch	SAN.	Sanitary sewer
PSIG	Pound-force per square inch gauge	SB	Shampoo bowl
PT	Plaster trap OR Potential transformer	SD	Shower drain OR Smoke detector
PVC	Polyvinyl chloride	SECT.	Section
R	Register	SHR.	Shower
R.A.	Return air	SHT.	Sheet
R.D.	Roof drain	SHT. MT'L.	Sheet metal
R/W	Right of way	SMACNA	Sheet Metal and Air Conditioning Contractors National Association
RAD.	Radius OR Radiation OR Radiator	SPD	Surge protection device
RECIRC.	Recirculating	SPEC.	Specification
REG.	Register	SQ.	Square
REINF.	Reinforced	SQ. FT.	Square feet
REL.	Relief	SQ. IN.	Square inch
REQ'D.	Required	ST	Sound trap
RH	Right hand	STAT	Thermostat
RHC	Reheat coil	STD.	Standard
RM.	Room	STL.	Steel

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STM.	Storm	VC	Vacuum cleaning
SUCT./S	Suction	VCP	Vitrified clay pipe
SW	Softened water	VERT.	Vertical
SW.	Switch	VIB. ISO.	Vibration isolator
T.W.	Tempered water	VOC	Volatile organic compound
T'STAT	Thermostat	VSP	Vitrified sewer pipe
TD	Temperature difference	W.	Width OR Water
TEMP.	Temperature	W.G.	Water gauge
THERM.	Thermometer OR Thermostat	W/	With
TIA	Telecommunications Industry Association	W/O	Without
TYP.	Typical	W/W	Wall to wall
UC	Undercut	WAT.	Water
UH	Unit heater	WB	Wet bulb
UL	Underwriters Laboratories, Inc.	WC	Water closet OR Water column
UR	Urinal	WH	Wall hydrant OR Water heater
USGBC	United States Green Building Council	WO	Waste oil
UV	Unit ventilator	WOG	Water/oil/gas
V.	Vent	XFMR	Transformer
V.T.R.	Vent through roof	YD.	Yard
VAC.	Vacuum	YH	Yard hydrant

END OF SECTION 22 05 00.40

SECTION 22 05 00.45 - TESTING AND INSPECTING SERVICES FOR PLUMBING

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 22 05 43, "Underground Plumbing Systems."

1.02 INSPECTIONS, TESTS, AND APPROVALS

- A. The Contract Documents, Laws, Ordinances, Rules, Regulations, or Orders of any AHJ may require the Work to be inspected, tested, or approved.
- B. Contractor shall bear all costs of such inspections, tests, or approvals.
- C. Contractor shall pay all professional engineering fees involved in required testing of equipment.
- D. Contractor shall arrange and pay for all inspections required to obtain a Certificate of Occupancy. Contractor shall include this cost in its bid.
- E. Arrange for inspection of the Work by the AHJ.

PART 2 PRODUCTS (NOT APPLICABLE)

PART 3 EXECUTION

3.01 TESTS AND INSPECTIONS

- A. Notify the A/E of all scheduled tests and adjustments at least (48) hours before they are scheduled, so that the A/E may witness same. If the Contractor performs any test or adjustment without the A/E present, or without proper notification, the Contractor shall perform the test or adjustment a second time, in the presence of the A/E. Coordinate all test schedules with the Owner to minimize inconvenience.
- B. Provide all required testing and obtain approvals. Secure required certificates of inspection, testing, or approval, and include them in the Operating and Maintenance Manual.
- C. After Work has been completed, but before pipe covering has been applied, test each system in accordance with the requirements of other Sections of this Specification. At the test pressures, the circulation shall be free and the piping shall be proven free of leaks.

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- D. Concealed lines shall be tested and approved before being concealed. If a leak appears during the final test, repair the line and any damage resulting from the leak.
- E. Coordinate the enclosure or concealment of the Work with the required inspections and tests so as to minimize the necessity of uncovering work for that purpose.
- F. Should any of the Work be covered up or enclosed prior to completion of all required inspections, tests, and approvals, uncover the Work and, after it has been completely inspected, tested, and approved, make all repairs and replacements with such materials and workmanship as are necessary to secure the approval of the A/E, and at no additional cost to the Owner.
- G. Furnish all test pumps, gauges, equipment, and personnel required, and test to demonstrate the integrity of the finished installation to the approval of the AHJ and the A/E.
- H. Check each piece of equipment for defects and verify that all parts are properly furnished and installed, that all items function properly, and that all adjustments have been made.
- I. Conduct such tests and adjustments of equipment to verify performance requirements or to meet the requirements of the A/E or AHJ. Submit data taken during such tests to the A/E.
- J. Test and check all control valves, drain valves, gauges, alarm devices, and waterflow and supervisory switches for proper operation. Submit a written record of these tests to the A/E.
- K. Demonstrate to the Owner that all Plumbing systems are operating properly.

3.02 FACTORY INSTALLATION AND START-UP

- A. For those items of equipment that are to be installed, tested, started up, and certified by a factory-trained Representative, furnish a letter from the Manufacturer to the A/E stating that this service shall be provided for this Project, describing the scope of services to be provided, and disclosing the name of the Representative assigned to provide the required services.

3.03 FINAL OPERATING TESTS AND PROCEDURES

- A. Prior to Contract completion, conduct system operational tests for a period of at least (5) days, not necessarily consecutive, as scheduled by the Owner, to demonstrate fulfillment of the requirements of the Contract. During this time, adjust equipment so that it will perform as the Manufacturer intended, and so that systems will function as designed.
- B. Operate each system in every mode of operation, and check the position of valves, switches, and other devices for proper closure, operation, and switching.

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3.04 CERTIFICATE

- A. Furnish approved Certificate of Final Inspection.

3.05 FOLLOW-UP INSPECTIONS

- A. Make an inspection within (90) days after occupancy of the building to make minor adjustments as needed to ensure that all equipment is operating properly and that satisfactory conditions are being maintained throughout, and to satisfy any unusual conditions. Schedule with the A/E. Consult with the Owner.
- B. A minimum of (1) month before the end of the guarantee period, contact the A/E and Owner to discuss system operation, and to plan for the future care and maintenance of the system.
- C. (1) month before the end of the guarantee period, contact the Owner and perform an inspection to review any items needing correction.
- D. Refer to Section 22 05 43 for additional inspection requirements related to excavation and backfill.

END OF SECTION 22 05 00.45

SECTION 22 05 00.50 - TEMPORARY FACILITIES AND CONTROLS FOR PLUMBING

PART 1 GENERAL

1.01 DESCRIPTION

- A. Provide and maintain temporary facilities and controls required for construction as long as needed for the safe and proper completion of the Work and remove upon completion of the Work.

1.02 QUALITY ASSURANCE

- A. Standards: Comply with the provisions of the following Codes:
 - 1. United States Department of Labor, Occupational Safety and Health Administration, Part 1926, "Safety and Health Regulations for Construction."
 - 2. Applicable Local Building Codes.

1.03 TEMPORARY CONSTRUCTION

- A. Materials shall be adequate in capacity for the required usage, shall not create unsafe conditions, and shall not violate applicable Codes and Standards.
- B. Contractor shall be solely responsible for the structural design of all temporary items that it uses in the construction of the building, or that become a permanent part of the building, including, but not limited to:
 - 1. Hoisting.
 - 2. Temporary bracing for structural steel.
 - 3. Shoring for concrete and masonry work.
 - 4. Shoring for cut earth banks.
 - 5. Suspended ceilings, equipment, walls, etc.
- C. Provide hoists, staging, rigging, scaffolding, runways, and necessary appurtenances required for the performance of the Work.
- D. Provide necessary tarps or other appropriate cover material to protect openings.
- E. Provide effective means of dust control. Dust enclosures shall extend tight to the structure and extend around all pipes, ducts, and similar items. Enclosures shall confine dust and

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dirt to the area of construction and completely seal off and isolate the area of construction from adjacent Owner-occupied areas.

- F. Completely remove temporary materials and equipment when their use is no longer required.
- G. Clean and repair damage caused by temporary installation or use of temporary facilities.

END OF SECTION 22 05 00.50

SECTION 22 05 00.60 - PRODUCT REQUIREMENTS FOR PLUMBING

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Division 00, "Procurement and Contracting Requirements."

1.02 DESIGN BASE MANUFACTURER STANDARD

- A. The Drawings and Specifications reflect a design based on the specific equipment requirements and configuration for a Design Base Manufacturer. Design coordination of equipment with the building and with other Trades has been made for this specific Model and Manufacturer of equipment. Where several Manufacturers are listed for an item of equipment or material, the first-named shall be considered the Design Base Manufacturer Standard.

1.03 OTHER ACCEPTABLE COMPONENTS (MANUFACTURERS)

- A. Other Manufacturers listed in the Specifications shall be acceptable; however, whenever the Contractor elects to furnish specified equipment or material manufactured by other than the Design Base Manufacturer, the Contractor shall be responsible for the cost and coordination of all modifications required to accommodate the selected equipment or material, including any Work of other Trades that might be affected. Where changes to other Trades' Work are required, the Contractor shall include the additional costs of all such Work in its bid.
- B. Any specified Equipment Manufacturer furnished by the Contractor, other than the Design Base Manufacturer, shall be, in the opinion of the A/E, equivalent in quality, design, features, performance, arrangement, and appearance to that of the Design Base equipment or material, including any special features or requirements.
- C. The inclusion of other Manufacturers in these Specifications shall not be construed to infer that each Manufacturer's product has been reviewed by the A/E for this application, or that it will provide all the features or capabilities required, or that its physical dimensions can be accommodated in the available space. Verify that the proposed equipment will provide all features and accommodations provided by the Design Base Manufacturer, and that it will fit in the available space.
- D. The A/E will not give consideration to any Manufacturer that the Contractor proposes as a Substitution, unless the A/E approves the Manufacturer during the bid period and specifically names the Manufacturer in the Contract Documents or Addenda thereto.

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- E. Where deemed necessary by the A/E, the Contractor shall, at no additional cost to the Owner, prepare layouts for these other brands of equipment that may have different dimensional or service requirements from the Design Base Manufacturer Standard. Submit these layouts to the A/E for review.
- F. Contractor shall reimburse the Owner for the cost of any design changes incurred by the A/E in the preparation of revised Drawings or Specifications to accommodate the use of any Manufacturer other than the Design Base Manufacturer.

1.04 SUBSTITUTIONS

- A. Refer to the Bid Documents for substitution requirements. If this Specification Section conflicts with the Bid Documents, comply with the Bid Documents.
- B. Contractor shall submit information for review no later than (10) days prior to the bid opening to the A/E on any other proposed equipment or material that the Contractor desires to use.
- C. Contractor shall be responsible for the same costs of coordination and modifications, etc., listed above in Article 1.03, "Other Acceptable Components (Manufacturers)."
- D. If the A/E determines that the proposed Manufacturer is acceptable, the A/E will issue an Addendum adding that Manufacturer to the Specifications and/or the Drawings prior to the bid opening.

1.05 EQUIPMENT SUITABILITY

- A. All equipment provided shall perform as intended. All items listed shall function properly, and as the Manufacturer intended. Install equipment according to the Manufacturer's recommendations. Properly attach equipment to the floor, wall, or structure. Each item of equipment shall be compatible with all other accessories or hook-ups, including piping, controls, wiring, and other equipment not furnished by the equipment Manufacturer, but required to achieve the intended function.

1.06 MISCELLANEOUS ACCESSORIES

- A. Provide any additional adapters, fittings, trim, structural steel angles, channels, strut channels, brackets, or other appurtenances to securely install all items of equipment specified or shown on the Drawings.
- B. All steel installed outside or exposed to moisture shall be hot-dipped galvanized.
- C. These accessories are required even though they may not be shown or detailed on the Drawings.

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- D. Installation shall be compatible with the building construction on which the item is to be located.
- E. Verify the type of construction prior to ordering the equipment item, so that all required accessories are included.

1.07 QUANTITIES

- A. Equipment may be referred to in these Specifications, or on the Drawings, as either singular or plural; the Contractor shall verify the exact number of items required to complete its Work.

END OF SECTION 22 05 00.60

SECTION 22 05 00.71 - EXAMINATION AND PREPARATION FOR PLUMBING

PART 1 GENERAL

1.01 EXAMINATION OF SITE

- A. Certain existing conditions may affect the manner or sequence of the performance of the Work.
- B. Before submitting its Bid, it is recommended that the Contractor visit the site of the proposed Project.
- C. After receipt of Bids, no adjustments to Bids will be permitted for lack of knowledge of Project conditions.
- D. Verify and reconcile Work required by the Contract Documents with existing conditions at the Site.
- E. Should the Contractor note any discrepancies during the Bidding Period, it shall notify the A/E immediately, in writing, to permit issuance of an Addendum to prevent misunderstandings at a later date.

1.02 PREPARATION

- A. Provide a Superintendent, on site, whose duties include the directing and supervising of work. Inform the A/E of the Superintendent's name and phone number, and the method of contact when the Superintendent is not at the site.
- B. As soon as possible after the Award of the Contract and the approval of shop drawings, the Contractor shall place orders for materials and equipment required for its Work. Immediately inform the A/E in writing as to any materials and equipment that cannot be obtained within the required time period, either due to conditions of the market or other governing factors.
- C. Inspect each item of materials or equipment immediately prior to installation. Reject damaged and defective items.
- D. Contractor shall determine erection procedures and sequence the construction to keep its work on schedule and to ensure the safety of the building and its occupants. This includes supplying any temporary bracing, guys, or tie-downs that might be required. Such materials shall remain the Contractor's property after completion of the Project.
- E. Provide attachment and connection devices and methods for securing work. Secure all work true to line and level and within recognized industry tolerances. Allow for expansion and building movement.
- F. Recheck measurements and dimensions of the Work as an integral step of starting each installation.

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- G. Where mounting heights are not indicated, individual units generally can be mounted at industry-recognized standard mounting heights for the particular application indicated. However, all mounting heights shall be approved by the A/E.

1.03 COORDINATION OF WORK

- A. Contractor shall coordinate its Work carefully with all other Contractors.
- B. Do not install work without first coordinating the layout, routing, etc., with other Contractors whose work may be affected.
- C. Coordinate location of all work with other Contractors and equipment.
- D. Consult all Contract Drawings that may affect the location of any equipment, apparatus, and piping, and make any other adjustments in location to secure coordination.
- E. Coordinate locations of all Plumbing Work with hangers, sprinklers, piping, ductwork, diffusers, grilles, registers, cable trays, occupancy/vacancy sensors, bus ducts, luminaires, speakers, smoke and heat detectors, conduits, and other equipment, and with ceilings, soffits, structural members, joists, purlins, metal decks, bracing, etc.
- F. Notify the A/E immediately of any changes in location or other conflicts that may affect the work of other Contractors.
- G. Contractor shall be responsible for the cost of additional engineering work required for changes to the work as shown or described, due to the relocation of items requested by the Contractor.
- H. Review all equipment nameplate ratings and advise the A/E immediately of any system design changes required to wire the equipment properly.

END OF SECTION 22 05 00.71

SECTION 22 05 00.73 - EXECUTION OF PLUMBING

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Division 00, "Procurement and Contracting Requirements."
- B. Division 01, "General Requirements."
- C. Division 09, "Finishes."

1.02 DESCRIPTION

- A. Contractor shall be solely responsible for construction means, methods, techniques, sequences, and procedures in connection with the Work.

1.03 QUALITY ASSURANCE

- A. Standard: National Fire Protection Association (NFPA). 2023 NFPA 262, "Standard Method of Test for Flame Travel and Smoke of Wires and Cables for Use in Air-Handling Spaces."

1.04 CONTRACT DOCUMENTS

- A. The Contract Drawings are schematic and show approximate locations, general arrangement, and general extent of Work. Not all items of work required for a complete and operating system are shown on the Drawings. Refer also to the Specifications for additional work that shall be provided. If a conflict occurs, the Contractor is responsible for the more restrictive requirement.
- B. The Contract Drawings are not intended to show every vertical or horizontal offset that may be necessary to complete the systems. Having piping and fittings fabricated and delivered in advance of making actual measurements shall not be sufficient cause to avoid making offsets or other changes as may be necessary to install piping and equipment.
- C. Verify exact locations in the field, and coordinate with all other Contractors.
- D. The A/E shall approve, in writing, significant deviations from the Drawings.
- E. The A/E reserves the right to make minor changes in location that do not require additional labor or material, up to the time of roughing-in, without additional cost. The A/E reserves the right to determine what is "significant" and what is "minor."
- F. If a conflict occurs between the Drawings and Specifications, or an overlap of work among the Trades become evident, immediately submit a written request for an interpretation or clarification from the A/E, who shall determine which interpretation has precedence. Refer to Division 00 General Conditions. In such event, none of the Contractors shall assume that it is relieved of the Work that is specified under its branch unless instructions are received, in writing, from the A/E.

1.05 UTILITIES AND OUTAGES

- A. Locate and identify all existing utilities prior to construction.
- B. Where necessary, make minor relocations to permit installation of the Plumbing Work.
- C. Advise the A/E immediately of major conflicts to permit modification of the Contract Documents prior to bidding.
- D. Conform to all Utility Company requirements.
- E. Closely coordinate any utility service shutdowns or outages, interruptions, or downtime with the A/E and Owner.
- F. Provide temporary service during the curtailment, if required by the A/E.
- G. Avoid inconveniencing the Owner.
- H. Notify the Owner at least (10) full working days in advance of commencing work in the area of existing utilities.
- I. Contractor shall alert occupants of nearby premises of any emergency conditions that arise as a result of its work in connection with existing utilities.

1.06 PROTECTION

- A. Unless equipment and material can be protectively stored in a manner acceptable to the A/E, they shall not be delivered to the site until the Work is ready to receive them.
- B. Protect all equipment and materials during construction from damage by weather, water, dirt, paint droppings, welding and cutting spatters, and other construction activities.
- C. All materials or equipment stored outside shall be elevated and protectively covered in a secured and locked area.
- D. Store materials and equipment sensitive to weather or construction conditions inside. Where necessary, store sensitive equipment in a heated area.
- E. During construction, cover all equipment and other items that are susceptible to damage until they can be installed in place.
- F. Immediately repair or replace damaged equipment or materials to the satisfaction of the A/E and at no additional cost to the Owner.
- G. Contractor shall protect the building and other Contractors' material and equipment from damage caused by its Work. Protect floors from cutting oil and chips.
- H. Use all means necessary to protect materials before, during, and after installation.
- I. Refer also to individual Specification Sections for specialized protection.

1.07 PLUMBING LUBRICATION AND PACKING

- A. Contractor shall lubricate all the equipment that it furnishes with the correct grade, type, and quantity of lubrication before placing equipment and valves in service.

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- B. Refer to the Equipment Manufacturer's recommended lubricants. Furnish equal lubricants as supplied by Castrol, Mobil, Shell, or the Owner's present supplier.
- C. Follow the Equipment Manufacturer's recommendations for specific lubricants and application schedule.
- D. Check each shaft or valve stem containing a packing gland for condition and examine for proper packing.
- E. When filling systems initially for hydrostatic pressure tests, adjust valve packing glands to finger-tight and allow packing to absorb water for (5) minutes prior to tightening packing nuts.
- F. Maintain lubrication and packing seals during construction and ensure that, at the time of final acceptance, all are operating properly.
- G. Contractor shall repair any damage caused by improper lubrication of the equipment that it furnished, at no expense to the Owner.
- H. Incorporate lubrication instructions in the Operating and Maintenance Manual.

1.08 EQUIPMENT ACCESS

- A. Locate all equipment to provide sufficient access to operate equipment or to service items requiring periodic maintenance. Coordinate with the structure and with all other Contractors' work.

1.09 PLUMBING ROUGH-IN AND FINAL CONNECTIONS

- A. Provide service rough-ins.
- B. Provide plumbing equipment shown on plumbing project documents, unless otherwise noted.
- C. Provide piping, valves, and specialties as specified under other Sections of these Specifications.
- D. Provide stops on supply connections to equipment not otherwise furnished with integral stops.
- E. Provide line-size indirect connections from sterilizers to hub drain. Coordinate sizes of drains with equipment installed.
- F. Refer to approved equipment drawings for exact rough-in sizes and locations.
- G. Coordinate rough-in locations with other Contractors and the Owner.
- H. Coordinate rough-in with all items of equipment.
- I. Make final connections to **new** equipment furnished by the **Lead Contractor** or others, unless otherwise noted.
- J. Food Service Equipment:
 - 1. Provide service rough-in for future service for kitchen sinks.

I wrote this, does it sound okay? What was there previously made it sound like the p.c. wasn't providing any fixtures at all, when they should be providing all but the kitchen fixtures

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2. **Lead contractor** shall furnish kitchen sink tailpieces.
3. **Lead Contractor** shall set kitchen equipment in place.
4. Receive, uncrate, and install kitchen sinks, disposers, trim, and faucets.
5. Install any other plumbing equipment furnished by the Equipment Contractor (i.e., vacuum breakers, gas valves, etc.).
6. Provide final plumbing connections to kitchen sinks, disposers, and faucets.

1.10 ACCESS PANELS

- A. **Lead** Contractor shall be responsible, unless otherwise noted, for all access panels shown on the Drawings, or where otherwise required, to permit convenient access to concealed plumbing equipment, boxes, connections, controls, or other devices requiring service, adjustment, or maintenance. Refer to the Architectural Drawings for location.
- B. Acceptable Manufacturers: Inryco-Milcor, Babcock-Davis, Bilco, Boico, Cesco, Inland-Ryerson, JL, Milco, Karp, Larsen's Manufacturing Co., Mifab, Mitco, or Zurn.
 1. Manufacturer and Model Number listed below shall be considered to be the Design Base Manufacturer and Model Number. Other Acceptable Manufacturers will be permitted, unless specifically excluded.
- C. Access Panels:
 1. Panels shall be 18 in. x 18 in., unless larger panels are shown or required. Panels shall be steel construction (except where aluminum or stainless steel is specified) with overlapping edges and concealed hinge door with screwdriver lock. Mounting frames shall be compatible with the building construction material in which they are installed. Each access panel in fire-rated shaft walls and in fire-rated ceilings shall be B Label.
 2. Ceiling access panels:
 - a. Gypsum: Inryco-Milcor Style DW. Prime-painted for finish painting with ceiling.
 - b. Plaster:
 - 1) Inryco-Milcor Style AP. With finish material same as ceiling material.
 - 2) Inryco-Milcor Style K. Prime-painted for finish painting with ceiling.
 - c. Acoustical concealed spline: Inryco-Milcor Style AT.
 - d. Fire-rated: Inryco-Milcor Style ATR.
 3. Wall access panels:
 - a. Gypsum: Inryco-Milcor Style DW. Prime-painted for finish painting with wall.
 - b. Plaster: Inryco-Milcor Style K. Prime-painted for finish painting with wall.

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- c. Masonry and tile:
 - 1) Inryco-Milcor Style M Standard. Prime-painted for finish painting with wall.
 - 2) Inryco-Milcor Style M. Stainless steel.
 - d. Fire-rated: Inryco-Milcor. Fire-rated access door, prime-painted for finish painting with wall.
 - 4. Floor access panels:
 - a. Finished: Babcock-Davis Type FT. Edging shall accommodate type floor covering required. Refer to Room Finish Schedule.
 - b. Interior, bare concrete floor: Babcock-Davis Type FC. Checkered plate cover.
 - c. Exterior, concrete slab: Babcock-Davis Type GT. Gutter and 1-1/2 in. drainage coupling, checkered plate cover.
 - D. Coordination:
 - 1. Coordinate closely with other Contractors to ensure that all panels furnished by the various Contractors shall be identical.
 - 2. Coordinate the location and installation of each access panel with the Lead Contractor, other Contractors, and equipment.
 - 3. Advise other Contractors of specific requirements relating to access panel installation, including sizes of openings to receive access panel and frame, as well as locations of supports, inserts, and anchoring devices.
 - E. Determine specific locations and sizes for access panels needed, but not indicated on the Drawings, to gain access to concealed equipment.
 - F. Plan panel locations to clear ceiling luminaires, grids, and obstructions, and to allow, wherever possible, full shoulder clearance beside the device to be inspected, adjusted, or repaired.
- 1.11 AIR-HANDLING PLENUMS
- A. Where space is used for air handling, such as above ceilings and elsewhere, do not install combustible or noxious materials.
 - B. All materials shall be listed for use in air-handling plenums. All wiring shall be listed with 2023 NFPA 262.
- 1.12 BLOCK COURSE COORDINATION
- A. Mounting heights are called out on the Drawings for many items. Adjust equipment mounting heights to accommodate brick or block coursing.
 - B. Coordinate installation of all items in a masonry wall with the Lead Contractor and A/E.

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1.13 CUTTING AND PATCHING

- A. All cutting and patching shall be accomplished in a neat and workmanlike manner, acceptable to the A/E.
- B. Cutting:
 - 1. Contractor shall perform all cutting required for installing its own Work unless otherwise noted.
 - 2. Cutting shall be done with such tools and methods so as to prevent unnecessary damage to surrounding areas and equipment.
 - 3. Use rotary drills to cut holes through concrete, brick, plaster, or tile.
 - 4. No cutting shall be done that will, in any way, reduce the structural strength of the building. Cutting of structural support beams, joists, plates, or other structural members is strictly prohibited without the specific written consent of the A/E and the Structural Engineer. Should such cutting be necessary, consult the A/E and do not proceed further without written approval of the A/E.
- C. Patching:
 - 1. Lead Contractor shall be responsible for patching except as otherwise noted.
 - 2. Only a qualified Finish Tradesman, skilled in the respective craft required, shall perform patching.
 - 3. Patching shall match adjacent surface construction.
 - 4. Materials and equipment used in the patching work shall comply with requirements of other Sections of this Specification.
- D. Perform all cutting and patching promptly. Make all repairs to leave the entire Work in good condition, including all cutting, fitting, and drilling of masonry, concrete, metal, wood, plaster, or other materials, as specified for proper assembly, fabrication, installation, and completion of the Work.
- E. Coordinate drilling, welding, etc., and method of attachment to columns, joists, beams, purlins, etc., with the Lead Contractor and A/E before proceeding.
- F. If a Contractor finds it necessary to cut into the Work of another Contractor, the cutting and patching shall be performed by such other Contractor at this Contractor's expense. With the written consent of the A/E, the Contractor shall be allowed to perform the cutting. Patching shall be similarly executed.
- G. Roofing:
 - 1. Lead Contractor shall be responsible for all cutting of roofing, modification of roofing, roof opening framing, or roofing repair work for plumbing items, including, but not limited to, plumbing vents, etc.
- H. Lead Contractor shall cut and frame all roof and exterior wall openings in new construction for plumbing items, including, but not limited to, plumbing vents, etc.

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- I. **Lead** Contractor, or other Tradesperson approved by the A/E, shall replace floors, ceilings, roofs, etc., damaged during construction. The cost of repairs shall be borne by the Contractor causing damage.
- J. **Lead Contractor shall provide** openings, with lintels if needed, for sleeves, piping, and other similar openings in **new** interior masonry walls and floors. Locate and dimension the opening prior to construction. Refer to Plumbing Drawings.
- K. Remove and replace masonry, wall tile, drywall, ceilings, and ceiling tiles in order to install Plumbing Work.
- L. Contractor shall be responsible for all cutting of the floor slab related to its own Work.
- M. Contractor shall be responsible for construction that is disturbed during the installation of piping, fixtures, or equipment.
- N. Cutting and patching includes remodeling and repairing of previously graveled or paved areas, walks, curbs, sod, floors, etc. Saw cuts shall be done in neat, straight lines.
- O. Contractor shall repair or replace any roads, sidewalks, or other items that its employees may damage during the performance of the Plumbing Work.

1.14 PAINTING AND RELATED WORK

- A. ALL PAINTING SHALL BE DONE BY A QUALIFIED TRADESPERSON SKILLED IN THE CRAFT.
- B. **Lead** Contractor shall be responsible for finish painting of walls, ceilings, and other Architectural items in the areas of construction.
- C. Previously Finished Areas: Contractor shall repaint any previously finished areas disturbed by its own cutting and patching. Painting of the patched area shall match the color of the adjacent construction in the general area of the patch. Repaint the entire wall-to-wall and floor-to-ceiling surface if, in the opinion of the A/E, a uniform appearance cannot be accomplished by touch-up.
- D. Factory-Finished Equipment: Clean, spot-prime with zinc chromate, and finish, equal to the original quality, any **new** factory-finished equipment that has rusted, has been damaged, or has deteriorated. Repaint the entire surface if, in the opinion of the A/E, a uniform appearance cannot be accomplished by touch-up.
- E. Support Steel / Ferrous Metal: Clean, remove rust from, and paint with zinc-chromate primer any plumbing support steel and bare ferrous metal that is not factory-painted, shop-painted, or galvanized, and that remains exposed to view in the finished areas of the building, **including mechanical/electrical rooms and storage rooms. Consult with the Lead Contractor to ensure** that the prime coat shall be compatible with the finish coat.
- F. Steel Hangers, etc.: Clean, remove rust from, and paint with zinc-chromate primer and aluminum-bronze paint all steel hangers, boxes, straps, and rods furnished under the Plumbing Work, that are not provided with rust-protective finish or are damaged in installation, and that remain exposed to view or are in unfinished and mechanical spaces.

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- G. Exterior Ferrous Metal: Paint with a prime coat any ferrous metal installed outside the building that is not factory-painted, shop-painted, or galvanized.
- H. Paint exposed piping, valves, etc.
 - 1. Exception: All exposed domestic water piping in the Kitchen is to be chrome-plated.
- I. Clean all new insulation coverings. If pre-sized insulation is not used, insulation coverings shall be sized if finish painting is required.
- J. Refer to Division 09 for additional requirements.

END OF SECTION 22 05 00.73

SECTION 22 05 00.74 - CLEANING AND WASTE MANAGEMENT FOR PLUMBING

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Division 01, "General Requirements."

1.02 CLEANING AND WASTE MANAGEMENT

- A. Contractor and its Subcontractor shall do their own cleanup, shall move materials that are in the way of constructing their work, shall repair and replace any damage they do, and shall do any other work of a similar nature which shall be done.

1.03 PROGRESS CLEANING

- A. Maintain all work areas in a neat and orderly manner, free of debris. Clean up all occupied travel areas at the end of each shift. Clean up material removed immediately after use.
- B. Clean up building and site thoroughly every day.
- C. Equipment cleaning and touch-up:
 - 1. Use only cleaning materials recommended by the Manufacturer of the surface to be cleaned.
 - 2. Clean interiors of all enclosures of dirt and debris before installing trim or covers.
 - 3. Brush-clean, prime, and paint-in-kind rust spots on any part.
 - 4. Clean galvanized piping in exposed areas with diluted acetic acid.
 - 5. Clean copper piping in exposed areas with fine emery cloth and solvent.
 - 6. Clean all gauges, thermometers, traps, dirt legs, strainers, and fittings.

1.04 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

- A. On-site rubbish collection containers shall be furnished and weekly rubbish removal service shall be provided or all rubbish and debris from the site shall be removed daily. Refer to Division 01.
- B. Do NOT use the Owner's dumpster for on-site rubbish and debris.

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- C. Disposal of material by removal:
 - 1. Prior to the completion of the work, remove from the construction area all material designated to be removed.
 - 2. All material to be removed shall become the property of the Contractor.
 - 3. Conduct cleaning and disposal operations in compliance with local ordinances and anti-pollution laws. Make all necessary arrangements with private parties and with local officials pertinent to locations and regulations for dumping. Contractor shall pay fees or charges required for dumping of materials.
 - 4. Contractor shall be responsible for locating a legal dump site and for all expenses due to length of haul.
 - 5. Do not burn or bury rubbish and waste materials on Project site. Do not dispose of waste into streams or waterways.
- D. Prior to completion of the work, remove from the vicinity of the work all temporary structures, unused materials, concrete forms, and other like materials. All work areas shall be graded and left in a neat manner conforming to the natural appearance of the landscape or as directed elsewhere in the Contract Documents.

1.05 FINAL CLEANING

- A. Repair, patch, and touch up all scratched or damaged surfaces to specified finish that matches adjacent surfaces, before final acceptance of the work. Repair dents and marred finishes to the satisfaction of the A/E. Prepare for finish painting, where painting is specified.
- B. Thoroughly clean all fixtures; material; finished surfaces of equipment; painted, enameled, or varnished work; and all other exposed finished surfaces. Remove all labels, stickers, marks, stains, fingerprints, spots, rust, oil, grease, dirt, dust, and other foreign materials, so that the Work is presented in a complete and finished condition, ready for acceptance and intended use.
- C. Verify that all drains are free of debris.
- D. Employ experienced workers or professional cleaners for final cleaning.

- E. Expedite the cleaning, washing, waxing, and polishing required within other Sections of these Specifications.
- F. Refer to Division 01 for additional requirements.

END OF SECTION 22 05 00.74

SECTION 22 05 00.77 - CLOSEOUT PROCEDURES FOR PLUMBING

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Division 01, "General Requirements."
- B. Section 22 05 00.33, "Shop Drawings Submittals for Plumbing."
- C. Section 22 05 00.45, "Testing and Inspecting Services for Plumbing."
- D. Section 22 05 00.73, "Execution of Plumbing."
- E. Section 22 05 00.74, "Cleaning and Waste Management for Plumbing."
- F. Section 22 05 00.78, "Closeout Submittals for Plumbing."
- G. Section 22 05 00.79, "Demonstration and Training for Plumbing."
- H. Section 22 05 53, "Identification for Plumbing Piping and Equipment."
- I. Section 22 08 00, "Commissioning of Plumbing."

1.02 DESCRIPTION

- A. Complete and submit the following items, which is a partial list of the items required prior to Contract Completion:

ITEM	REFER TO SECTION
Shop drawing submittals	22 05 00.33
Operational inspection, testing, adjusting, and balancing of all equipment and systems	22 05 00.45, as well as other Sections
Protection	22 05 00.73, as well as other Sections
Painting	22 05 00.73
Cleaning	22 05 00.74
Operating and Maintenance Manuals	22 05 00.78

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ITEM	REFER TO SECTION
All required certifications and labels	22 05 00.78, as well as other Sections
Guarantees and equipment warranties	22 05 00.78, as well as other Sections
Spare parts	22 05 00.78, as well as other Sections
As-Built Documents	22 05 00.78
Instruction of the Owner's personnel	22 05 00.79, as well as other Sections
Equipment and piping identification; valve tagging	22 05 53

B. Refer to Section 22 08 00 for additional items.

C. Provide or perform all of the above items before Contract Completion.

1.03 PUNCH LIST PROCEDURE

A. Refer to Division 01.

END OF SECTION 22 05 00.77

SECTION 22 05 00.78 - CLOSEOUT SUBMITTALS FOR PLUMBING

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Division 00, "Procurement and Contracting Requirements."
- B. Division 01, "General Requirements."
- C. Section 22 05 00.77, "Closeout Procedures for Plumbing."
- D. Section 22 05 00.79, "Demonstration and Training for Plumbing."

1.02 OPERATING AND MAINTENANCE MANUALS

- A. An Operating and Maintenance Manual, approved by the A/E, is required for Contract Completion. Compile and submit a draft of the Operating and Maintenance Manual to the A/E (90) days before Contract Completion.
- B. Submit the final corrected Operating and Maintenance Manual (7) days before Contract Completion.
- C. Upon approval, provide:
 - 1. (1) Operating and Maintenance Manual in the physical format described below.
 - 2. A Portable Document Format (PDF) file containing a copy of the Operating and Maintenance Manual saved on a Universal Serial Bus (USB) flash drive.
- D. The following items, together with any other pertinent data, shall be included in each Operating and Maintenance Manual. This list is not necessarily complete and shall be used only as a guide. Format of manual is to be as follows:
 - 1. Operating and Maintenance Manual binders shall be heavy-duty, loose-leaf, 3-ring, vinyl-covered binders, no larger than 11 in. wide x 12 in. high, with pocket inside the covers to receive folded oversized sheets.
 - 2. Identify each binder on front and spine with the printed title "OPERATION AND MAINTENANCE MANUAL," project name, and subject matter of contents.
 - 3. Material shall be typewritten or printed and shall be fully legible.

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4. Inside cover:
 - a. Title of Project.
 - b. Date of Project completion.
 - c. Name and address of the Owner.
 - d. Name and address of the Prime Contractor.
 - e. Date of submittal.
 - f. Name and address of the Contractor.
 - g. Name and address of the Architect.
 - h. Name and address of the Engineer.
5. Second Page: Index.
6. Each section shall be divided by labeled tabs.
7. First Section: A copy of each shop drawing and approved submittal with an index at the beginning of the section.
8. Second Section:
 - a. A list of all equipment used on the job.
 - b. Complete list of all devices installed and cross-referenced to room number or location.
 - c. Parts list with numbers of replaceable items, including sources of supply.
 - d. Model and serial numbers of components of systems installed.
 - e. Suppliers and Manufacturers List:
 - 1) Typed statement of material to be furnished.
 - 2) Manufacturers' and nearest Factory Representatives' names, addresses, telephone numbers, and e-mail addresses.
 - 3) Routine and 24-hour emergency service/repair information:
 - a) Name, address, telephone number, and e-mail address of servicing agency.
 - b) Names of personnel to be contacted for service arrangements.
9. Third Section:
 - a. Description of systems.

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- b. Manufacturer's literature describing each piece of equipment, including the following:
 - 1) Operating and maintenance instructions.
 - 2) Lubrication instructions.
 - 3) Start-up and shutdown procedures.
 - 4) Routine and emergency servicing instructions.
- c. Valve charts.
- d. Special wrenches, keys, etc.
- e. Copies of all testing, adjusting, and balancing reports.
- f. Prints of all system wiring and control diagrams.
- g. All certifications and related information.
 - 1) Typed verification of compliance with certification requirements.
 - 2) Health Department Certificate (Commercial Kitchens).
- h. Copies of all written warranties.
- i. The Owner's receipt of spare parts.
- j. The Owner's statement of completion of instruction. Refer to Section 22 05 00.79.

E. Refer to Division 01 for additional requirements.

1.03 GUARANTEES AND WARRANTIES

- A. Guarantee: Contractor shall guarantee its equipment, workmanship, and materials for a period of (1) year from the date of Contract Completion. Should defects develop within this period, the Contractor shall, at no cost to the Owner, remedy the defects and reimburse the Owner for all damage to other Work caused either by the defects or as a result of the work of correcting the same.
- B. Extended Warranties: The table below contains a list of items with warranty requirements and their specification sections. These extended warranties are over and above the 1-year guarantee specified above. This list may not be a complete list of all required warranties. Include copies of all written warranties in the Operating and Maintenance Manual, whether listed or not.

ITEM WITH WARRANTY REQUIREMENTS	SPECIFICATION SECTION
Thermostatic Mixing Valves	22 11 19
Grease Interceptors	22 13 23
Storage Tanks	22 33 33
Water Heaters	22 34 36
Water Heaters, Storage Tanks	22 34 36.13
Water Heaters, Heat Exchangers, Thermal Shock	22 34 36.26
Disposers (Food Waste Grinders)	22 40 00

C. Post-Warranty Service:

1. The products and Manufacturers identified throughout these Specifications represent reputable suppliers that generally have an inventory of spare parts, and a network of authorized service providers that can support the products installed on this Project.
2. Contractor shall, for a reasonable charge, assist the Owner for a period of (7) years in obtaining parts and service on the products installed.
3. Once the Contractor's 1-year guarantee has expired, the Owner will pay the cost of these parts and service.
4. If the item cannot be repaired, the Contractor shall recommend an equivalent replacement item at a reasonable cost to the Owner.

D. Provide a copy of a certificate of warranty commencement signed by the Contractor, the Owner, and the A/E in each Operating and Maintenance Manual. The A/E will prepare the correct warranty form for signatures.

E. Refer also to Division 00, "General Conditions," Division 01, and other Specification Sections that define the starting date of the guarantee period, or that discuss either additional warranty requirements, or extended equipment warranties beyond the standard period.

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1.04 PROJECT AS-BUILT DOCUMENTS

- A. Contractor shall maintain, at the job site, (1) copy of the As-Built Documents, including the following items related to its Trade:
 - 1. Contract Drawings.
 - 2. Specifications.
 - 3. Addenda.
 - 4. Reviewed Shop Drawings.
 - 5. Change Orders.
 - 6. Other modifications to Contract.
 - 7. Field Test Records and Start-Up Reports.
 - 8. Minutes of Coordination and Progress Meetings.
 - 9. (1) set of the Contract Drawings that will be used for As-Built Drawings. This set shall be used exclusively for documenting and recording the exact location of all installed Work.
 - 10. Option: With the approval of the A/E and Owner, As-Built Documents may be made available in an electronic format via an Internet-accessible server.
- B. Do not use As-Built Documents for construction purposes.
- C. Store As-Built Documents apart from documents used for construction at the Contractor's temporary field office, or in another location readily available to the A/E.
- D. Provide files and racks for storage of As-Built Documents, as appropriate, with As-Built Documents identified.
- E. Maintain As-Built Documents in clean, dry, legible condition.
- F. Use red ballpoint pen for all marking of As-Built Documents.
- G. As-Built Documents shall be available at all times for inspection by the A/E and Owner.
- H. As-Built Drawings:
 - 1. As-Built Drawings shall be legibly marked to record actual "as-built" construction with the following information (some of these items may not be applicable):
 - a. Horizontal and vertical location of above-ground and underground utilities and appurtenances referenced to permanent surface improvements.

- b. Location of internal utilities and appurtenances concealed in construction referenced to visible and accessible features of structure.
 - c. Field changes of dimension and detail.
 - d. Changes made by Change Order.
 - e. Details not on the original Contract Drawings.
- 2. Do not permanently conceal any work until the required information has been recorded.
- 3. Record the location of all concealed utilities on the As-Built Drawings.
- 4. Record deviations in locations of concealed piping, valves, equipment, and all buried or concealed utility lines and services, etc., dimensioned from a fixed control point, including the following:
 - a. Depth of bury.
 - b. Invert elevations at start of storm or sanitary line.
 - c. At each change of direction.
 - d. At each change of slope.
 - e. Additional dimensions in order to fully define location.
- 5. Minor piping variations need not be recorded.
- 6. Record locations of abandoned piping, including exterior lines.
- 7. Record deviations made necessary to incorporate equipment different from the Design Base equipment.
- 8. Record all Addendum Items.
- 9. Mark locations of access panels on the As-Built Drawings.
- 10. Add valve tag numbers to the As-Built Drawings.
- I. Include a complete list of all devices installed and cross-referenced to room number or location.
- J. At completion of project, the Contractor shall deliver (1) printed copy of its As-Built Documents to the A/E.
- K. Contractor shall accompany its As-Built Documents with a transmittal letter, in duplicate, containing the following information:
 - 1. Date.

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2. Project title and number.
3. Contractor's name and address.
4. Title and number of each As-Built Document.
5. Certification that each As-Built Document as submitted is complete and accurate.
6. Signature of the Contractor or its Authorized Representative.

1.05 SPARE PARTS

- A. Provide a receipt, signed by the Owner, for (1) complete set of all the operating appurtenances and loose items related to the operation and maintenance of the completed Project for the Owner's use after the guarantee period expires. Insert the receipt in the Operating and Maintenance Manual. These appurtenances and items include, but not are limited to, the following:
1. Keys to panels, door and window hardware, and other devices not directly provided to the Owner from the Manufacturer.
 2. Operating handles, levers, cranks, specialized wrenches or drivers, remote controls, and similar items.
 3. Extra materials (e.g., attic stock) listed below. This list may not be a complete list of all spare parts. Provide all spare parts, whether listed or not.

SPARE PART AND QUANTITY	SPECIFICATION SECTION
Division 11 items	
Spare air filter — (1) for each ice machine	11 46 83
Division 22 items	
Spare pressure gauge	22 05 19
Spare thermometer for thermostatic mixing valve(s)	22 11 19
Spare trap seals — (5) for each type	22 11 19
Set of spare mechanical seals — (1) set for each pump	22 11 23.13
Domestic water pressure gauge with hose end connection	22 11 23.23
Set of gaskets — (1) set for each pump	22 11 23.23
Set of spare mechanical seals — (1) set for each pump	22 11 23.23

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SPARE PART AND QUANTITY	SPECIFICATION SECTION
Set of gaskets — (1) set for each manhole or handhole	22 13 23
Set of spare mechanical seals — (1) set for each pump	22 13 29
Any other parts mentioned elsewhere	

END OF SECTION 22 05 00.78

SECTION 22 05 00.79 - DEMONSTRATION AND TRAINING FOR PLUMBING

PART 1 GENERAL

1.01 DESCRIPTION

- A. Provide training schedule and training outline for approval (45) days prior to Contract Completion.
- B. Provide a minimum of (4) hours of in-service training for the system's operators. Provide all applicable user manuals and related training documentation.
- C. After all system operational tests have been completed, schedule an instruction period with the Owner. Schedule well in advance, so that all the Owner's personnel may attend if they desire. Coordinate with the Owner and A/E for date and time of training sessions.
- D. Participate in training sessions. Instruct the Owner in the operation and maintenance of all systems and equipment. Use Operating and Maintenance Manuals to familiarize the Owner with equipment and procedures. Allow sufficient time for this instruction.
- E. Record a video of all Owner training, instructions, and equipment start-up demonstrations on a digital video disc (DVD). Turn over a copy of the DVD to the Owner.
- F. Instruction shall include the following:
 - 1. Location of equipment and explanation of what it does (function).
 - 2. Reference to operating instruction manuals for record and clarity.
 - 3. Coordination of written and verbal instructions, so that the operation of each system is fully understood by operating personnel.
 - 4. Complete review of items contained in Operating and Maintenance Manuals.
 - 5. Discussion of maintenance procedures that are required be followed by the Owner, including cleaning procedures and what items to check.
 - 6. Complete demonstration and explanation of each System.
- G. Provide a copy of the Owner's "Certificate of Equipment Demonstration," signed by the Contractor, the Owner, and the A/E, in each Operating and Maintenance Manual. If the Owner does not have a standard demonstration form, the A/E will provide one.

END OF SECTION 22 05 00.79

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SECTION 22 05 16 - EXPANSION FITTINGS AND LOOPS FOR PLUMBING PIPING

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 22 05 48, "Vibration and Seismic Controls for Plumbing Piping and Equipment."
- B. Section 22 11 19, "Domestic Water Piping Specialties."

1.02 DESCRIPTION

- A. Provide thermal expansion control for all piping (not shown on the Drawings).
- B. Provide expansion joints and accompanying anchors and guides where expansion cannot be provided for with loops and offsets (not shown).
- C. Isolate all piping for both noise and vibration transmission.
- D. To prevent damage to building, equipment, joists, hangers, and piping, provide expansion loops, guides, anchors, and offsets in piping systems to accurately control pipe movement due to equipment operation or thermal gradients.
- E. Anchor points are not indicated on the Drawings. Install anchor points as needed to properly control piping movement. Typically, piping shall be anchored at tees, elbows, and bends. Piping shall also be anchored in each direction.

PART 2 PRODUCTS

2.01 EXPANSION JOINTS

- A. Acceptable Manufacturers: Metraflex or approved equal by Flexicraft, Flexonics Products, Hyspan Precision Products Inc., Minnesota Flexible Corp., Twin City Hose Inc., or Uniroyal General Rubber.
- B. Metraflex. Stainless steel bellows and elements; cast iron equalizing rings, tie-rods, and pipe connections; 250 PSIG working pressure. Use packless, internally guided type for lines 2 in. and smaller.
- C. 2 in. and Smaller Lines: Metraflex HPMF. Stainless steel with copper sweat ends, packless, internally guided expansion compensator type.

EXPANSION FITTINGS & LOOPS FOR PLUMBING PIPING

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D. 3 in. and Larger Lines: Metraflex MC. All wetting parts stainless steel, including bellows and elements. Vanstone flanges, cast iron equalizing rings, steel cover over bellows. Pipe connections, 250 PSIG working pressure. Size each bellows end for 3 in. expansion.

E. Seismic/Thermal Expansion Joints: Metraflex Metraloop®.

2.02 PIPE ALIGNMENT GUIDES

A. Acceptable Manufacturers: Metraflex or approved equal by Flexicraft, Flexonics Products, Hyspan Precision Products Inc., Minnesota Flexible Corp., Twin City Hose Inc., or Uniroyal General Rubber.

B. Metraflex. Painted steel cylinder guide assembly with welded mounting brackets, 2-piece pipe clamp "Spider" assembly.

PART 3 EXECUTION

3.01 INSTALLATION

A. Line Expansion:

1. U-Bends: Install U-bends or loops in all piping subject to expansion and contraction.
 - a. Option: Provide Metraflex Metraloop, sized in accordance with the Manufacturer's recommendations.
2. Expansion Joints: Where unable to provide U-bends, or where specifically shown or specified, provide expansion joints. Install according to the Manufacturer's instructions.

B. Branch Connections: Branch connections to mains shall be made with a minimum of (2) 90-degree elbows, and shall incorporate at least (1) change of direction in the horizontal plane and (1) change of direction in the vertical plane, before connecting to equipment or fixtures or before dropping in or rising in a wall.

C. Guides: Supplement all loops and expansion joints with adequate guides as close to loops and joints as possible, and at recommended intervals from loops and joints, to preserve alignment and pitch. Guides shall be rigidly secured to the structure and shall permit axial movement only. Follow the Manufacturer's instructions where applicable.

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- D. Anchors: Install pipe anchors where required to secure the pipe and eliminate movement.
Attach pipe anchors securely to the structure.

END OF SECTION 22 05 16

SECTION 22 05 17 - SLEEVES AND SLEEVE SEALS FOR PLUMBING PIPING

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Division 01, "General Requirements."
- B. Division 07, "Thermal and Moisture Protection."

1.02 DESCRIPTION

- A. Provide sleeves and sleeve seals, including all accessories required for a complete installation, at penetrations of exterior floor and wall assemblies. Sleeve where piping:
 - 1. Passes through equipment room walls.
 - 2. Passes exposed through non-fire-rated construction walls.
 - 3. Passes exposed through non-fire-rated construction above-grade floors.
 - 4. Passes through smoke-rated or fire-rated separations. Apply firestopping to plumbing penetrations of fire-rated floor and wall assemblies to maintain or restore original fire-resistance rating of assembly according to Division 07.

1.03 TERMINOLOGY

- A. Firestopping: Material or combination of materials used to retain integrity of fire-rated construction by maintaining an effective barrier against the spread of flame, smoke, and hot gases through penetrations in fire-rated wall and floor assemblies.

1.04 QUALITY ASSURANCE

- A. Standards:
 - 1. American Society for Testing & Materials (ASTM):
 - a. ASTM E 84, Standard Test Method for Surface Burning Characteristics of Building Materials."
 - b. ASTM E 814, "Standard Method of Fire Tests of Through Penetration Fire Stops."
 - 2. Factory Mutual Global (FMG).

3. National Fire Protection Association (NFPA):
 - a. 2023 NFPA 70, "National Electrical Code® (NEC)."
 - b. 2021 NFPA 101, "Life Safety Code."
4. Underwriters Laboratories, Inc. (UL) 1479, "Fire Tests of Through-Penetration Firestops."

1.05 SUBMITTALS

- A. Submit Manufacturer's product data, installation instructions, and details for each required product.

PART 2 PRODUCTS

2.01 SLEEVE SEALS — NON-FIRE-RATED PLUMBING PENETRATIONS

- A. Piping Sleeve Material:
 1. Up to 8 in. diameter: Machine-cut Schedule 40 black steel pipe.
 2. 8 in. and larger: Schedule 30 black steel pipe.
 3. Bare copper piping: Copper sleeves.
 4. Option: Schedule 40 plastic pipe may be used for piping sleeves, where adjacent ceiling space is not used for return air plenum. Refer to the Drawings.
- B. Sizing: Sleeves shall be large enough for insulation to be continuous, or for sleeve seals to be installed. Size to allow 1/2 in. minimum clearance all around piping or piping insulation.

2.02 SLEEVE SEALS — FIRE-RATED PLUMBING PENETRATIONS

- A. Acceptable Manufacturers: Either provide sleeves and sleeve seals from the Manufacturers listed in Division 07 or provide sleeves and sleeve seals from the following Manufacturers: 3M Fire Barrier caulking or putty, BP Fyre Putty, Carborundum Fibersil, Chase Foam, CSD Sealing Systems "FSP," Dow Corning 3-6548, Hilti Fire Stop, International Protective Coatings Corporation "Flame-Safe," Johns-Manville Cerafiber, KBS Mortar Seal, or Nelson Fire Stop.
- B. ASTM E119 and E814 Silicone RTV foam, UL-approved.
- C. Sleeves shall be large enough for fire-rated sleeve seals to be installed. Size to allow 1/2 in. minimum clearance all around.

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2.03 FIRESTOP PILLOWS

- A. Acceptable Manufacturers: KBS or approved equal by 3M or Specified Technologies Inc.
- B. KBS SEALBAG™: UL-listed, FMG-approved.

2.04 SLEEVE SEALS — WATERTIGHT PLUMBING PENETRATIONS

- A. Provide sleeves with waterstop anchor flange at midpoint where penetrating structure at or below grade.
 - 1. Acceptable Manufacturers: Calpico, 3M, or Hilti.
- B. Option: Compression Seals: GPT Industries Link-Seal. Stainless steel bolts and nuts. Provide correct size seal and coordinate with sleeve size.
 - 1. Acceptable Manufacturers: GPT Industries or approved equal by Metraflex, Wayne, Michigan, or Calpico.
- C. Sleeves shall be large enough for watertight sleeve seals to be installed. Size to allow 1/2 in. minimum clearance all around.

PART 3 EXECUTION

3.01 COORDINATION

- A. Coordinate location of sleeves, core-drilled holes, and other holes required with the Lead Contractor, Structural Engineer, equipment, and other Trades.

3.02 INSTALLATION — GENERAL

- A. Indicated sizes and locations are based on early Project information and should be verified by the Contractor.
- B. Chases and constructed openings that are not indicated on the Drawings shall be provided at the expense of the Contractor requiring the chase or constructed opening.
- C. If it becomes necessary to cut into the work because of the failure of the Contractor to notify the Lead Contractor, then the Lead Contractor shall do any cutting and patching required later, at the Contractor's expense.
- D. Provide dimensions and locations of openings required in walls for sleeves, piping, and similar items, and deliver sleeves, to the Lead Contractor, so as not to delay construction.

- E. BEFORE CONSTRUCTION OF MASONRY OR CONCRETE WALLS IS STARTED, give the Lead Contractor locations and sizes of all openings required for the installation of equipment or sleeves.
- F. If sleeves are not installed due to the fault of the Contractor, the Lead Contractor shall core-drill holes through masonry or concrete construction at the expense of the Contractor.
- G. In New Drywall, Masonry, or Concrete Wall Construction, as applicable to this Project:
 - 1. Furnish sleeves for plumbing items to the Lead Contractor for installation where directed by the Contractor.
 - 2. Carefully coordinate and check locations of sleeves immediately before and after each concrete pour and/or masonry installation.
 - 3. Option for above ceiling only: Lead Contractor shall omit concrete block(s) where required for penetrations and shall provide square-finished opening. Do not omit concrete blocks in sound barrier walls, under bearing points, or in fire-rated or smoke-rated walls.
 - 4. Coordinate with other Contractors, Structural Engineer, and A/E.
- H. Sleeves are not required in floor slabs on grade or in core-drilled openings not requiring firestopping or watertightness.
 - 1. Exception: Sleeves are required at core-drilling through hollow-core pre-cast concrete slabs and through concrete block walls to facilitate containment of required firestopping material.
- I. Furnish all conduit sleeves and caulking for openings in new walls and turn them over to the Lead Contractor for installation and caulking.
- J. Piping shall not bear on sleeves. Install sleeves plumb and true to line, grade, and position.
- K. Cutting:
 - 1. Cut off sleeves through walls flush with each surface.
 - 2. Cut off sleeves 1/8 in. above finished floors and 3 in. above floors in kitchens, utility rooms, equipment rooms, and shafts. Cut off bottom of sleeve flush with surface.
 - 3. Refer to Division 01 for specific requirements regarding approved cutting methods.
- L. All piping sleeves and escutcheons shall have ample clearance for pipe and covering and shall have chrome-plated wall and floor escutcheons over the pipe in finished areas. Provide chrome-plated or prime-coated escutcheons in other rooms.

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- M. Unused sleeves shall be plugged and fire-packed. Provide finish that matches finish of adjacent surface.

3.03 INSTALLATION — FIRE-RATED PLUMBING PENETRATIONS

- A. Verify which walls, ceilings, or floors are fire-rated, if any, and provide approved fire-blocking or fire-protective devices.
- B. Use approved, UL-listed, fire-retardant sealants, backing, and packing to maintain fire rating of the structure penetrated.
- C. Spray UL-listed foam sealant around exposed piping entering and leaving fire-rated wall or floor structures.
- D. To ensure fire blocking, close space around piping and other plumbing items passing through walls and floors. Seal space up to a 1/2 in. gap with sealant or caulking. Close off space greater than a 1/2 in. gap with sheet metal and seal airtight.
- E. For larger openings, provide firestop pillows.
- F. Pack all fire-rated or sound-rated separation sleeves with glass fiber, high-temperature mineral wool, aluminum-silica fiber, fire-retardant rope, calcium silicate, or other noncombustible material to maintain fire rating of structure, and finish with fire-rated sleeve seals.
- G. Fill space around all sleeves extending into exposed areas with material compatible with adjacent construction and finish.
- H. All openings shall have been UL-tested utilizing the proper rated penetrations to be equal to, or greater than, the barrier assembly in which the penetration occurs.
- I. Install all penetrations and openings in accordance with the Manufacturers' instructions.
- J. The fire-blocking assembly shall maintain the required fire-resistance rating of the wall or floor in which it is placed. Apply a further sealant, if necessary, to attain a smoke-tight condition. Close openings without sleeves in the same manner.
- K. Refer to Division 07 for additional requirements.

3.04 INSTALLATION — WATERTIGHT PLUMBING PENETRATIONS

- A. Sleeving with absolutely watertight seal is required for plumbing work passing through exterior walls above grade, and underground foundation walls and other below-grade penetrations into building.
- B. Piping sleeves penetrating outside wall or slabs on grade shall have welded intermediate flange imbedded in masonry. Space around piping shall be clamped, packed, and caulked to make an absolutely watertight seal within sleeve. Caulk around outside of sleeve.
 - 1. Option: Install compression seals.

END OF SECTION 22 05 17

SECTION 22 05 19 - METERS AND GAUGES FOR PLUMBING PIPING

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 22 05 00.78, "Closeout Submittals for Plumbing."
- B. Section 22 10 00, "Plumbing Piping."
- C. Section 22 11 16, "Domestic Water Piping."

1.02 DESCRIPTION

- A. Provide meters and gauges as shown or required to complete the installation of all hydronic piping systems installed on this Project.

1.03 SPARE PARTS

- A. Provide spare pressure gauge. Refer to Section 22 05 00.78.

PART 2 PRODUCTS

2.01 GAUGES, PRESSURE

- A. Acceptable Manufacturers: Weiss Instruments or approved equal by Ashcroft Inc., Marsh, Marshalltown, H.O. Trerice Co., or Weksler Instruments Corp.
- B. Weiss 4PG-1. Standard single Bourdon tube, black die cast aluminum case, 4-1/2 in. diameter dial, chrome removable slip ring, gauge cock and pigtail, $\pm 1\%$ accuracy. Range shall be approximately 50% in excess of maximum pressure anticipated. Provide gauge cock.

2.02 THERMOMETERS

- A. Acceptable Manufacturers: H.O. Trerice Co. or approved equal by Ashcroft Inc., Marsh, Marshalltown, Weiss Instruments, Weksler Instruments Corp., or Winters Instruments.
- B. Trerice "Industrial Type." Blue reading, organic liquid, 9 in. scale, cast aluminum, epoxy powder-coated case, front double-strength window, straight or adjustable angle pattern, brass separable well, range of 30 degrees F. to 180 degrees F. Provide insulation extension.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Gauges, Pressure:
 - 1. Where feasible, install and adjust gauge to permit reading from standing position on floor level.

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2. Provide pressure gauge on the water service entrance riser, on inlet and outlet of recirculating pump, and where indicated on the Drawings.
3. Install pressure gauge on well pump discharge, downstream of check valve.
4. Install a 3/8 in. ball valve at each pressure gauge.

B. Thermometers

1. Install and adjust thermometers to permit them to be read from floor level.
2. Provide on the inlet and outlet piping at each water heater, on inlet and outlet of each mixing valve, on each hot water storage tank, and on hot water recirculating pump suction line, where shown on the Drawings.

END OF SECTION 22 05 19

SECTION 22 05 23 - GENERAL-DUTY VALVES FOR PLUMBING PIPING

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 22 05 53, "Identification for Plumbing Piping and Equipment."

1.02 DESCRIPTION

- A. Provide valves and accessories for plumbing piping system(s):
 - 1. In locations accessible to qualified personnel for operation, removal, maintenance, or servicing.
 - 2. As shown on the Drawings and as discussed in other Sections of these Specifications.
 - 3. To facilitate the maintenance, operation, isolation, and servicing of all plumbing piping systems.

1.03 QUALITY ASSURANCE

- A. Standards:
 - 1. American National Standards Institute (ANSI).
 - 2. American Society for Testing and Materials (ASTM).
 - 3. American Society of Mechanical Engineers (ASME).
 - 4. American Society of Sanitary Engineers (ASSE).
 - 5. American Water Works Association (AWWA).
 - 6. California Assembly Bill (AB) 1953.
 - 7. Manufacturer's Standardization Society of the Valve and Fittings Industry (MSS).
 - 8. National Science Foundation (NSF) 61.

PART 2 PRODUCTS

2.01 MANUFACTURERS AND MODEL NUMBERS

- A. Manufacturer and Model Number listed below shall be considered to be the Design Base Manufacturer and Model Number. Other Acceptable Manufacturers will be permitted, unless specifically excluded.
- B. All valves of the same type used on the Project shall be by the same Manufacturer.

2.02 BALANCING VALVES (2 IN. AND SMALLER)

- A. Acceptable Manufacturers: Bell & Gossett CB Series (Xylem Inc.), Flo Pac Model MB, Nibco Fig. 1710, or Taco ACUF Series, or approved equal by Apollo, Hammond, Kitz, Milwaukee, Red-White, or Watts.
- B. 200 PSIG, ASTM B-62 bronze body/brass ball, threaded or solder joint ends, differential pressure read-out ports, 1/4 in. drain port, memory stop feature to allow valve to be closed for service and reopened to setpoint without disturbing balance position, and calibrated nameplates to ensure specific valve settings.

2.03 BALANCING VALVES, THERMOSTATIC (2 IN. AND SMALLER)

- A. Acceptable Manufacturers: ThermOmegaTech or approved equal by Caleffi or Acorn.
- B. ANSI/AWWA C800. NSF 61 and California AB1953 for lead content. Inline valve with integral check valve and spring-operated thermal actuator. Type 303 stainless steel body. Tapered female threaded, NPT, end connections: Factory setting range 105 degrees F. to 180 degrees F. 200 psig maximum working pressure. 300 degrees F. maximum working temperature.

2.04 BALL VALVES (3 IN. AND SMALLER)

- A. 150 lb. SWP, bronze body, threaded or solder joint ends, TFE seat and seals, stainless steel ball, blow-out-proof stem, extended handle with vapor seal, adjustable memory stop, and protective sleeve.
 - 1. ONLY Acceptable Manufacturers and Model Numbers: Milwaukee Model BA-100 or BA-150 or approved equal by Watts No. B-6000 or B-6001, Apollo 70-100 or 70-200, Hammond 8501 or 8511, Nibco 580-70 or 585-70, Kitz 68 or 69, or Red-White 5044F or 5049F.

- B. With Lockable Handles: 150 lb. SWP, bronze body, threaded ends, TFE seat and seals, stainless steel ball, extended handle with vapor seal, adjustable memory stop, and protective sleeve.
 - 1. Acceptable Manufacturers and Model Numbers: Milwaukee Model BA-100 LD, Apollo No. 70-100-27, or Nibco 580-70-LH, or approved equal by Hammond, Kitz, Red-White, or Watts.

2.05 CHECK VALVES

- A. Acceptable Manufacturers: Milwaukee or approved equal by Apollo, Hammond, Kitz, Nibco, Red-White, or Watts.
- B. 2 in. and Smaller: Milwaukee Fig. 509 or 1509. 125 lb. SWP, ASTM B-62 bronze body, swing check, threaded or solder joint tubing ends, with renewable bronze disc and integral seat.
- C. 2-1/2 in. and Larger (swing): Milwaukee Fig. F-2974. 125 lb. SWP, iron body, bronze trim, with flanged ends.

2.06 DRAIN VALVES

- A. Acceptable Manufacturers: Milwaukee or approved equal by Nibco or Walworth.
- B. Milwaukee BA-100H or BA150H. 3/4 in. bronze hose-end ball valve, with cap and chain (where concealed). Provide 3/4 in. garden hose thread adapter on outlet of drain valve, with cap.

2.07 GATE VALVES

- A. 2 in. and Smaller: 125 lb. SWP, ASTM B-62 bronze, single wedge disc, inside screw, rising stem, union bonnet, threaded or solder joint tubing ends, designed for unrestricted flow, with back seat features for repacking under full pressure.
 - 1. Acceptable Manufacturers and Model Numbers: Milwaukee Model 148 or 149, or Hammond IB645 or IB647, or approved equal by Apollo, Kitz, Nibco, Red-White, or Watts.
- B. 2-1/2 in. and Larger (rising stem): Milwaukee Fig. F-2885. 125 lb. SWP, iron body, bronze trim, with flanged ends.
 - 1. Acceptable Manufacturers: Milwaukee or approved equal by Apollo, Hammond, Kitz, Nibco, Red-White, or Watts.

C. 4 in. and Larger:

1. Option: Butterfly valve. Keystone Fig. 222. 250 lb. SWP double-seated, bronze disc, stainless steel shaft, acetal bushings, cast iron lug body, 200 degrees F. maximum water temperature, bubble-tight shutoff, and EPDM seat. Provide extended shaft necks to clear insulation where required.
 - a. Valves 4 in. and smaller: Furnish lever handle.
 - b. Valves 6 in. and larger: Furnish manual crank-operated worm gear operator.
 - c. Acceptable Manufacturers: Keystone or approved equal by Milwaukee.

2.08 GLOBE VALVES

- A. Acceptable Manufacturers: Milwaukee or approved equal by Apollo, Hammond, Kitz, Nibco, Red-White, or Watts.
- B. Milwaukee Fig. F-2981. 125 lb. SWP, iron body, bronze trim, with flanged ends.

2.09 PLUG VALVES, LUBRICATED

- A. 2 in. and Smaller: 150 lb. SWP (200 lb. WOG), semisteel body, threaded ends with TFE stem seal and seat, square stem, and permanently attached lever handle.
 1. Acceptable Manufacturers: Homestead Fig. 611, Powell Fig. 2200, Rockwell 142, or Nordstrom 142.
 2. Option: Ball valve. Milwaukee "Butterball," Model BB2-100. 175 lb. SWP, bronze body, viton seat, threaded ends, stainless steel disc and stem, and lever handle.
 - a. Acceptable Manufacturers: Milwaukee or approved equal by Watts, Apollo, or Nibco.
- B. 2-1/2 in. and Larger: 150 lb. SWP (200 lb. WOG), semisteel body, flanged ends with TFE stem seal and seat, square stem, wrench-operated. Furnish wrench.
 1. Acceptable Manufacturers: Homestead Fig. 612, Powell Fig. 2201, Rockwell 143, or Nordstrom 143.

2.10 WATER-PRESSURE REDUCING VALVES

- A. Acceptable Manufacturers: Watts or approved equal by Apollo, Hammond, Kitz, Milwaukee, Nibco, or Red-White.

- B. Watts Model #LF223-S: Tested and approved in accordance with ASSE Standard 1003. Lead-free copper alloy body, reinforced Buna-N diaphragms, brass piston and cylinder, composition seat disc insert for tight shutoff, integral stainless steel strainer, built-in back-pass check, threaded ends, and line size.

2.11 WATER-PRESSURE REGULATORS (WATER-PRESSURE REDUCING VALVES)

our static pressure at the street is 66 psi, do we need any PRVs?

- A. Watts Model (LF26A (3-way)) (LF26A (2-way)) (ANSI 112.76.2): Lead-free copper alloy body and spring chamber, internal stainless-steel screen, threaded ends, line size.
1. Acceptable Manufacturers: Watts or approved equal by Zurn-Wilkins, Amtrol, Bell & Gossett (Xylem Inc.), Braukmann, Cla-Val Co., Mifab, or Cash Acme.
- B. PRV, High-Capacity (ASSE 1003), 2-1/2 in. to 3 in.: Watts Model LFN223B. Lead-free copper-silicon alloy body, reinforced Buna-N diaphragm, replaceable stainless-steel alloy seat, integral stainless-steel screen, and built-in bypass controls.
1. Acceptable Manufacturers: Watts or approved equal by Apollo, Hammond, Kitz, Milwaukee, Nibco, or Red-White
- C. PRV, Standard-Capacity (ASSE 1003, ANSI A112.26.2), (1/2 in. to 1 in.) (1-1/4 in. to 2 in.): Watts Model LF25AUB-Z3. Lead-free copper-silicon alloy body, reinforced EPDM with PTFE wetted surface diaphragm, (1/2 in. to 1 in. replaceable engineer polymer seat) (1-1/4 in. to 2 in. replaceable stainless-steel seat disc), integral stainless-steel screen, EPDM valve disc, and built-in bypass controls.
1. Acceptable Manufacturers: Watts or approved equal by Zurn-Wilkins, Amtrol, Bell & Gossett (Xylem Inc.), Braukmann, Cla-Val Co., Mifab, or Cash Acme.

2.12 WATER-PRESSURE RELIEF VALVES

- A. Acceptable Manufacturers: Bell & Gossett (Xylem Inc.), Watts, or Wilkins, or approved equal by Apollo, Hammond, Kitz, Milwaukee, Nibco, or Red-White.
- B. ASTM B-62 bronze body, diaphragm-activated, phosphor-bronze diaphragm and renewable seat disc, with adjustable pressure settings, 1/2 in. male inlet and female outlet. Refer to the Drawings for pressure setting.

2.13 SEALANT TAPE

- A. Polytetrafluoroethylene (PTFE) tape.

PART 3 EXECUTION

3.01 COORDINATION

- A. Coordinate valves and accessories with structure and with mechanical and electrical equipment.
- B. Coordinate all access panel locations, if required, with the Lead Contractor.

3.02 INSTALLATION

- A. Install valves:
 - 1. At service connections to equipment not provided with valves.
 - 2. At branch lines from main lines.
 - 3. At low points for draining.
- B. Install valves with handwheel at or above centerline of pipe.
- C. Install valve at inlet and outlet to water heater.
- D. Thermostatic Balancing Valve (TBV):
 - 1. Install TBV at the end of each domestic hot water supply just before the return line. Refer to the Drawings for locations.
 - 2. Provide a shut-off valve before and after the TBV with a union after the TBV.
 - 3. Provide a strainer upstream of each valve.
 - 4. Install valve with proper flow direction as marked on the valve body.
 - 5. Provide pipe thread sealant rated for domestic water applications on all male pipe threads except on the stainless-steel union threads, which have an integrated O-ring that fully seals when hand-tightened.
 - 6. If using PTFE tape, do not apply tape to first thread to avoid cutting tape and causing obstructions to flow.
 - 7. Closing temperature shall be 125 deg. F. except for 140 deg. F. kitchen domestic hot water which shall have a 140 deg. F. closing temperature.
- E. Ball Valves:
 - 1. Install on hot- and cold-water branches serving more than (1) fixture.

2. Install on food service equipment connection.
 3. Install in domestic water supply lines to any equipment or fixtures not provided with stops.
- F. Drain Valves: Install 3/4 in. valves at low points in piping to provide complete drainage of all systems.
- G. Water-Pressure Reducing Valves: Provide ball valves.
- H. Water-Pressure Regulators (Water-Pressure Reducing Valves):
1. For main domestic water service, provide (2) regulators in parallel — the smaller regulator sized to handle 10% to 20% of system capacity, and the larger sized to handle the balance. Set outlet pressure for 75 PSI.
 2. If water pressure to the building exceeds 75 PSI, provide a multiple pressure regulator assembly set for 75 PSI outlet pressure for the entire building domestic water system.
 3. Kitchen dishwasher: Install regulator for kitchen dishwasher furnished with the Food Service Equipment Work. Set for 20 PSI (adjustable) outlet pressure. Coordinate size and capacity with equipment supplied.
 4. Install strainer on upstream line to inlet of each valve if not integral to regulator.
- I. Water-Pressure Relief Valves: Extend drain from valve to 6 in. above floor, or to floor drain.
- J. Open valves fully before soldering to piping.
- K. Do NOT install solder joint valves on systems where the fluid temperature may soften the solder.
- L. During soldering or brazing, remove valve internals and protect the remaining element of sweat end valves against heat damage.
- M. Do NOT install any valves where the fluid operating pressure exceeds 80% of the valve's pressure rating.
- N. Do NOT use bronze valves on iron fittings or pipe without dielectric unions.
- O. Tighten packing glands before placing valves in service.
- P. Tag all valves. Refer to Section 22 05 53.

**GENERAL-DUTY VALVES FOR
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3.03 TESTING

- A. Test all valves for tightness.
- B. TBV: Verify proper operation.
- C. Test-operate all valves at least once from closed-to-open-to-closed positions while valve is under pressure.

END OF SECTION 22 05 23

SECTION 22 05 29 - HANGERS AND SUPPORTS FOR PLUMBING PIPING AND EQUIPMENT

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Division 03, "Concrete."
- B. Section 22 05 16, "Expansion Fittings and Loops for Plumbing Piping."
- C. Section 22 05 43, "Underground Plumbing Systems."
- D. Section 22 05 48, "Vibration and Seismic Controls for Plumbing Piping and Equipment."
- E. Section 22 07 00, "Plumbing Insulation."

1.02 DESCRIPTION

- A. Hangers and Supports for Plumbing Equipment and Systems: Provide for equipment and piping installed under the Plumbing Work.
- B. Supports for Equipment:
 - 1. Provide all structural supports between building structural framing members to support major plumbing equipment, including water heaters, air compressors, etc.
 - 2. Attach all items of equipment securely to structure, floor, or concrete pad. Means of attachment shall be approved by the A/E and shall not degrade the structure.
 - 3. For any variation in supports needed for equipment actually purchased, the Contractor shall bear all costs of modifications, including redesign costs.
- C. Support Steel: Attach to building structure to support piping and smaller plumbing equipment items. Provide supports between structural members, except where specifically called out to be provided as part of the General Trades Work. Obtain prior written approval of the A/E.
- D. Hangers for Piping: Install all necessary pipe hangers, supports, clamps, wall brackets, and attachments in order to:
 - 1. Properly support piping from building structure.
 - 2. Allow controlled thermal movement of piping systems.
 - 3. Facilitate action of expansion joints, loops, and bends, where used.

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4. Provide slope where indicated.
- E. Concrete Work: A qualified Tradesperson shall provide all concrete work.
- F. Provide all necessary inserts, expansion shields, beam clamps, piping floor supports, auxiliary steel, angles, clips, mechanical fasteners, reinforcement steel, structure drilling, welding to structure, etc.
- G. Components shall be selected and matched to the load imposed on them.
- H. Refer to Structural and Architectural Drawings and Specifications to determine what supports are provided as part of the General Trades Work.

1.03 TERMINOLOGY

- A. Abbreviations:
 1. Chlorinated polyvinyl chloride (CPVC).
 2. Cross-linked polyethylene (PEX).
 3. Polyvinyl chloride (PVC).

1.04 QUALITY ASSURANCE

- A. Standards:
 1. American Concrete Institute (ACI).
 2. American National Standards Institute (ANSI)
 3. American Society for Testing and Materials (ASTM).
 4. American Society of Mechanical Engineers (ASME), including:
 - a. ASME B31.1, "Code for Power Piping."
 - b. ASME B31.9, "Building Services Piping."
 5. Factory Mutual Global (FMG).
 6. Manufacturers Standardization Society of the Valve and Fittings Industry (MSS).
 - a. Standard Practice 58.
 - b. Standard Practice 69.
 - c. Standard Practice 89.
 7. National Fire Protection Association (NFPA).

**HANGERS & SUPPORTS FOR
PLUMBING PIPING &
EQUIPMENT**

8. Underwriters Laboratories, Inc. (UL).

PART 2 PRODUCTS

2.01 STRUCTURAL STEEL

- A. Steel plates, shapes, and bars; steel beams; strut channels; standard-weight black steel pipe; or steel angle. ASTM A36 / ASTM A36M, black and galvanized, of adequate size and strength to safely carry the weight of equipment item.

2.02 HANGERS FOR PIPING

- A. Acceptable Manufacturers: Anvil International, B-Line Systems Inc., Carpenter & Patterson Inc., ERICO, Globe Pipe Hanger Products Inc., Klo-Shure, National Pipe Hanger Corp., or PHD Manufacturing Inc.
- B. Uninsulated Piping:
1. Steel, cast iron, or plastic; 1/2 in. to 8 in.: Anvil International Fig. 69. Galvanized carbon steel, adjustable swivel ring.
 2. Steel, cast iron, or plastic; 10 in. and larger: Anvil International Fig. 260. Adjustable clevis, carbon steel yoke, U-strap, rod, and hex nuts.
 3. Copper, 1/2 in. to 4 in.: Anvil International Fig. CT-65. Copper-plated hangers, adjustable tubing ring, carbon steel ring, and malleable iron adjusting nut.
- C. Insulated Piping:
1. Steel, plastic, or copper; all sizes: Anvil International Fig. 260. Adjustable clevis, carbon steel yoke, U-strap, rod, and hex nuts.
 2. Cast iron, all sizes: Anvil International Fig. 590. Adjustable clevis, carbon steel yoke, U-strap, rod, and hex nuts.
 3. Thermal protector: Thermal Pipe Shields Model T2000. Vapor barrier, galvanized steel sheet metal jacket, cellular glass thermal insulation (ASTM C552) same thickness as pipe insulation. Refer to Section 22 07 00 for Insulation dimensions.
 - a. Acceptable Manufacturer: Thermal Pipe Shields Inc.
- D. Vent Piping:
1. Plastic: Anvil International Fig. 69. Galvanized carbon steel, adjustable swivel ring.
 2. Metal: Anvil International Fig. 97. Adjustable pipe ring, steel band, malleable iron adjusting nut.

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3. Metal: Anvil International Fig. 69. Adjustable pipe ring, steel band, malleable iron adjusting nut.
 4. Glass: Anvil International Fig. 97C. Coated adjustable pipe ring, plastic-coated carbon steel band, malleable iron adjusting nut.
- E. PEX and CPVC piping:
1. Install vinyl-coated hangers for PEX piping with the following maximum horizontal spacing and minimum rod diameters:
 - a. 1 in. and smaller: 32 in. with 3/8-in. rod.
 2. Install hangers for vertical PEX piping every 48 in.
 3. Install vinyl-coated hangers for CPVC piping with the following maximum horizontal spacing and minimum rod diameters:
 - a. 1 in. and smaller: 36 in. with 3/8 in. rod.
 - b. 1-1/4 in. to 2 in.: 48 in. with 3/8 in. rod.
 - c. 2-1/2 in. to 3-1/2 in.: 48 in. with 1/2 in. rod.
 - d. 4 in.: 48 in. with 5/8 in. rod.
 4. Install supports for vertical CPVC piping every 60 in. for 1 in. and smaller, and every 72 in. for 1-1/4 in. and larger.
 5. Support piping and tubing not listed above according to MSS SP-69 and the Manufacturer's written instructions.
- F. For Piping Indicated to be Installed "Tight to Ceiling or Wall":
1. Uninsulated, 1/2 in. to 4 in.: Anvil International Fig. 262. Carbon steel.
 - a. Option: PowerStrut channels with PowerStrut Model PS 3126 hold-down clamp. Use with rubber cushion for copper pipe.
 2. Insulated, 3/4 in. to 8 in.: Anvil International Fig. 103. Carbon steel, offset pipe clamp.
- G. Vertical Piping:
1. Anvil International Fig. 261. Friction riser clamp with 2-point bearing.
 2. Use Anvil International Fig. CT-121 copper-plated riser clamp for copper pipe.

2.03 HANGER RODS, CLAMPS, AND BRACKETS FOR PIPING

- A. Acceptable Manufacturers: Anvil International, B-Line Systems Inc., Carpenter & Patterson Inc., Globe Pipe Hanger Products Inc., ERICO, Miro, National Pipe Hanger Corp., PHD Manufacturing, Inc., or Klo-Shure.
- B. Hanger Rods: Solid steel, threaded ends or all-thread rods of diameter listed below, with double nut attachment to the hanger and at the hanger attachment. Sizes are as follows:

PIPE SIZES	ROD DIAMETER
Up to 2 in.	3/8 in.
2-1/2 in. and 3 in.	1/2 in.
4 in. and 5 in.	5/8 in.
6 in.	3/4 in.
8 in. to 12 in.	7/8 in.
(3/8 in. rods are acceptable for plastic vent lines)	

- C. Hanger Clamps:
- Beam clamps (up to 8 in. diameter pipe): Anvil International Fig. 218, Fig. 228, or Fig. 292. Top beam clamp, steel jaw, hook rod with nut, and spring washer steel eye-bolt, with Anvil International Fig. 157 or Fig. 290 extension piece for appropriate rod diameter.
 - Option: Anvil International Fig. 86, Fig. 93, or Fig. 94. Malleable iron clamp, hardened steel cup point set screw with Anvil International Fig. 89 steel retaining clip.
 - Purlin clamps:** PHD Manufacturing Fig. 290. Malleable iron with jam nut and pointed steel set screw. Confirm type to be used with the Lead Contractor and Metal Building Manufacturer.
- D. Hanger Brackets:
- Acceptable Manufacturers: Anvil International, B-Line Systems Inc., Carpenter & Patterson Inc., Globe Pipe Hanger Products Inc., ERICO, National Pipe Hanger Corp., PHD Manufacturing Inc., Klo-Shure, Unistrut Corp., or ABB (Thomas & Betts) Superstrut®.
 - Wall brackets: Carbon steel, back plates, and bolts.
 - Anvil International Fig. 194. Lightweight 750 lb. load.

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- b. Anvil International Fig. 195. Medium-weight 1,500 lb. load.
- c. Option: Unistrut or ABB Superstrut® A1200.

2.04 SPRING HANGERS FOR PIPING

- A. Acceptable Manufacturers: Anvil International, or Kinetics Noise Control, or approved equal by B-Line Systems Inc., Carpenter & Patterson Inc., ERICO, Globe Pipe Hanger Products Inc., Klo-Shure, National Pipe Hanger Corp., or PHD Manufacturing Inc.
- B. Anvil International Fig. B268, Type A or B. Steel, adjustable pre-compressed variable spring.
- C. Kinetics Noise Control Model SH. Refer to Section 22 05 48.

2.05 TRAPEZE HANGERS FOR PIPING

- A. Acceptable Manufacturers: Unistrut Corp., or ABB (Thomas & Betts) Superstrut®, or approved equal by Anvil International, B-Line Systems Inc., ERICO, GS Metals Corp., National Pipe Hanger Corp., Powerstrut, or Wesanco Inc.
- B. Unistrut P1000 or ABB Superstrut® A1200. Angle iron, or channel of sufficient length to support pipes and insulation on individual hangers, roller supports, or saddles with insulation protectors.

2.06 HANGER COVERS FOR PIPING

- A. Acceptable Manufacturers: Buckaroos Inc. or approved equal by Pipe Shields Inc. or PHS Industries.
- B. Buckaroos Wraparoos. 2-piece PVC cover. Snap-on design, adhesive seals, to 205 degrees F. temperature. Includes 18 in. long PVC hanger rod cover. Available in multiple colors. The A/E shall approve all color selections for plumbing equipment.

2.07 PIPE-INSULATING SUPPORT

- A. Piping 1-1/2 in. and Smaller: Anvil International Fig. 167. 12 in. long, 18-gauge galvanized or stainless steel sheet metal half round piping shield inside clevis hanger.
 - 1. Acceptable Manufacturers: Anvil International or approved equal by Buckaroos Inc., Carpenter & Patterson Inc., PHS Industries Inc., Pipe Shields Inc., Rilco Manufacturing Co. Inc., or Value Engineered Products Inc.

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- B. Piping 2 in. and Larger: Insul-Cooustic Co. Insul-Shield. Pre-compressed fiberglass load-bearing segment, integral fire-retardant vapor barrier jacket and saddle.
 - 1. Acceptable Manufacturers: Insul-Cooustic Co. or approved equal by Pipe Shields Inc., Buckaroos Inc., or ERICO.

2.08 PIPE ROLL STANDS

- A. Acceptable Manufacturers: Anvil International or approved equal by B-Line Systems Inc., Carpenter & Patterson Inc., Globe Pipe Hanger Products Inc., ERICO, National Pipe Hanger Corp., PHD Manufacturing Inc., or Klo-Shure.
- B. Anvil International Figure 271. Cast iron roll, stand with holes for anchoring, sized for use with protection saddle.

2.09 FINISH

- A. Unless otherwise noted, all steel hangers and supports shall be standard black.
 - 1. Exception: Hangers and supports for exposed exterior applications shall be galvanized.

2.10 CONCRETE

- A. All concrete shall be 5,000 PSI minimum and placed in accordance with applicable ACI Standards. **Refer to Division 03.**
- B. Class II with a 28-day compressive strength of 5,000 lbs., and a maximum water-to-cement ratio of 0.50.
- C. Concrete bases and pads: Granular material conforming to **Division 03** requirements.
- D. **Concrete Encasement of Underground Piping:** Material conforming to **Division 03** requirements. Minimum of 3 in. thick and containing red marker dye. Concrete shall be pre-mixed by the concrete supplier; do NOT trowel in color additive by hand.

2.11 INSERTS

- A. **In Masonry with Voids:** Master-Carr #94255A100. Steel plated with yellow chromated zinc, female threaded toggle.
 - 1. Acceptable Manufacturers: Master-Carr or approved equal by Vertical Solutions or Hohmann & Barnard.

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- B. **In Poured Concrete:** Anvil International Fig. 282. Malleable iron, universal mounting, lateral adjusting.
1. Acceptable Manufacturers: Anvil International Inc. or approved equal by B-Line Systems Inc., Carpenter & Patterson Inc., Globe Pipe Hanger Products Inc., ERICO, Klo-Shure, National Pipe Hanger Corp., PHD Manufacturing Inc., or Unistrut.

PART 3 EXECUTION

3.01 INSTALLATION

- A. All piping supports, parts, and installation shall conform to the latest requirements of ASME B31.1 and MSS SP-58, SP-69, and SP-89, and shall not exceed the maximum pipe deflections allowed by ASME B31.9, except as supplemented or modified by the requirements of this Specification.
- B. Install all piping systems with adequate provisions made for expansion and to prevent stresses on valves and equipment. Provide adequate pipe anchors, guides, and supports from building structure to properly control pipe movement. Method shall suit job conditions. Refer to Section 22 05 16.
- C. Provide proper steel angles and channel systems, etc., between structural members for hanger supports where necessary. Weld to steel structural members where noted. Consult with the A/E regarding this procedure.
- D. All hangers installed shall be appropriate for type of building structure indicated.
- E. Do not support hangers from roof deck **or floor above**. Span from structural members with supplementary steel where direct attachment to structural members is not practical.
- F. Install hangers and supports with all necessary inserts, bolts, rods, nuts, washers, and other accessories, according to the Manufacturer's recommendations. Hangers shall be double-nutted.
- G. Use a separate hanger for each pipe. Do not hang one pipe from another.
- H. Provide additional supports required to maintain piping alignment.

- I. Spacing of Piping Supports: Maximum spacing of piping supports shall be as follows:

PIPE SIZE	STEEL PIPE	COPPER PIPE
3/4 in. or smaller	6 ft.	5 ft.
1 in. through 1-1/4 in.	6 ft.	6 ft.
1-1/2 in.	6 ft.	8 ft.
2 in.	10 ft.	8 ft.
2-1/2 in.	10 ft.	9 ft.
3 in. and larger	10 ft.	10 ft.

1. Cast iron: 5 ft. maximum spacing (10 ft. spacing permitted where 10 ft. pipe lengths are installed) behind every hub, at branch connections, and at each change of direction.
 2. Plastic: Locate and space in accordance with the Manufacturer's recommendations. (4 ft. maximum spacing on plastic soil and vent piping.)
 3. Glass: 8 ft. to 10 ft. maximum spacing behind every joint, at branch connections, and at each change of direction.
- J. Support piping near elbows, such that the total cantilevered length for both sides of the elbow does not exceed 3 ft.
- K. Support piping at pumps and equipment from floor, structure, or walls, so that piping weight is not supported by pumps or by equipment.
- L. Vertical Stacks: Rest vertical storm and waste stacks firmly on masonry footings and support them firmly at each floor.
- M. Structural Steel: Verify size required with the Structural Engineer.
- N. Insulated Piping Hangers:
1. Adjust pipe hangers and hanger rods to proper elevation in a plumb position before pipe insulation is installed.
 2. Hanger size shall be sufficient to accommodate pipe and insulation without compressing the insulation. Size hangers for outer diameter of insulation so that hangers pass freely around the insulation.

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3. Install 12 in. long, 18-gauge half-round shields inside clevis pipe hangers at all hangers (regardless of size). Verify method to be used.
 4. Install insulating saddles of non-compressible insulation with a continuous vapor barrier on pipes 2 in. and larger during the period that the insulation is being applied.
 5. Do not apply insulation to piping until insulation shields and saddles are in place at each pipe hanger.
 6. Refer to Section 22 07 00.
- O. Option: Trapeze hangers may be used where parallel runs of pipe occur.
- P. Vertical Piping Hangers:
1. Support vertical risers at the floor with friction riser pipe clamps.
 2. Shorten legs to conceal riser clamp within pipe chase and still provide adequate support.
- Q. Hanger Clamps:
1. DO NOT USE C-CLAMPS WITHOUT RETAINING CLIPS ON PIPING 1-1/2 IN. AND LARGER.
 2. Beam Clamps: Use top flange beam clamps to avoid burning metal deck.
- 3. Purlin Clamps:**
- a. Provide additional auxiliary steel and fasteners to hang from purlins.
 - b. Where hanging from Z-purlins, use purlin clamps.
 - 1) Option: Drill vertical portion of purlin and provide angle bolted to purlin and threaded for hanger rod. Obtain approval of the Lead Contractor, Metal Building Manufacturer, and A/E for attachment method before proceeding. Comply with the Metal Building Manufacturer's recommendations.
- R. Metal Fabrication:
1. Fit connections together to form hairline joints.
 2. Field-weld connections that cannot be shop-welded because of shipping size, space limitations, and field conditions.
 3. When field-welding, comply with AWS D1.1 procedures for shielded metal arc welding, appearance and quality of welds, and methods used in correcting welding work.

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- S. Food Processing Areas: Provide hanger covers to cover and protect steel clevis type pipe insulation hangers and shields from insects, bacteria, moisture, dust, and dirt.
- T. Roofs:
 - 1. Roof integrity and warranty shall be preserved.
 - 2. Coordinate installation of supports and roof penetrations.
 - 3. Provide pipe supports for piping installed on roof under Plumbing Work.
 - 4. Seal all penetrations through roof.
 - 5. Coordinate with the Lead Contractor BEFORE cutting.
- U. Ceilings:
 - 1. Do NOT support piping from ceiling and ceiling grid systems.
 - 2. Piping, and each ceiling and ceiling grid system, shall be a separate installation. Independently support each from the building structure.
 - 3. Where interferences occur, in order to support piping, install trapeze-type hangers or supports. Locate them where they do not interfere with access to valves and other plumbing equipment items.
- V. Painting:
 - 1. Touch-up:
 - a. Clean field welds, bolted connections, and abraded areas of all shop paint.
 - b. Paint exposed areas immediately after erecting hangers and supports.
 - c. Use same paint materials as used for shop painting.
 - d. Apply paint by brush or spray to provide a minimum dry film thickness of 2.0 mils.
 - 2. Galvanized surfaces: Clean and apply galvanizing repair paint to comply with ASTM A780.

3.02 INSTALLATION — SPRING HANGERS FOR PIPING

- A. Install on inlet and outlet piping of new pumps. Refer to Section 22 05 48.

HANGERS & SUPPORTS FOR PLUMBING PIPING & EQUIPMENT

3.03 INSTALLATION — TRAPEZE HANGERS FOR PIPING

- A. Design support system. Provide support capacity, support spacing, trapeze hanger rod diameter, and quantity to support total piping load.
 - 1. All rods on trapeze hangers shall be 1/2 in. minimum.
 - 2. Clamp each pipe individually.
 - 3. Provide individual U-bolt clamps for insulated pipes.
 - 4. Clamp size shall accommodate insulation and shield.
 - 5. Provide individual piping attachment to each strut channel hanger, angle, or channel.
- B. Obtain approval of a Structural Engineer licensed in Illinois. Submit details to the A/E for approval.

3.04 INSTALLATION — CONCRETE

- A. Provide timely notification and carefully coordinate locations, sizes, and extent of all concrete foundations and supports for plumbing equipment with the Lead Contractor and Structural Engineer.
- B. For any equipment item to be supported, provide dimensions for rough-in to the Lead Contractor.
- C. Construction shall conform to Division 03 requirements.

DESIGNER — EDIT TABLE. Add “furnish,” “install,” or “provide”:

Item	Lead Contractor	Plumbing Contractor
Anchor bolts		Provide
Concrete bases for _____		
Concrete blocks for _____		
Concrete curbs around pipe sleeves		
Concrete encasement of underground piping under footings and foundations		
Concrete equipment pads for _____		

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Item	Lead Contractor	Plumbing Contractor
Concrete fill for inertia base frames		
Concrete pour for grease interceptor anti-buoyancy backfill		
Concrete slabs with mitered edges		
Concrete surrounding _____		
Concrete thrust blocks and supports for underground piping installation		
Steel reinforcing bars and mesh	Provide	

- D. Differences in elevation between (2) points shall not exceed 0.16 in. in 1 ft. and 0.38 in. in 5 ft. Slab area shall be moist-cured to prevent excessive moisture loss.
- E. Concrete Encasement of Underground Piping:
1. Clean around underground piping prior to encasement.
 2. Where required, concrete encasement shall extend 6 in. around piping or sleeve for 1 ft. each side of footings or foundations, and up to footing bearing elevation. If piping is less than 2 ft. below grade to top of piping, provide 6 in. concrete encasement.
 3. Backfill around piping after concrete has properly hardened.
- F. A Testing Laboratory Representative shall be present during the placement of all concrete unless this requirement is waived by the A/E. Provide ample notice to the Testing Laboratory and maintain records of all tests, indicating exact location of the structure represented by each test.
1. Obtain concrete for required tests at point of placement.
 2. Perform (1) strength test for each 50 yards, or fraction thereof, placed in any (1) day.
 3. Determine slump for each strength test.
 4. Determine air content for each strength test.
 5. Determine concrete temperature for each strength test when air temperature is less than 40 degrees F. or will be within (24) hours.
 6. Do not place concrete if slump, air content, or air temperature vary from required limits.

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G. Curing:

1. Concrete may be either moist-cured or receive an application of curing compound.
 - a. Exception: When concrete above grade is placed in the open and the air temperature exceeds 60 degrees F., the concrete shall be moist-cured for the first (24) hours.
2. Whichever curing method is used, it shall commence immediately after disappearance of water sheen and continue for at least (7) days. Do not allow curing to be delayed overnight.
3. Prevent excessive moisture loss from formed surfaces. If forms are removed before (7) days have elapsed, cure the formed surfaces by moist-curing or application of curing compound for the remainder of the curing period.

H. **Lead Contractor shall keep** all concrete pads moist for (5) days.

I. Concrete Testing: When concrete fails to meet the acceptance criteria specified in ACI 301, Section 17.2, the A/E may order further testing of in-place concrete, in accordance with Section 117.3. When such tests are ordered, the Lead Contractor shall pay cost of testing.

J. Lead Contractor shall bear all costs of correcting rejected work.

K. Refer to Division 03 for additional requirements regarding concrete work.

3.05 INSTALLATION — INSERTS

- A. Avoid cutting concrete or masonry by using inserts.
- B. Refer to the Manufacturer's recommendations for correct selection as to piping size and loading.
- C. Confirm with the Lead Contractor that the concrete used is of a sufficient strength to hold the insert.
- D. Provide the correct size and type of inserts, along with information on their correct location, to the Lead Contractor, so as not to delay construction.
- E. Masonry Construction with Voids: Install appropriately sized all-thread rods within a 2-component polyester resin adhesive used in conjunction with a galvanized metal screen. Install rods in holes drilled with carbide-tipped drill bits.
- F. New Poured Concrete: Before pouring, the Contractor shall **furnish** inserts (when used or required) for its Work to the Lead Contractor, who shall install where directed by the Contractor.

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SECTION 22 05 43 - UNDERGROUND PLUMBING SYSTEMS

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 22 05 00.35, "Special Procedures for Plumbing."
- B. Section 22 05 00.73, "Execution of Plumbing."
- C. Section 22 05 29, "Hangers and Supports for Plumbing Piping and Equipment."

1.02 DESCRIPTION

- A. Provide:
 - 1. Excavation and backfill.
 - 2. Underground service identification tape.
 - 3. Paving.
- B. Concrete Work (pads, bases, slabs, bollards, encasement of piping, etc.): Refer to Section 22 05 29.

1.03 QUALITY ASSURANCE

- A. Standards:
 - 1. Associated General Contractors of America (AGCA) "Manual of Accident Prevention in Construction".
 - 2. Occupational Safety and Health Administration (OSHA) requirements.
 - 3. State Department of Transportation (DOT) construction specifications.
 - 4. State and Local requirements.

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PART 2 PRODUCTS

2.01 BACKFILL

A. Typical backfill conditions:

1. Piping where bedding is not specified or required by Code: Previously excavated material.
2. All underground piping: Crushed stone (DOT Item 304), pea gravel, or coarse sand from 4 in. below piping to slab or 4 in. below grade.
3. Other piping outside building: Pea gravel from 4 in. below piping to 12 in. minimum above pipe, then previously excavated material.
4. Piping under drives, streets, and parking lots: Pea gravel from 4 in. below piping to 12 in. minimum above pipe, then previously excavated material. Piping less than 2 ft. below grade to top of pipe shall be encased in 6 in. of concrete.
5. Piping in trenches cut through rock: Tamped backfill shall be used for at least 6 in. under and around pipe and for at least 2 ft. above the pipe.
6. Piping under footings and foundations: Concrete encasement.
7. Underground grease interceptor, oil interceptor, sump pump, sewage ejector, lift station, or valve boxes: Pea gravel.

B. Materials:

1. Crushed stone: (DOT Item 304).
2. Pea gravel: 1/8 in. minimum to 3/4 in. maximum diameter.
3. Sand: Clean, dry, coarse or medium.
4. Washed gravel: 3/4 in. size.

C. Materials Prohibited from Being Used for Backfill:

1. Material containing large rocks (over 2 in.), building materials, masonry debris, cinders, rubbish, wood, or other material subject to decay.
2. Any material which may cause damage to piping.
3. Frozen earth.

2.02 UNDERGROUND SERVICE IDENTIFICATION TAPE

- #### A. Acceptable Manufacturers: Seton or approved equal by Brady, Ideal, or Emedco.

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- B. Seton. 6 in. wide metallic underground warning/detection tape detectable with metal detector.

2.03 PAVING

- A. Asphalt Paving:
 - 1. Aggregate base: DOT Item 304, 8 in. total, compacted in (2) 4 in. layers.
 - 2. Base course: DOT Item 402, 1-3/4 in. thickness.
 - 3. Surface course: DOT Item 404, 1-1/4 in. thickness.
 - 4. Tack coat: DOT Item 407.
 - 5. Prime coat: DOT Item 408.
- B. Sealer: DOT Item 409.

PART 3 EXECUTION

3.01 EXCAVATION

- A. Existing Soil Conditions:
 - 1. Bidders are strongly urged to visit the site and investigate existing soil conditions themselves.
 - 2. Rock excavation is not anticipated. Contractor will be entitled to additional compensation for excavating any rock that is not indicated on the Contract Drawings.
 - a. Rock excavation is defined as the excavation of any buried boulders and rock, in excess of 1/2 cu. yd. in volume, which requires systematic drilling with jackhammer or use of other special equipment.
 - b. If rock or boulders are encountered, immediately stop all excavation. Notify the A/E, and submit a cost per cubic yard, over and above Base Bid, for excavation and backfill of rock areas.
 - c. All other excavation, including soft shale and similar sedimentary soil formations, shall be regarded as "earth" excavation, and shall be included in the Contractor's Bid.
 - d. Do not proceed with rock excavation without prior written authorization of the A/E.
 - e. DO NOT USE EXPLOSIVES.

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3. If available, refer to Soils Report.
 4. Whenever, in the opinion of the A/E, the soil is unsuitable for supporting piping, provide proper foundation after receiving written approval of the A/E. The Contract price will be adjusted accordingly.
- B. Obtain approval from the A/E for bearing conditions.
- C. Existing Underground Utilities:
1. Contact the Julie Before You Dig Utility Service (800-892-0123) well in advance of the start of any excavation to determine if any of the Utility Companies have underground utilities in, or near, the project area.
 2. Contact the Owner and the Local Water and Sewer Department, Gas Company, Electric Company, Telephone Company, etc., regarding the possibility of encountering existing utilities. Maintain the integrity of all existing utilities.
 3. Locate and identify existing underground utilities, if any, in the areas of new Work. If utilities are to remain in place, provide adequate means of protection during earthwork operations. Should uncharted, or incorrectly charted, piping or other utilities be encountered during excavation, notify the A/E immediately for instructions before proceeding. Cooperate with other Contractors, Utility Companies, and the Owner in keeping services and the facilities in operation.
 4. Protect existing utilities encountered during excavation work in a manner acceptable to the Owner of the Utility. Excavation damage to, or interruption of service of, any underground utility shall be a liability of the Contractor, whether or not plotted on the Drawings. Contractor shall promptly repair or replace, at its expense, all damage to any existing utility to the full satisfaction of the Utility Company, the Owner of the Utility, and the A/E.
 5. All existing utility service piping, conduits, wiring, etc., which are uncovered, whether within or beyond the property lines, shall be suitably supported, protected, and maintained in operation, and shall be protected against settlement when excavations are refilled.
- D. Coordinate timing of excavations in advance with other Contractors, the A/E, and the Owner, where required by AHJ, or where otherwise specified herein.
- E. Excavations shall be open-cut from the surface. No undercuts will be permitted, except where specifically directed by the A/E.
- F. Remove surplus and/or unsuitable-for-fill excavated material from the site. Deposit any usable excavated material where directed by the A/E.

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G. Excavation Safety:

1. ALL EXCAVATION WORK SHALL COMPLY WITH REQUIREMENTS OF THE LATEST OSHA STANDARDS.
2. Excavation and trench wall supporting, shoring, sloping, cribbing, stepping of excavations, and other steps required for safety shall be in strict accordance with OSHA and Local Code requirements.
3. Slope sides of excavations to comply with requirements of OSHA and AHJ. Shore and brace where sloping is not possible because of space restrictions or instability of material exposed.
4. Maintain sides and slope of all excavations in a safe condition, until completion of backfilling.
5. Provide materials for shoring and bracing, such as sheet piling or sheathing, uprights, stringers, and cross-braces, in good serviceable condition required to safely support walls of the excavation. Maintain shoring and bracing in excavations regardless of time period excavations will be open. Carry down shoring and bracing as excavation progresses.
6. Barricade open excavations occurring as part of this Work, and post with warning lights. Erect warning lights in accordance with OSHA and AHJ requirements. Consult with the A/E regarding additional requirements.
7. Maintain in place adequate barricades, guards, planking, plating, signage, warning lights, etc., at and around excavations. Maintain in a safe condition until backfilling is completed.
8. Install steel cover plates, safety barricades, and coverings for holes in walks, pavements, streets, or grassy areas.
9. Protect existing structures that are to remain, such as utilities, sidewalks, pavements, and other facilities, from damage caused by settlement, lateral movement, undermining, washout, and other hazards created by earthwork operations.
10. Refer to AGCA's "Manual of Accident Prevention in Construction" and to the safety regulations of the appropriate State Agency.
11. Refer to Section 22 05 00.35 for additional safety requirements.

H. Provide and operate pumping equipment to keep excavation free of water at all times.

I. Protect excavations from frost by covering and heating.

J. Trenches:

1. Coordinate the use of a joint trench with all Utility Companies sharing the trench.

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2. Do not excavate utility trenches closer than 5 ft. parallel to building or column footings, except with prior written approval of the A/E. When parallel trenches deeper than the building footings are required, the horizontal distance from the footing shall be equal to, or greater than, 1-1/2 times the vertical distance below the footing, but in no case shall the horizontal distance be less than 5 ft., except with the written approval of the A/E.
 3. Coordinate routing of underground lines with the A/E to minimize disruption and damage to trees, shrubs, walks, drives, and other landscape features.
 4. Mark trench location. Hold trench width to a minimum. Cover as soon as possible.
 5. Minimize time trenches are kept open. Fill as soon as possible.
- K. Roadways and Public Right-of-Way:
1. **Lead Contractor shall saw-cut** any street, drive, sidewalk, curb, parking lot paving, or other permanent hard surface that is encountered in the path of the excavation.
 2. Avoid the need to cut concrete or paving by roughing in piping BEFORE concrete or paving is installed.
 3. Cut and replace any exterior paving or sidewalks to install underground work.
 4. Excavation, backfill, surface repair, and traffic control within the public right-of-way shall be in accordance with governing agency rules and regulations.
 5. Contractor shall pay any fee associated with alteration of a roadway.
- L. Refer to Section 22 05 00.73 for cutting and patching responsibilities.
- M. Backfilling:
1. The bedding and backfill requirements listed in this Section represent a minimum requirement. If State or Local requirements are in excess of listed requirements, such requirements shall supersede.
 - a. Reference datum: All dimensions or elevations are relative to finished grade (not existing grade).
 2. ALL BACKFILL WORK SHALL COMPLY WITH REQUIREMENTS OF THE LATEST OSHA STANDARDS.
 3. Deposit fill in layers of thickness required by the nature of the soil, or as directed, but not exceeding 6 in. compacted thickness, to a point 24 in. above pipe, and 12 in. thickness above this point. Compact each layer to a uniform solid mass. Place fill in horizontal layers, beginning with lowest areas and building up until entire area to be filled is at a uniform elevation. Compaction shall be a minimum of 90 lbs. per cu. ft., laboratory dry weight.

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4. Do not place fill material on frozen ground, or use fill in a frozen condition.
5. Backfill only when exact locations of lines and equipment have been recorded, and all tests and inspections have been completed.
6. Control moisture content of compacted fill to ensure maximum density by adding water and working soil prior to compacting.
7. Use machine tampers around perimeter of foundation walls or areas inaccessible to large equipment and rollers. Do not use rolling equipment in areas adjacent to foundations.
8. Provide clay bulkheads, minimum 3 ft. long, across the trench at 100 ft. intervals to impede the natural movement of ground water. Top of bulkheads shall extend 6 in. above top of pipe.
9. Backfill to finish grade or to levels suitable for finish grading.
10. **Lead Contractor shall provide** finish grading.
11. Where excavations have not been properly filled or where settlement occurs, refill, compact, smooth off, and finish excavations to a final condition acceptable to the Owner and A/E.
12. Install continuous underground service identification tape above buried outside piping service at 18 in. above pipe elevation on pipe centerline to identify all the underground Plumbing Work.
13. Mark location of trenches and lines on As-Built Drawings.
14. (11) months after Contract Completion, review the excavated areas and add any backfill required due to settling, etc. Re-seed or re-sod.
15. **Lead Contractor shall provide** seed grass cover to reestablish landscape to previous condition where disturbed by Plumbing Work. This includes replacement of existing plants, shrubs, and related items.
16. **Lead Contractor shall repair** and restore plants, shrubs, trees, sodded areas, paving, streets, curbs, and walks. Match existing in the area where excavations are made.
17. **Lead Contractor shall maintain** seeded lawns not less than (11) months after Contract Completion and add any backfill required due to settling, etc. If seeded in non-growing season and not given full (11) months of maintenance, or if not considered acceptable at that time, continue the following growing season until an acceptable lawn is established, as determined by the A/E.

3.02 INSTALLATION — UNDERGROUND PLUMBING PIPING

- A. Provide excavation and backfilling or directional boring for the Plumbing Work.

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B. Bedded Piping:

1. Provide pipe bedding for all gas, storm sewer, and sanitary sewer lines located outside of the building.
2. Provide pipe bedding for all sanitary sewer lines located below the floor slab.
3. Mechanical excavation shall extend 4 in. to 6 in. below final elevation of pipe.
4. In trenches cut through rock, mechanical excavation shall extend 6 in. below and around final elevation of pipe.
5. Bed pipes in pea gravel, minimum 4 in. below pipe, 6 in. above pipe.

C. Unbedded Piping:

1. Mechanical excavation shall be held to 4 in. above final elevation. The remainder shall be trenched by manual excavation, so that piping is fully supported on undisturbed soil.
2. DO NOT SHORE PIPING IN TRENCH.
3. If excavation should go deeper than required, fill with bank run gravel and tamp firmly to achieve final elevation. Wherever material is encountered in the bottom of the excavation that is not capable of supporting piping properly, remove such material to depth required. Backfill with crushed stone and tamp firmly to achieve final elevation.

D. Place no backfill until underground lines have been tested. Compact backfill in 6 in. deep layers.

E. Contractor shall be responsible for any other repairing of excavated areas.

F. Coordinate with the Lead Contractor.

3.03 PAVING

A. **Lead Contractor shall repave** all areas affected by plumbing construction work. Paving shall either match existing pavement or comply with the requirements listed below.

B. Preparation:

1. Proof-roll the sub-grade. Roll and compact to obtain an acceptable, firm, even sub-grade surface. Fill and consolidate dented or depressed areas. Remove incompatible material, replace with clean fill, and compact solidly.
2. Coordinate junction of new and existing pavement. Meet existing surface levels and maintain drainage slopes.

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C. Installation:

1. Base Materials: Install 8 in. aggregate base material and compact in accordance with DOT Item 304 until a uniformly smooth, hard surface, conforming to the lines, grades, elevations, and cross-section shown, has been established. Moisture may be added at job site to aid compaction.
2. Surface Materials:
 - a. Apply prime coat uniformly to aggregate base in accordance with DOT Item 408, at rate of 0.25 gallons to 0.30 gallons per sq. yd. Allow to dry and cure.
 - b. Install asphalt materials in (2) layers and compact each layer in accordance with DOT Item 401. Provide a hard, smooth, unyielding surface, true to thickness and elevation shown. Roll until no roller marks are visible.
 - c. After final rolling, do not permit vehicular traffic on the pavement until it has cooled and hardened, i.e., for at least (8) hours.

D. Final Sealing:

1. Prior to sealing, thoroughly clean and sweep area. Block all traffic to areas to be sealed. Provide striping as shown on the Drawings. Maintain barriers until sealing and striping is complete.
2. Apply (2) coats of sealer to all asphalt surfaces (30) days after paving installation.

END OF SECTION 22 05 43

SECTION 22 05 53 - IDENTIFICATION FOR PLUMBING PIPING AND EQUIPMENT

PART 1 GENERAL

1.01 DESCRIPTION

- A. Identify the following plumbing items by labels, tags, or stencils:
1. Equipment:
 - a. Domestic water heaters, domestic hot water recirculating pump, control cabinets, and similar items.
 - b. All shock absorbers, whether accessible or not.
 - c. Trap primers, including description of what floor drains are served.
 2. Equipment located above an accessible ceiling: Every location of a hose bib, etc.
 3. Piping:
 - a. Exposed in equipment rooms, trenches, crawl spaces, tunnels, accessible service areas, or accessible chases.
 - b. Running above accessible ceiling construction.
 4. All valves, including control valves.
 - a. Exception: Valves or stops serving only (1) fixture need not be tagged.
 5. Valves located above an accessible ceiling: Every location of a shut-off valve, etc.
- B. Do NOT use "Dymo" or similar tape type labels.

1.02 QUALITY ASSURANCE

- A. Standards:
1. American National Standards Institute / American Society of Mechanical Engineers (ANSI/ASME):
 - a. ANSI/ASME A13.1, "Scheme for the Identification of Piping Systems."
 - b. ANSI Z535.1, "Safety Color Code."
 2. Occupational Safety and Health Administration (OSHA).

PART 2 PRODUCTS

2.01 ACCEPTABLE MANUFACTURERS

- A. Seton, Brady, Emedco, or MSI.

2.02 EQUIPMENT IDENTIFICATION

- A. Label: Aluminum, brass, laminated multi-color rigid plastic, or stainless steel, with holes for attachment to equipment.
 - 1. Size: Sufficient for text, minimum 2-1/2 in. wide by 3/4 in. high.
 - 2. Text characters: Engraved or stamped. Equipment tag text characters sized for normal viewing distance, minimum 1/4 in. high, maximum 3/4 in. high. Equipment specification number text characters approximately 3/4 size of tag characters.
 - 3. For laminated plastic labels, match background and character colors in system identification schedules in Part 3. If exact matches are not available, match background colors from schedules, and provide character colors in contrasting white on black.
- B. Location Identification for Equipment Located above Accessible Ceilings: Multi-color, pressure-sensitive-adhesive labels, or pressure-sensitive-adhesive equipment tag.

2.03 PIPING IDENTIFICATION

- A. 1 in. wide, gloss enamel color band. Stencils, same color as band.
- B. For first color, 1 in. wide color band. For second color, 1/2 in. wide color band. Stencils.
- C. Labels: Preprinted, color-coded plastic, with text characters indicating pipe service and arrows indicating flow direction. Include pressure rating where applicable.
 - 1. Pre-tensioned: Pre-coiled and semi-rigid label formed to cover full circumference of pipe (including insulation and jacket) and to adhere to pipe without fasteners or adhesives.
 - 2. Self-adhesive: Flexible label long enough to cover full circumference of pipe (including insulation and jacket) with permanent contact adhesive backing.

2.04 VALVE IDENTIFICATION

- A. Tags: Aluminum, brass, laminated multi-color rigid plastic, or stainless steel, with hole for attachment to valve.
 - 1. Size: Minimum 2 in. diameter round or 1-1/2 in. square.

2. Text characters: Engraved or stamped. 1/4 in. high indicating pipe service and 1/2 in. high indicating valve number.
 3. Fasteners: Metal bead chain, metal "S" hook, or nylon tie.
- B. Brass or aluminum tags with brass or aluminum chain.
- C. Tags: Aluminum, brass, laminated multi-color rigid plastic, or stainless steel, with hole for attachment to valve.
1. Size: Minimum 2 in. diameter round or 1-1/2 in. square.
 2. Text characters: Engraved or stamped. 1/4 in. high indicating pipe service and 1/2 in. high indicating valve number.
 3. Fasteners: Metal bead chain, metal "S" hook, or nylon tie.
- D. Location Identification for Valves Located above Accessible Ceilings: Multi-color, pressure-sensitive-adhesive labels, or pressure-sensitive-adhesive valve tag.

2.05 STENCILS

- A. Card stock or metal pre-cut stencil with equipment tag, or piping service and flow direction arrow.
- B. Paint: Exterior grade, gloss alkyd enamel. Colors in accordance with schedules listed in Part 3.

PART 3 EXECUTION

3.01 PREPARATION

- A. Clean surfaces of substances which could impair bond of identification device adhesive, where applicable.

3.02 COORDINATION

- A. Coordinate to ensure that the identification used by all Trades is uniform in type, style, and appearance.
- B. Equipment: Coordinate the exact nomenclature to be used on labels with the Owner and A/E.

- C. Equipment Located above Accessible Ceilings: Coordinate with the Owner regarding color-coded equipment location identification system. If color-coded equipment system does not exist, refer to Identification Schedule.
- D. Piping: Coordinate with the Owner regarding piping identification system. If piping identification system does not exist, refer to Identification Schedule.
- E. Valves: Coordinate with the Owner regarding valve numbering system. If valve-tagging system does not exist, refer to Identification Schedule.
- F. Valves Located above Accessible Ceilings: Coordinate with the Owner regarding color-coded valve location identification system. If color-coded valve system does not exist, refer to Identification Schedule.
- G. Coordinate label, tag, and fastener materials with expected conditions. Do not use plastic materials in locations where expected temperatures are outside of the Manufacturer's recommended limits.

3.03 INSTALLATION

- A. Equipment:
 - 1. Provide a label for each equipment item that has a unique tag. Do not provide a label if multiple items of equipment have the same tag.
 - 2. Locate equipment label where it will be accessible and visible when project is complete. For equipment that is installed exposed in finished space, locate tag in an accessible concealed location, such as behind an access door.
 - 3. Attach label with stainless steel screws.
 - a. Exception: Use compatible adhesive where screws might damage equipment.
- B. Equipment Located above Accessible Ceilings:
 - 1. In addition to attaching identification on the equipment, attach equipment location identification to the accessible ceiling grid at the point of equipment access.
 - 2. Apply pressure-sensitive adhesive labels or tags to a clean, dry surface.
 - 3. Provide schedule.
- C. Piping:
 - 1. Apply piping identification only after any finish painting is completed.
 - 2. Provide labels for piping at MAXIMUM 25 FT. intervals.

3. Identify piping at connections to equipment, at valves, at branches from main, and at each riser.
4. Where a pipe passes through a wall, label the pipe where it enters and exits the wall.
5. Option: Stencils may be used instead of manufactured labels. Apply stencil over background, in colors indicated. When paint is dry, provide a clear coat finish over stencil. Stencils shall be legible from a standing position.

D. Valves:

1. Provide a unique tag on each valve. Tags shall be stamped with service designation and numbered consecutively for each service. Attach tags to valves in such a manner that valve shall be operable without removing or damaging tag. Tags shall be readable from a standing position when valve is in normal position. Add valve tag numbers to As-Built Drawings.
2. Valves located above accessible ceilings:
 - a. In addition to attaching identification on the equipment, attach equipment location identification to the accessible ceiling grid at the point of equipment access.
 - b. Apply pressure-sensitive adhesive labels or tags to a clean, dry surface.
3. Prepare a valve tag directory with charts showing locations, designations, and sizes of valves. Laminate under plastic and mount as directed by the A/E. Include additional copies of valve charts in the Operating and Maintenance Manuals.

3.04 PIPING IDENTIFICATION SCHEDULE

A. Text character size shall be as follows:

Overall Pipe/Insulation Size	Minimum Character Height
Up to 1-1/4 in. diameter	1/2 in. high
1-1/2 in. to 2 in. diameter	3/4 in. high
2-1/2 in. to 6 in. diameter	1-1/4 in. high
8 in. to 10 in. diameter	2-1/2 in. high
12 in. and larger diameter	3-1/2 in. high

is this typically an
owner preference
thing or should i just
pick one

B. Identify piping as follows:

Type of Service	Band Color	Designation
Domestic Cold Water Supply	Light Blue	DCWS
Domestic Hot Water Supply (140 degrees F.)	Dark Blue	DHWS-140
Domestic Hot Water Return	Dark Blue	DHWR
Natural Gas	Yellow	GAS
Drain	(Room Color)	—

C. Identify piping as follows:

Type of Service	Background Color	Striping Color	Designation
Domestic Cold Water	Green	White	DCW
Domestic Hot Water Supply	Yellow	Black	DHWS
Domestic Hot Water Supply Return	Yellow	Black & White	DHWR
Sanitary Waste	Purple	Purple	SAN
Storm Water	Purple	Aluminum	STORM

D. Identify piping as follows:

Type of Service	Background Color	Character Color	Designation
Domestic Cold Water Supply	Green	White	DCW
Domestic Hot Water Supply (140 degrees F.)	Yellow	Black	DHW-140
Domestic Hot Water Return (140 degrees F.)	Yellow	Black	DHWR-140
Storm	Green	White	STM
Sanitary	Green	White	SAN
Sanitary Vent	Yellow	Black	V

Type of Service	Background Color	Character Color	Designation
Grease (Kitchen Grease Waste)	Yellow	Black	GRS
Natural Gas	Yellow	Black	GAS

3.05 VALVE IDENTIFICATION SCHEDULE

A. Identify valves as follows:

Type of Service	Valve Tag Designation
Water Service	W
Domestic Cold Water Supply	DCW
Domestic Hot Water Supply (140 degrees F.)	DHW-140
Domestic Hot Water Return (140 degrees F.)	DHWR-140
Natural Gas	GAS

END OF SECTION 22 05 53

SECTION 22 05 76 - FACILITY DRAINAGE PIPING CLEANOUTS

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 22 05 29, "Hangers and Supports for Plumbing Piping and Equipment."
- B. Section 22 13 00, "Facility Sanitary Sewerage."
- C. Section 22 13 23, "Sanitary Waste Interceptors."
- D. Section 22 14 00, "Facility Storm Drainage."

1.02 DESCRIPTION

- A. Provide cleanouts as shown on the Drawings.

1.03 QUALITY ASSURANCE

- A. Standard: American National Standards Institute (ANSI).

1.04 SUBMITTALS

- A. Floor Cleanouts: Each shop drawing shall indicate the ANSI load rating. If the rating is not indicated, the shop drawing will be rejected.

PART 2 PRODUCTS

2.01 CLEANOUTS

- A. Acceptable Manufacturers: Mifab or approved equal by Jay R. Smith Mfg. Co.®, Josam, WADE (McWane Inc.), Watts, or Zurn Industries LLC.
- B. All cleanouts shall be line size (6 in. size on 6 in. or larger line).
- C. Floor Cleanouts (F.C.O.):
 - 1. Carpeted floors: Mifab Model C1100-RC-6-34. Adjustable cast iron body, heavy-duty load rating, round stainless steel scoriated cover with vandal-resistant carpet marker, plastic plug, line size, and caulk or push-on outlet.
 - 2. Floors with ceramic tile or quarry tile finish: Mifab Model C1100-S-6-34. Adjustable cast iron body, heavy-duty load rating, vandal-resistant, square stainless steel scoriated cover, plastic plug, line size, and caulk or push-on outlet.

3. Floors with vinyl tile finish:
 - a. Mifab Model C1100-R-6-34. Adjustable cast iron body, heavy-duty load rating, vandal-resistant, round stainless steel cover, plastic plug, line size, and caulk or push-on outlet.
 - b. Mifab Model C1100-T-6-34. Adjustable cast iron body, heavy-duty load rating, vandal-resistant, round stainless steel cover with 1/8 in. tile recess, plastic plug, line size, and caulk or push-on outlet.
 4. Bare concrete floors: Mifab Model C1100-XR-6-34. Adjustable cast iron body, heavy-duty load rating, vandal-resistant, round stainless steel scoriated cover, plastic plug, line size, and caulk or push-on outlet.
 5. Vehicular traffic: Mifab Model C1100-XR-4-6-34. Adjustable cast iron body, heavy-duty load rating, vandal-resistant, round cast iron scoriated cover, plastic plug, line size, and caulk or push-on outlet.
 6. Each floor cleanout required to have a vandal-resistant top shall have at least (3) screws for attachment to top.
 7. Provide a membrane clamp on each floor cleanout having a waterproofing membrane.
- D. Wall Cleanouts (W.C.O.): Mifab Model C1400-RD-6. Round polished stainless-steel access cover, and vandal-resistant screw. Diameter of cover based on cleanout and wall opening, nominal 6-inch diameter.
- E. Exterior Cleanouts (E.C.O.): Mifab Model C1230-1. Cast iron body, round 1/2 in. thick gasketed nickel-bronze cover/plug, line size, and no-hub or push-on outlet. Provide Mifab Model C1300-MF-4-6 access housing, vandal-resistant, with anchor flanges, lacquered cast iron heavy-duty housing body, and extra-heavy-duty ductile iron access cover.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Coordinate location, elevation, and installation of cleanouts with the Lead Contractor and the A/E.
- B. If not shown on Architectural Drawings, confirm elevation and proper pitch to cleanouts with the A/E before roughing in.
- C. Install cleanouts in accessible locations.
- D. Use graphite on all cleanout plug threads.
- E. Install top of cleanouts flush and level with wall and finished floor.

- F. Provide exposed cleanouts without cover plate in unfinished areas or in above-ceiling installations.
- G. Where tile is to be installed, adjust top of cleanout to be flush with the finished surface.
- H. Wall Cleanouts: Install at the base of all stacks at 18 in. above finished floor where indicated on the Drawings.
- I. Floor Cleanouts:
 - 1. Install square cleanouts in ceramic tile floors and carefully align edges parallel to room walls.
 - 2. Install duct tape over floor cleanout covers to provide protection from scratching and collection of dirt and debris during construction. Remove tape just prior to final inspection.
- J. Exterior Cleanouts: Install exterior cleanouts in the center of an 18-inch square, 6-inch thick concrete slab. Cleanout and slab shall be installed flush with finished grade or pavement. Top of concrete shall slope to drain. Edges of concrete shall be mitered. Refer to Section 22 05 29 regarding concrete work.
- K. REMOVE AND REPLACE ALL ITEMS NOT INSTALLED AT PROPER ALIGNMENT, AT PROPER ELEVATION, FLUSH, AND LEVEL, AT NO COST TO THE OWNER.

END OF SECTION 22 05 76

SECTION 22 07 00 - PLUMBING INSULATION

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 22 05 29, "Hangers and Supports for Plumbing Piping and Equipment."

1.02 DESCRIPTION

- A. Provide insulation for new work.
- B. Insulate the following items:
 - 1. Piping:
 - a. Domestic cold-water supply, regardless of length or location.
 - b. Domestic hot-water supply, regardless of length or location.
 - c. Domestic hot-water return, regardless of length or location.
 - d. Make-up water, regardless of length or location.
 - e. Domestic water supply and drain piping under all lavatories and sink fixtures, including non-ADA fixtures, that could come in contact with wheelchair occupants.
 - f. Each above-ground drain that receives cold water wastes to a point where it connects to the main drain.
 - g. Above-ground trap primer drain piping.
 - h. Above-ground floor drain traps having trap primers, and first 3 ft. of piping.
 - i. Above-ground storm piping (both horizontal and vertical), roof sump, and roof drain bodies.
 - j. Underground domestic hot-water supply and return piping.
 - k. Underground sanitary waste piping from kitchen equipment and fixtures to grease interceptor.
 - l. Note: Insulation is not required on for limited area sprinkler piping.
 - 2. Equipment:
 - a. Hydro-pneumatic tank.
 - b. Hot water storage tanks (if not factory-insulated).

- c. Note: The following equipment shall be factory-insulated to comply with the State Energy Code and IECC.
 - 1) Water heaters.
 - 2) Water heating boilers and storage tanks.
 - 3) Hot water storage tanks.

1.03 TERMINOLOGY

- A. "Concealed" insulation: Either embedded in masonry or other construction, or installed above ceilings, below floor slab, behind wall furring, in trenches or tunnels, or within chases, soffits, walls, double partitions, or crawl spaces.
- B. "Exposed" insulation: In full or partial view; not "Concealed" as defined above.
- C. Abbreviations:
 - 1. All-service jacket (ASJ).
 - 2. Chlorofluorocarbons (CFCs).
 - 3. Foil/scrim/kraft (FSK).
 - 4. Hydrochlorofluorocarbons (HCFCs).
 - 5. Polyvinyl chloride (PVC).
 - 6. Polyvinylidene chloride (PVDC).
 - 7. Volatile organic compounds (VOCs).

1.04 QUALITY ASSURANCE

- A. Standards:
 - 1. American Society of Heating, Refrigerating, and Air Conditioning Engineers (ASHRAE) 90.1, Current Enforced Edition, "Energy Standard for Buildings Except Low-Rise Residential Buildings."
 - 2. American Society for Testing and Materials (ASTM).
 - 3. Americans with Disabilities Act (ADA).
 - 4. International Association of Plumbing and Mechanical Officials (IAPMO).
 - 5. International Code Council, Inc. (ICC):
 - a. International Building Code (IBC).
 - b. International Energy Conservation Code (IECC), Current Edition.

6. Underwriters Laboratories, Inc. (UL), including UL 723/ASTM E84, "Test for Surface Burning Characteristics of Building Materials."
 7. Code of Federal Regulations (CFR) Title 40, "Protection of Environment," Part 59, "National Volatile Organic Compound Emission Standards for Consumer and Commercial Products," Subpart D, "National Volatile Organic Compound Emission Standards for Architectural Coatings."
 8. U.S. Environmental Protection Agency (EPA).
 9. U.S. Military Standards (MIL).
 10. Illinois Administrative Code (IAC):
 - 1) IAC Title 35: Environmental Protection.
 - 2) IAC Title 41: Fire Protection.
 - 3) IAC Title 71: Public Buildings, Facilities, and Real Properties.
 - 4) IAC Title 77: Public Health.
 - 5) IAC Title 83: Public Utilities.
- B. Insulation shall be in accordance with IECC and OAC 4101:2.
1. Pipe insulation thickness shall comply with ASHRAE 90.1 "Minimum Pipe Insulation Thickness" Table.
 2. Water heater insulation R-value shall comply with ASHRAE 90.1.
 3. Storage tank insulation shall provide a maximum allowable heat loss of 15 BTUH per sq. ft. of tank surface.
- C. Fire and Smoke Hazard Ratings: Insulation and related components, such as coverings, coatings, tapes, and cloths, shall not exceed 25 Flame Spread, 50 Smoke Developed, and 50 Fuel Contributed on above ground insulation, according to UL 723/ASTM E84.
- D. Installation shall be done by Tradesmen specializing in insulation work in strict accordance with Manufacturer's recommendations.

PART 2 PRODUCTS

2.01 MANUFACTURERS AND MODEL NUMBERS

- A. Manufacturer and Model Number listed below shall be considered to be the Design Base Manufacturer and Model Number.
- B. Other Acceptable Manufacturers will be permitted, unless specifically excluded.

2.02 PIPING INSULATION

A. Insulation for Above-Ground Piping:

1. Indoor: Owens Corning Fiberglas 25 ASJ/SSL/II. 1-piece, heavy-duty, fiberglass insulation with all-purpose vapor barrier jacket of white kraft bonded to aluminum foil and reinforced with fiberglass yarn, longitudinal pressure-sensitive tape for sealing of lap joint, 0 degrees F. to 800 degrees F. temperature rating. Thermal conductivity: 0.22 to 0.28 BTU per hour per inch thickness per cubic foot per degrees F.
 - a. Insulation dimensions:
 - 1) Domestic cold water, make-up water, well water, electric water cooler waste trap, and trap primer drain piping:
 - a) 1 in. and smaller: 1/2 in. thick minimum, 6 in. long.
 - b) 1-1/4 in. to 10 in.: 1 in. thick minimum, 6 in. long.
 - c) 12 in. and larger: 1 in. thick minimum, 9 in. long.
 - d) Option: 3/4 in. thick fire-rated Armaflex may be used in walls (not in chases).
 - 2) Domestic hot water and domestic hot water recirculating piping:
 - a) 2 in. and smaller: 1 in. thick minimum, 6 in. long.
 - b) 2-1/2 in. to 6 in.: 1-1/2 in. thick minimum, 6 in. long.
 - c) 8 in. and larger: 1-1/2 in. thick minimum, 9 in. long.
 - 3) Above-ground drains, floor drains with trap primers, and storm piping: 1 in. thick, minimum.
 - b. Acceptable Manufacturers: Owens Corning Fiberglas or approved equal by Armacell, CertainTeed Corp., Imcoa, Johns-Manville, Knauf, Manson, or Rubatex.
2. In-wall piping: Armacell fire-rated "Armaflex." Closed cell, foamed plastic, tubular, 1/2 in. minimum wall thickness, fire-rated, 220 degrees F. temperature rating. Pipe and joints shall be sealed with Manufacturer's approved adhesive.
 - a. Acceptable Manufacturers: Armacell or approved equal by K-Flex, Kaiflex, Aeroflex, Imcoa, Rubatex, Owens Corning, Johns Manville, Knauf, or Manson.
 - b. Option: Indoor pipe insulation, listed in Paragraph A. 1. above, may be used in walls.
3. Domestic hot water piping insulation shall provide a maximum allowable heat loss of (35) (25) BTUH per sq. ft. of pipe surface area for above-ground piping.

DESIGNER — CHOOSE (B) OR (C):

- B. Insulation for Piping to Lavatories Accessible to Handicapped: Conforms to IBC with testing standard of ASTM E84 Class A material (25 Flame/450 Smoke) criteria; IAPMO PS 94-2001A-listed. Protectors:
1. Antifungal, antimicrobial, molded closed cell PVC.
 2. 1-piece sleek design with smooth, soft texture, and custom-fitted, extended tailpiece.
 3. Weep hole at bottom of J-bend; for condensation drainage and air circulation.
 4. PVC Velcro fasteners for full enclosure. Do NOT provide tie fasteners.
 - a. Option: Secured by heavy-duty interlocking snap fasteners.
 5. Encloses stops and all supply and exposed drain piping, in accordance with ADA 4.19.4 or IBC.
 6. Acceptable Manufacturers: Plumberex Trap Guard, Buckaroos, or Owens Corning.
- C. Wall-Mounted Lavatory Trap / Supply Enclosures: Truebro Lav-Shield Model #2018-AS-L. High-impact, stain-resistant, molded PVC, single-piece construction, china white, with stainless steel vandal-resistant fasteners. Encloses all supply and drain piping, and electronic faucet equipment.
1. Acceptable Manufacturers: Truebro or approved equal by Brocar, McQuire, or Plumberex.
- D. Insulation for Underground Piping:
1. Indoor underground domestic hot water piping: Armacell fire-rated "Armaflex," Rubatex, or Imcoa. Closed cell, foamed plastic, tubular, 1/2 in. minimum wall thickness, fire-rated, 220 degrees F. temperature rating.
 - a. Domestic hot water piping insulation shall provide a maximum allowable heat loss of (35) (25) BTUH per sq. ft. of pipe surface area for underground piping.
 - b. Seal pipe and joints with Manufacturer's approved adhesive.
 - c. Acceptable Manufacturers: Armacell, Rubatex, or Imcoa, or approved equal by CertainTeed Corp., Johns-Manville, Knauf, Manson, or Owens Corning.
 2. Underground sanitary waste piping (kitchen equipment and fixtures to grease interceptor): Armacell Armaflex. Tubular construction, 1 in. minimum wall thickness, fire-rated, closed cell, foamed, sponge, or expanded rubber material, 220 degrees F. temperature rating. Pipes and joints shall be sealed with Manufacturer's approved adhesive.
 - a. Acceptable Manufacturers: Armacell or approved equal by CertainTeed Corp., Imcoa, Johns-Manville, Knauf, Manson, Owens Corning Fiberglas, or Rubatex.

- E. Pipe Fittings Insulation: Owens Corning TIW-1. Wrapped 1 lb. density fiberglass blanket with reinforced foil and kraft vapor barrier facing or molded fiberglass segments with pre-molded, 1-piece PVC covers. Wrapped fiberglass blanket shall be installed to same thickness as pipe insulation.
 - 1. Acceptable Manufacturers: Owens Corning or approved equal by Armacell, CertainTeed Corp., Imcoa, Johns-Manville, Knauf, Manson, or Rubatex.
- F. **Pipe Fittings Insulation Covers**: Pre-molded, 1-piece PVC covers, with maximum 25 Flame Spread and 50 Smoke Developed ratings. Shall be listed and certified for installation in ceiling plenum spaces.
 - 1. Acceptable Manufacturers: Ceelco, Johns-Manville, Knauf, Proto, Speedline, or Zeston.
- G. **PVC Jacket over Piping Insulation**: Ceelco 25/50. UL-approved 1-piece molded PVC jacket covers, solvent weld joints, minimum 20 mil thickness. UV-resistant for outdoor applications.
 - 1. Acceptable Manufacturers: Ceelco or approved equal by Johns-Manville, Knauf, Proto, Speedline, or Zeston.

2.03 EQUIPMENT INSULATION

- A. Rubatex. Flexible sheet, 3/4 in. thick closed cell, foamed plastic, smooth outer skin, 220 degrees F. temperature rating.
 - 1. Acceptable Manufacturers: Rubatex or approved equal by Armacell, CertainTeed Corp., Imcoa, Johns-Manville, Knauf, Manson, or Owens Corning Fiberglas.
- B. Owens Corning Series 703. Flexible 3 lb. density fiberglass sheet with all-purpose jacket of white kraft bonded to aluminum foil and reinforced with fiberglass yarn, 450 degrees F. temperature rating.
 - 1. Acceptable Manufacturers: Owens Corning Fiberglas or approved equal by Armacell, CertainTeed Corp., Imcoa, Johns-Manville, Knauf, Manson, or Rubatex.

2.04 MASTICS, ADHESIVES, SEALANTS, AND FINISHES

- A. Acceptable Manufacturers: Aerobol, Benjamin-Foster, ICC, or Vimasco.
- B. Mastics, adhesives, sealants, and finishes shall be approved for their intended use by the Insulation Manufacturers.
- C. Do NOT use vegetable or wheat paste.

D. Materials Free of HCFCs, CFCs, VOCs, etc.

1. Mastics: Materials shall be compatible with insulation materials, jackets, and substrates. Comply with MIL-C-19565C, Type II. For indoor applications, use mastics with an acceptable VOC content when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
2. Adhesives:
 - a. Calcium silicate: Fibrous, sodium-silicate-based adhesive with a service temperature range of 50 degrees F. to 800 degrees F. For indoor applications, use adhesive with a VOC content of 80 grams/liter or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
 - b. Cellular-glass, phenolic, polyisocyanurate, and polystyrene: Solvent-based resin adhesive, with a service temperature range of -75 degrees F. to +300 degrees F. For indoor applications, use adhesive with a VOC content of 50 grams/liter or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
 - c. Flexible elastomeric and polyolefin: Comply with MIL-A-24179A, Type II, Class I. For indoor applications, use adhesive with a VOC content of 50 grams/liter or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24)
 - d. Mineral-fiber: Comply with MIL-A-3316C, Class 2, Grade A. For indoor applications, use adhesive with a VOC content of 80 grams/liter or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
 - e. ASJ adhesive, and FSK and PVDC jacket adhesive: Comply with MIL-A-3316C, Class 2, Grade A for bonding insulation jacket lap seams and joints. For indoor applications, use adhesive with a VOC content of 50 grams/liter or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
 - f. PVC jacket adhesive: Compatible with PVC jacket. For indoor applications, use adhesive with a VOC content of 50 grams/liter or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
3. Lagging Adhesives: Comply with MIL-A-3316C Class I, Grade A. Shall be compatible with insulation materials, jackets, and substrates. For indoor applications, use lagging adhesives with an acceptable VOC content when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
4. Sealants:
 - a. Joint sealants for polystyrene products: For indoor applications, use sealants with a VOC content of 250 grams/liter or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

- b. FSK and metal jacket flashing sealants: For indoor applications, use sealants with a VOC content of 250 grams/liter or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
- c. ASJ flashing sealants, and vinyl, PVDC, and PVC jacket flashing sealants: For indoor applications, use sealants with a VOC content of 250 grams/liter or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

PART 3 EXECUTION

3.01 DELIVERY, STORAGE, AND HANDLING

- A. Protection: Leave insulation boxed and stored until time for use. Elevate and cover material to avoid moisture contamination and physical abuse.

3.02 INSTALLATION — PIPING INSULATION

- A. Coordinate installation of insulation with pressure testing of piping. Piping shall be pressure-tested and accepted before piping insulation is installed.
- B. Complete installation of insulation and sealing shall be in strict accordance with the Manufacturer's recommendations.
- C. Do not use damaged or water-soaked insulation.
- D. Maintain continuity of all insulation. No voids in insulation or vapor barrier will be permitted.
- E. Insulation shall be continuous through sleeves and hangers and through walls where no sleeves are required.
- F. Make application on clean, dry pipes and tanks, with all joints butted firmly together.
- G. Leave no "raw" ends on insulation. Bevel insulation terminations, seal with insulating cement or mastic, and cover ends same as pipe insulation covering.
 - 1. Option: In lieu of beveling, use PVC caps over straight-cut ends that have been vapor-sealed.
- H. Insulation of Pipe Hangers and Supports:
 - 1. Insulate any pipe hanger, hanger rod, or pipe support that has direct contact with any hot or cold piping or any item that is subject to sweating.
 - 2. Set all hangers perpendicular to the axis of the piping before any insulation is applied. Do not apply insulation until insulation saddles are in place.
 - 3. Install insulating saddles of non-compressible insulation with a continuous vapor barrier on pipes 2 in. and larger.

4. Install pipe insulating support at all hangers (regardless of size), unless insulating-type hangers are used. Verify method to be used.
 5. Refer to Section 22 05 29 for additional information on pipe hangers, insulation shields, insulating saddles, and pipe insulating support.
- I. Where piping is installed above the ceiling, and below ceiling insulation, ensure that ceiling insulation has no insulation voids above piping.
 - J. Install plenum-rated PVC fitting covers in all air-handling ceiling plenum spaces, and on all piping fittings.
 - K. Do not install foam insulation in air-handling plenum spaces.
 - L. Insulate piping where concealed in walls or chases.
 - M. Insulate piping located in walls with 1/2 in. thick fire-rated foam insulation or indoor pipe insulation.
 - N. All exposed insulated piping and fittings, from floor up to 7 ft.-6 in. above floor, shall have a PVC cover applied over pipe insulation and solvent welded together. Material shall be same as PVC fitting covers.
 - O. Exposed insulation shall be finished smooth, sized if required, and left ready for painting.
 - P. Attach labels to all piping insulation every 50 ft. stating "NON-ASBESTOS CONTAINING."
 - Q. Contractor shall replace all damaged insulation at its own expense.
 - R. Water pipes of differing temperatures shall be insulated individually and not bundled together.

3.03 INSTALLATION — PIPE FITTINGS INSULATION

- A. Insulate valves and flanges for all services, unless otherwise noted. Insulate all couplings. During bidding, the Contractor shall verify the extent of these connections and shall prepare its bid accordingly.
- B. Insulation of Fittings:
 1. Insulate fittings with factory-formed mitered sections of pipe insulation or 1 lb. density fiberglass blanket insulation wrapped under compression to full thickness of the pipe insulation.
 2. Secure mitered sections and blanket insulation with double wrapping of tape over the entire fitting.
 3. Apply vapor barrier mastic on concealed fittings to maintain vapor barrier.

4. Cover with molded PVC cover.
- C. Insulation of Unions and Flanges:
1. Do not cover unions and flanges initially.
 2. Bevel insulation and apply cement to ends.
 3. Then add insulation over these items, so that it can be removed without disturbing adjacent pipe insulation.
 4. Cover with molded PVC cover.
- D. Insulation of Fittings, Unions, Flanges, and Valve Bonnets:
1. Wrap with oversized pipe insulation to provide the same insulation thickness as pipe insulation.
 2. Use oversized insulation sections to form a collar between the (2) insulation sections with blanket insulation used to fill the gaps.
 3. Cover with PVC jacket.
- E. Where insulation terminates, taper insulation down to the pipe surface complete with jacket.
- F. **PVC Jacket:** In addition to insulation jacketing and fitting covers specified above, install PVC jacket over all exposed insulation below 7 ft.-6 in. above floor. Jacket shall have overlapping joints and shall be sealed with suitable adhesive. Butt joints shall be sealed with matching 3 in. wide ASJ pressure-sensitive tape.
- 3.04 EQUIPMENT INSULATION THICKNESSES
- A. Water Storage Tank: Install 3/4 in. thickness. Seal and finish.

END OF SECTION 22 07 00

SECTION 22 08 00 - COMMISSIONING OF PLUMBING

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 01 91 00, "Commissioning."
- B. Section 01 91 51, "Mechanical/Electrical Systems Manual."

1.02 DESCRIPTION

- A. This section describes the process for commissioning of the various building systems and outlines the duties of parties involved.
- B. The commissioning process may be applied to all equipment, components, and systems to be commissioned as listed in Part 3 of this Section, including specified interfaces to and from equipment and systems provided under the other Divisions of this Specification.

1.03 COMMISSIONING AGENT

- A. The Commissioning Authority (CxA) is independently retained by the Owner and shall have the authority to grant final acceptance of each system commissioned.

1.04 THE COMMISSIONING TEAM

- A. In addition to the CxA, the Commissioning Team shall consist of:
 - 1. Owner.
 - 2. Plumbing Contractor.
 - 3. HVAC Contractor.
 - 4. Electrical Contractor.
 - 5. Test and Balance Subcontractor.
 - 6. Temperature Controls Contractor.
 - 7. Mechanical Design Engineer.
- B. The appropriate Contractors shall cooperate and coordinate their work during the commissioning process. They shall perform all work required for the installation of the components and systems, and for operation during the commissioning process. They shall furnish all necessary resources to accomplish the installation and the commissioning.

1.05 COMMISSIONING DOCUMENTATION

- A. Commissioning Plan: CxA shall create a plan during the design phase of the project to identify scope of commissioning for the project and a preliminary schedule of activities for use during the project by members of the Commissioning Team.
- B. Meeting Minutes: CxA shall generate and issue to members of the Commissioning Team after each commissioning progress meeting.
- C. Design Phase: CxA shall review plumbing drawings at Design Development and Construction Document Phases, shall recommend improvements to correct deficiencies in design, and shall specifically focus on plumbing controls.
- D. Construction Checklists: CxA shall provide a list of items to inspect for each piece of equipment that is to be commissioned. Contractor shall complete checklist prior to equipment start-up and functional testing.
- E. Start-up Plan: Contractor shall submit to identify methods to be used for equipment pre-checks, start-up procedures, start-up schedule, and sample reports to document completion. Contractor shall document all equipment deficiencies and corrections made in the field as part of start-up report.
- F. Test Reports: Contractor shall generate reports to document system/equipment testing included in the Contract that is not dictated by the CxA (i.e., air balance report, etc.).
Review As-Built Drawings.
- G. Commissioning Issues Log: CxA shall update and issue log to identify system deficiencies found through the commissioning process.
- H. Functional Tests: CxA shall create, and the Contractors shall review, for use during functional testing of each system.
- I. Training Plan: Contractor shall submit plan identifying personnel providing training and their qualifications, training supplemental materials, and training session agendas, for review by the CxA.
- J. Final Commissioning Report: CxA shall provide report summarizing results, status of remaining operating deficiencies, and future actions and nonactions.
- K. Commissioning Complete: Formal Commissioning Project Complete document with signoff to add finality to project.

PART 2 PRODUCTS

2.01 TESTING EQUIPMENT AND INSTRUMENTATION

- A. Contractor shall provide all industry standard test equipment required for performing the specified tests. Any proprietary vendor-specific test equipment shall be provided by that Vendor or Manufacturer.
- B. Contractor's instrumentation shall be of sufficient quality and accuracy to test and measure system performance within the tolerances required. Instrumentation shall be calibrated at the Manufacturer's recommendation intervals with calibration tags permanently affixed to the instrument. Instrumentation shall be maintained in good repair and operating condition throughout the duration of use on this project and shall be immediately re-calibrated or repaired if dropped or damaged in any way during use on the project.

PART 3 EXECUTION

3.01 COMMISSIONING RESPONSIBILITIES

- A. Within (30) days of the award of Contract, the Contractor shall submit the names of all the Trades people who will be part of the commissioning process. Contractor, and all its Sub-Trades and Suppliers, shall cooperate with the CxA in the commissioning process. Contractor's responsibilities shall include:
 - 1. Review the project schedule and identify the milestone commissioning activities at the initial commissioning scope meeting, to be held within (90) days of contract award. Milestone commissioning activities shall include, but are not limited to:
 - a. Equipment start-ups.
 - b. System start-ups.
 - c. Testing activities performed by the Contractor.
 - d. Readiness of each major system.
 - e. System functional testing as part of the commissioning process.
 - 2. Attend commissioning meetings, and complete action items arising from them, to allow the commissioning process to proceed on schedule.
 - 3. Complete and provide all system readiness documentation required by the commissioning process.
 - 4. Provide a start-up plan for each piece of equipment and system that is identified to be commissioned. Notify the CxA a minimum of (7) calendar days before start-up of major equipment and systems.

5. Provide personnel and testing instrumentation required to operate and test equipment and systems as part of functional performance testing. Testing may include calibration verification of system devices. Testing shall take place under the direct supervision of the CxA. Contractor shall reimburse the Owner and CxA for costs associated with retesting of systems that fail initial testing.
6. Contractor shall respond in writing to the CxA and Owner at least as often as commissioning meetings are being scheduled concerning the status of each apparent outstanding discrepancy identified during commissioning. Discussion shall cover explanations of any disagreements and proposals for their resolution. Contractor shall not consider any required re-testing a justified reason for a claim of delay or for a time extension.

3.02 COMMISSIONING PROCESS

- A. The commissioning process depends upon proper coordination between all Commissioning Team members, strict adherence to schedule, and completion of all required documentation. Responsibilities of each team member are described in this and other sections of the contract.
- B. Construction Phase:
 1. Commissioning Meetings: An initial Commissioning Scope Review / kick-off meeting will be held with all members of the Commissioning Team at the beginning of the project (generally within (90) days of award of contract). CxA will schedule periodic Commissioning Team progress meetings to review progress of commissioning work and coordinate activities. (Commissioning progress meetings will be scheduled to coincide immediately before or after the regular weekly construction progress meeting.) Contractor shall anticipate at least (1) commissioning meeting for every month of the construction phase (additional required meetings will not be considered additional work to this contract).
 2. Commissioning Schedule: Contractor shall assist the CxA in the development of a written schedule that integrates the commissioning activities into the construction schedule specified in Division 01, "General Requirements." Update the commissioning schedule to reflect changes in the work. The commissioning schedule shall include at least the following dates:
 - a. Submission of Operation and Maintenance information for systems to be commissioned.
 - b. Schedule for systems, subsystems, and equipment start-up, including services of Manufacturers' authorized service representatives, and performance of pretest checks.
 - c. Schedule for functional performance testing, including seasonal testing.
 - d. Schedule for the Owner's operating personnel training.

3. Prerequisite to Start-Up: Test and Balance report shall be signed off by the CxA prior to final commissioning of the respective systems. CxA will verify and correct the Testing and Balancing work in progress. It is expected that all relevant and known punch-list items are addressed prior to that agenda commissioning activity.
4. Equipment and System Start-Up: Before any equipment or system is started, the Start-up Plan, including all pre-start check documentation provided by the Equipment Manufacturer and construction checklists provided by the CxA, shall be submitted. Contractor shall notify the CxA of the scheduled start-up a minimum of (7) days prior to the start-up, and give the CxA the opportunity to witness part or all of the start-up work and conduct its own pre-check inspection. After startup is completed, the Contractor shall submit a completed start-up report for each piece of equipment.
5. Initial Operation: Once the Contractor completes the start-up, testing, balancing, and calibration of all components and systems, it shall operate all systems through the specified modes of operation, and test the system responses to specified abnormal or emergency conditions. Contractor shall complete the system and perform this functional and performance pre-check before the Commissioning Team performs the functional acceptance testing.
 - a. Functional acceptance testing included in the commissioning process is verification that the Contractor has provided a complete and functioning system per the Contract requirements. It is not an opportunity for the Contractor to determine deficiencies and work remaining.
 - b. Contractor shall ensure that a qualified technician(s) is available and present during the agreed-upon schedules and of sufficient duration to complete the necessary tasks, tests, adjustments, and problem resolution.
 - c. Functional testing of the system shall be terminated and re-scheduled if it is deemed by the CxA that the system is not ready for functional testing and that the Contractor has not fully completed the required initial operation pre-check. Contractor shall reimburse the Owner, CxA, Construction Manager, and A/E for costs associated with the additional time and resources required to reschedule and repeat testing due to a lack of system readiness by the Contractor.

C. Acceptance Phase:

1. Functional Acceptance Testing: Systems identified for commissioning shall be operated through the entire specified sequence of operations, as directed by the CxA for verifying acceptable operation. Contractor shall provide all testing instrumentation required and operate the system during the tests, and by this, the Contractor shall ensure that the systems are not operated beyond their limits as installed.
2. System Deficiencies: All system operational deficiencies identified during the functional acceptance testing will be recorded by the CxA for correction by the Contractor. Work to correct the deficiencies will be under the direction of the Owner,

the Owner's Representative, or the Construction Manager. Final acceptance of the system shall not be granted until all deficiencies identified are corrected.

3.03 SYSTEMS/EQUIPMENT TO BE COMMISSIONED

- A. The following plumbing systems shall be commissioned:
 - 1. Domestic water boilers.
 - 2. Hot water circulating pumps.
 - 3. Thermostatic mixing valves.
- B. The equipment listed below shall be provided a formal start-up. (Refer to other Sections of the Project Specification for additional start-up requirements.)
 - 1. Domestic water boilers.

END OF SECTION 22 08 00

SECTION 22 10 00 - PLUMBING PIPING

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 22 05 23, "General-Duty Valves for Plumbing Piping."
- B. Section 22 05 29, "Hangers and Supports for Plumbing Piping and Equipment."
- C. Section 22 05 53, "Identification for Plumbing Piping and Equipment."
- D. Section 22 07 00, "Plumbing Insulation."
- E. Section 22 11 16, "Domestic Water Piping."
- F. Section 22 11 19, "Domestic Water Piping Specialties."
- G. Section 22 11 23.23, "Domestic-Water In-Line Pumps."**
- H. Section 22 13 00, "Facility Sanitary Sewerage."
- I. Section 22 14 00, "Facility Storm Drainage."

1.02 DESCRIPTION

- A. Provide piping and fittings as shown or required to completely install all plumbing systems on this Project, including the following:
 - 1. Domestic water piping system. Refer also to Section 22 11 16. For specialties, refer to Section 22 11 19
 - 2. Sanitary sewer piping system. Refer also to Section 22 13 00.
 - 3. Storm sewer piping system. Refer also to Section 22 14 00.
- B. Valves: Refer to Section 22 05 23 and to the Drawings.
- C. Pumps: Refer to Section 22 11 23.23.

1.03 QUALITY ASSURANCE

- A. Standards:
 - 1. American National Standards Institute (ANSI).

2. American Society for Testing and Materials (ASTM):
 - a. ASTM B88, "Standard Specification for Seamless Copper Water Tube."
 - b. ASTM E84.
 - c. ASTM B32/NSF 61.
3. American Society of Mechanical Engineers (ASME):
 - a. ASME/ANSI B31.1, "Code for Power Piping."
 - b. ASME Boiler and Pressure Vessel Code, Section IX, "Welding and Brazing Qualifications."
4. American Welding Society (AWS).
5. National Sanitation Foundation (NSF).
6. National Fire Protection Association (NFPA): 2023 NFPA 70, "National Electrical Code® (NEC)," Articles 110 and 408.
7. Underwriters Laboratories, Inc. (UL).
8. Illinois Administrative Code (IAC):
 - 1) IAC Title 35: Environmental Protection.
 - 2) IAC Title 41: Fire Protection.
 - 3) IAC Title 71: Public Buildings, Facilities, and Real Properties.
 - 4) IAC Title 77: Public Health.
 - 5) IAC Title 83: Public Utilities.

PART 2 PRODUCTS

2.01 PIPING AND FITTINGS

- A. Cast Iron Threaded or Flanged Fittings: 125 lb. SWP or 250 lb. SWP (ANSI B16.4 and ANSI B16.1).
- B. Malleable Iron Threaded Fittings: 150 lb. SWP or 300 lb. SWP (ANSI B16.3).
- C. Ductile iron grooved fittings (ASTM A536, Grade 65-45-12).
- D. Factory-made wrought steel, butt-weld fittings (ANSI B16.9).
- E. Butt-welding ends for pipe, valves, flanges, and fittings (ANSI B16.25).

- F. Steel pipe flanges and flanged fittings (ANSI B16.5).
- G. Forged steel fittings, socket-welded and threaded (ANSI B16.11).
- H. Wrought copper and bronze solder-joint pressure fittings (ANSI B16.22).
 - 1. Option: "T-drill" mechanically formed tee connections, brazed joints.
- I. Nipples: Same weight and material as connected pipe, except shoulder nipples shall be extra-heavy. Do not use short nipples.

2.02 JOINTS, UNIONS, AND CONNECTIONS

- A. Threaded Piping Joints:
 - 1. National Pipe Thread (NPT) taper, general purpose (ASME B1.20.1).
 - 2. TFE tape, applied to male thread only.
 - 3. Option: Thread compounds by Permatex.
- B. Filler Materials:
 - 1. Solder: ASTM B32 alloy Sn94 and Sn95. 95/5 tin-antimony, Grade 95.
 - a. Acceptable Manufacturers: Engelhard, J. W. Harris, or Oatey.
 - 2. Brazing: AWS A5.8. Classification BCUP-3 or BCUP-4.
 - 3. Welding: AWS D10.12 for sizes 8 in. or less.
- C. Copper Pipe Unions: Cast bronze, Class 150, ground joint, solder end. Do NOT use wrought copper unions.
 - 1. Acceptable Manufacturers: Chase, Crane, Dart, Epco, Flagg, Mueller, Nibco, or Rockwell.
- D. Steel Pipe Unions: Black malleable iron, Class 250, bronze ground ball joint.
 - 1. Acceptable Manufacturers: Chase, Crane, Epco, Fairbanks, Grinnell, Mueller, or Stockham.
- E. Dielectric Unions: Metal union with dissimilar materials electrically separated by gasket. Union materials, end connections, and pressure rating compatible with piping materials, system fluid, system pressures, and system temperatures.
 - 1. Acceptable Manufacturers: Capitol, Dart, Epco, Vogt, or Watts.
- F. Flanges: Cast iron or cast bronze, 125 lb. or higher. ASME B16.5 flat-face or raised-face metal flange plates with matching bolt holes, gasket, bolts, nuts, and washers. Flange

materials, end connections, and pressure rating compatible with piping materials, system fluid type, system pressures, and system temperatures.

- G. Dielectric Flanges: Flange with electrically isolating gasket, bolt sleeves, and bolt washers. Flange materials, end connections, and pressure rating compatible with piping materials, system fluid type, system pressures, and system temperatures.

1. Acceptable Manufacturers: Capitol, Crane, Dart, Epco, or Vogt.

- H. Gaskets:

1. Nonmetallic. ASME B16.21, 1/8 in. minimum thickness, fibrous type with double coat of graphite, asbestos-free.
2. High-temperature piping systems: ASME B16.20, Type 304 stainless steel with carbon steel guide.

- I. Bolts and Other Hardware:

1. Bolts for steel, cast iron, brass, and bronze, for 250 PSI working pressure and for 450 degrees F. or below: ASME B18.2.1 carbon steel bolts, with American Standard regular, square heads. ASME B18.2.2 American Standard heavy-hexagon, semi-finished nuts. ASME B16.22.1 plain washers and ASME B18.21.1 lock washers.
 - a. Option: ASME B18.2.1 tee-head, high-tensile steel bolts and nuts may be used in mechanical joint pipelines.
2. Where equipment is installed outside, all hardware and bolts shall be stainless steel.

2.03 ROLL GROOVED COUPLINGS AND FITTINGS:

- A. Acceptable Manufacturers: Victaulic Co., Anvil International "Gruvlok", or Tyco Grinnell.

- B. Rigid couplings with 300 PSI pressure rating and 180 degrees F. temperature rating:

1. Victaulic Style 607 (slant bolt pad) with bolt pads metal-to-metal.
2. Tyco-Grinnell Figure 672 with even-gap bolt pads.
3. Anvil-Gruvlok Figure 6400 with even-gap bolt pads.
4. Option: Grooved couplings may be installed on 2 in. to 8 in. roll grooved standard weight pipe.

- C. Install copper tubing in accordance with Coupling Manufacturer's instructions.

- D. Pre-lubricated, EPDM gasket, Grade EHP material, rated for maximum 250 degrees F. Gaskets used on potable water systems shall be UL-classified in accordance with ANSI/NSF 61.
 - 1. Gruvlok gaskets NSF 61 approved ONLY for cold water, up to 86 degrees F., and shall be lubricated, inside and out, with Gruvlok Xtreme Lubricant in any copper application.
- E. DO NOT USE ANY OTHER TYPE OF ROLL GROOVED COUPLINGS.
- F. Option for branch: Installing Contractor may use a ductile iron bolted branch outlet with locating collar, Grade E pressure-responsive gasket seal, and carbon steel bolts and nuts.
- G. Grooved valves and accessories are acceptable.

2.04 PIPE BOOTS

- A. Acceptable Manufacturers: PortalsPlus or approved equal by Pate.
- B. PortalsPlus "Quadraseal" or "Pipe Boot." Molded 1/16 in. thick minimum EPDM pipe boot, ozone-resistant, ultraviolet-resistant, with 3 in. base flange, conically shaped, watertight, fully compatible with roof construction.
- C. Shingled Roofs: Oatey, Dektite, or Gibraltar Building Products.

2.05 ROOFING MEMBRANE SEALANT

- A. Pourable Sealant: Carlisle.

PART 3 EXECUTION

3.01 INSTALLATION

- A. These installation instructions apply to all piping systems, including domestic water, sanitary sewer, storm sewer, and compressed air, except where otherwise noted.
- B. Coordinate pipe elevations and final locations with the work of all other Contractors before installation.
- C. Provide information on chases, sleeves, and openings required for Plumbing Work to other Contractors.
- D. All piping materials and installation shall comply with State and Local Building Code requirements.

- E. Provide all fittings and accessories required to complete the installation.
- F. Personnel Passages and Exit Paths: Do not install piping where it will impede personnel using exit paths.
- G. Structural Members: Do not install piping where it will interfere with structural members. Do not cut structural members for piping installation without written approval from the Structural Engineer.
- H. Equipment, Such as Cranes: Do not install piping where it will interfere with equipment operation.
- I. Roofs:
 - 1. Coordinate penetration sizes and locations, sleeves, and weatherproofing.
- J. Protect non-ferrous metallic piping from contact with steel framing members, concrete, masonry, and corrosive soil via pipe sleeves or plastic sheathing.
 - 1. Furnish pipe boots and flashing. Lead Contractor shall provide piping penetrations through roof and install pipe boots and flashing. Lead Contractor shall provide copper pitch box filled with roofing membrane sealant and secured to roof deck. Install piping. Lead Contractor shall "roof in," making watertight after piping is in place.
 - 2. Provide weather-tight curb cap if piping cannot be put in place immediately.
 - 3. Piping subject to freezing may be installed in unheated ceiling spaces below roofs only where specifically shown immediately above the ceiling of a heated space and with building insulation closely packed above the piping to protect it from freezing. Ensure that vulnerable piping is protected against freezing.
- K. Walls:
 - 1. Coordinate penetration locations, sleeves, and weatherproofing (where applicable).
 - 2. For masonry unit walls, locate piping penetrations at masonry joints where feasible.
 - 3. Do not install piping containing fluids subject to freezing in exterior walls, regardless of wall insulation location. Where unavoidable, obtain prior approval of the A/E, and install piping so that the wall insulation is between the pipe and the outside wall.
- L. Doors: Do not install piping where it will restrict access to or operation of doors.
- M. Skylights: Do not install piping above or below skylights.
- N. Windows: Do not install piping in front of windows or where it will restrict access to or operation of windows.

- O. Floors:
 - 1. Coordinate penetration locations, sleeves, and weatherproofing (where applicable).
 - 2. 3 in. high concrete curbs shall be provided around pipe sleeves where pipes penetrate floors in kitchens, utility rooms, equipment rooms, shafts, or other areas potentially subject to flooding. Refer to Section 22 05 29 regarding concrete work.
- P. Access Doors or Panels: Do not install piping where it will restrict access to or operation of doors or panels.
- Q. All piping installed exposed in mechanical closets (at water heaters, for example) shall comply with ASTM E84 flame spread/smoke developed requirements for plenum-rated piping.
- R. Shafts: Do not run piping through a shaft **unless it uses the shaft for vertical offset**.
- S. Elevator Equipment Rooms: Do not extend piping through elevator equipment rooms.
- T. Pipe Chases or Pipe Spaces. Where feasible, install piping near walls and either near floor or at least 6 ft. above floor to maximize access.
- U. Piping, Conduit, Cable Trays, and Wiring: Coordinate all runs. Group runs to minimize space usage and maximize access.
- V. Plumbing Fixtures and Equipment: Do not install piping where it will restrict access to or operation of fixtures or equipment.
- W. HVAC Ductwork: Do not penetrate ductwork with piping without written approval of A/E (unless the Drawings specifically indicate the penetration).
- X. HVAC Equipment: Do not install piping where it will restrict access to or operation of equipment.
- Y. Electrical Equipment:
 - 1. Do not install piping in an electrical equipment room unless it specifically serves that room.
 - 2. Do not install piping above, within 36 in. of front of, or in the immediate vicinity of, electrical panels, electrical switchgear, or other electrical equipment, as described in 2017 NEC Articles 110 and 408.
 - a. If compliance with these requirements is not possible, obtain approval in writing from the AHJ and A/E to provide a sheet metal drip pan under the piping to prevent water and condensation from running or dripping onto electrical items. Provide drip pan at no additional cost to the Owner.

- Z. Luminaires: Do not install piping across, in front of, or directly below luminaires.
- AA. Interferences: If any system piping is placed where it causes or contributes to interference, relocate it. Obtain instructions from the A/E. The A/E's decision shall be final in regard to the arrangement of equipment, ductwork, piping, etc. Contractor causing the conflict shall assume all costs and responsibilities resulting from improper coordination of work for the corrective work of all Contractors.

3.02 PIPING LAYOUT

- A. The Drawings are schematic and are not intended to precisely show piping locations.
- B. Arrange all piping in accordance with the best standards of the Trade.
- C. Install piping in a neat and workmanlike manner. Install horizontal piping parallel or perpendicular to building lines, unless indicated otherwise. Install vertical piping plumb, unless indicated otherwise.
- D. Securely support piping from structure with approved hangers, rods, anchors, guides, brackets, and accessories.
 - 1. Support piping in accordance with the Manufacturer's recommendations.
 - 2. Where multiple piping services run parallel to each other in the same area, trapeze hangers may be used for racking the pipes.
 - 3. Independently support piping from building structural members. Do not support piping from ceilings, ducts, other pipes, cable trays, conduit, wiring, etc.
 - 4. Refer to Section 22 05 29.
- E. Install piping systems with adequate provisions for expansion, contraction, vibration, and prevention of stresses on valves and equipment.
- F. Adjust locations and provide additional offsets, drains, and air vents needed to properly install piping system without interferences.
- G. Exercise due and particular caution to determine that all parts of the work are quickly and easily accessible.
- H. Where special conditions are encountered in field, the A/E shall decide arrangement and alignment of piping.
- I. Changes in Piping Direction:
 - 1. Make changes in piping direction with fittings. Do not bend or spring piping.
 - 2. Offset or reroute piping to resolve interferences that develop in the field.

3. Provide offsets for the proper operation of the system.
 4. Offset piping around columns, beams, and other obstructions.
 5. Offset vents immediately above connection to waste piping.
- J. Conceal pipes, except in equipment rooms, in rooms without ceilings, or where specifically indicated otherwise.
- K. Install unions and valves in accessible locations. Use removable ceiling pads or provide access panels in ceilings, walls, and floors for service access. Coordinate installation of access panels.
- L. Keep piping level, except where a slope is required.
- M. Slope piping to drainage points with minimum slope of 1 in. in 40 ft. (0.2%).
- N. Sanitary and vent lines shall be continuously sloped.
- O. Avoid trapping piping.
1. Provide a 3/4 in. ball drain valve with capped hose end connection at each low point in the piping.
 2. Provide a manual air vent with 1/4 in. ball valve at each high point in pressurized piping.
- P. Do not trap drain piping except at equipment connections.
- Q. Do NOT trap sanitary and vent lines.
- R. Make reductions in piping lines with reducing couplings or weld fitting reducers.
- S. For horizontal piping, use eccentric reducer to keep top of pipe level.
- T. For vertical piping, use concentric reducer to keep pipe plumb.
- U. For horizontal piping, connect branches at or above centerlines of mains.
- V. DO NOT USE BULLHEAD CONNECTIONS IN ANY PIPING SERVICE.
- W. If equipment connection size is smaller than branch piping to connection, make reduction at equipment connection point. Do not reduce size of branch piping.
- X. Valves:
1. Install isolation valves for any piped equipment item not equipped with factory-provided isolation valves.

2. Install isolation valves where lines serving more than one equipment item branch from main lines.

3. Refer to Section 22 05 23 for valve requirements.

Y. Insulation:

1. Install pipes, valves, fittings, etc., requiring insulation with a minimum of 1/2 in. clearance between the finished covering and other work and between the finished coverings of parallel, adjacent pipes.

2. Install pipe insulation continuously through all openings in fire-treated lumber through which copper pipes will pass, to protect the copper pipe from deterioration from fire treatment salts.

3. Refer to Section 22 07 00 for additional insulation requirements.

Z. Provide piping identification. Refer to Section 22 05 53.

3.03 PIPING AND FITTINGS FABRICATION

A. Cut pipe accurately, and work into place without forcing or springing.

B. Clean piping components of loose material.

C. Protect all piping, fittings, and valves from excessive tool marks.

D. Joints, Unions, and Connections:

1. Ream pipe or tubing ends and remove burrs to full inside diameter after cutting.

2. Threaded connections:

a. Threads shall conform to ASME B1.20.1.

b. Provide union adjacent to each threaded valve (unless union is integral to valve body).

3. Soldered or brazed connections:

a. Soldering and brazing work shall comply with ASME/ANSI B31.1.

b. Mating surfaces of joint shall be smooth and clean.

c. 100% lead-free solder shall be used on water piping, joints, and fittings.

d. Open valves fully before soldering. Protect or remove parts of valves and specialties subject to deterioration by heat.

4. Welded connections:
 - a. An AWS-certified welder shall perform all welding and brazing in accordance with ASME/ANSI B31.1.
 - b. A qualified welder, certified in accordance with the State Building Code, shall install welded pressure piping.
 - c. Do not torch-cut pipe.
 - d. Use backing rings on all welded joints.
 - e. Butt-welded joints shall have substantially full penetration and recommended bead reinforcement.
 - f. Slip-on, socket, and fillet welds shall have geometry in accordance with ASME/ANSI B31.1.
 - g. Welds shall be stamped at each joint or fitting.
 - h. Remove slag, weld scale, and other welding residue as work proceeds and at completion.
5. Dissimilar material connections:
 - a. Install dielectric connectors at all connections of:
 - 1) Copper and ferrous metal piping.
 - 2) Copper piping and equipment having ferrous metal connection.
 - b. Isolate direct contact between pipe, fittings, valves, flanges, and hangers of dissimilar metal by use of dielectric unions, shims, gaskets, coatings, or other approved methods.
 - c. Use dielectric gaskets where joining dissimilar piping material.
6. Flanged connections:
 - a. Flanged joints shall be faced, square, and true.
 - b. Provide gaskets suitable for the chemical composition, operating temperature, and operating pressure of the piped fluid or gaseous medium.
7. Connections to equipment or devices:
 - a. Make connections to equipment as recommended by the Manufacturer or as detailed on the Drawings.
 - b. Provide union or flange near each connection arranged to permit removal of parts and equipment without disconnecting piping beyond the union or flange. Provide isolation valve on piping system side of each equipment connection union or flange to permit equipment removal without shutting down piping system.

- c. Provide union **or flange** in final piping connection to each device or item of equipment, and elsewhere, to makeup or disconnect piping.
- d. Do not make welded connections to equipment **or valves**.

E. Grooved Connections:

1. A factory-trained representative of the Coupling Manufacturer shall provide on-site training for the Contractor's field personnel in the use of grooving tools, application of groove, and installation of grooved joint products.
 - a. The Manufacturer's Representative shall periodically visit the job site and review installation.
 - b. Provide documentation of the site visits by the Manufacturer's Representative in the Operating and Maintenance Manuals.
 - c. Include verification of proper gasket lubricants, uniform gap spacing of couplings, proper torque requirements, and notation of any installation deficiencies.
2. Grooved Piping Installing Contractor shall have prior experience installing at least (5) grooved mechanical piping systems.
3. Install grooved joint couplings, gaskets, and fittings in accordance with the Coupling Manufacturer's written installation instructions.
4. Each grooved pipe end shall be clean and free from indentations, projections, and roll marks in the area from pipe end to groove and on both sides of groove.
5. Provide elastomer gasket suitable for the intended service. The Coupling Manufacturer shall produce gasket.
6. Verify and use the Coupling Manufacturer's recommended gasket lubricants, including any specific lubricants to enhance the temperature rating of the coupling. Lubricate gasket both inside and out where required by the Coupling Manufacturer. Submit lubricant data prior to installation and provide proof that the lubricant is being used on the job site.
7. Alternate tightening bolts **on opposite sides of coupling** until properly torqued in accordance with the Coupling Manufacturer's specifications.
8. If the Coupling Manufacturer specifies torque requirements for its fittings, test random couplings, at the A/E's discretion, to verify that proper torque requirements were achieved during installation.
9. Verify that there are no bolt pad gaps on metal-to-metal type couplings, or that equal gaps are maintained on both sides of the coupling, in accordance with the Coupling Manufacturer's written installation instructions.
10. Remove and replace any joints deemed improperly installed.

F. Escutcheons:

1. Where exposed pipes pass through walls, floors, or ceilings of finished rooms, provide chrome-plated escutcheons.
 - a. Option: Prime-coated steel or rough brass escutcheons may be provided in unfinished rooms.
2. Protect escutcheons from tool marks.

G. At the end of each day's work, cap or plug open ends in installed piping for protection.

3.04 INSTALLATION — PIPE BOOTS

- A. Coordinate all locations and sizes of roof penetrations for the Plumbing Work.
- B. **Furnish** pipe boots **and flashing** for all piping penetrations through roof.
- C. **Lead Contractor shall install** pipe boots and flashing.
- D. **Lead Contractor shall make** watertight.

END OF SECTION 22 10 00

SECTION 22 11 19 - DOMESTIC WATER PIPING SPECIALTIES

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 22 05 00.78, "Closeout Submittals for Plumbing."
- B. Section 22 05 29, "Hangers and Supports for Plumbing Piping and Equipment."
- C. Section 22 10 00, "Plumbing Piping."
- D. Section 22 11 23.13, "Domestic-Water Packaged Booster Pumps."
- E. Section 22 11 23.23, "Domestic-Water In-Line Pumps."

1.02 DESCRIPTION

- A. Provide plumbing piping specialties or accessories as shown or required to complete the installation of all plumbing systems installed on this Project.
 - 1. Backflow preventers (BFPs).
 - 2. Connection box for ice maker.
 - 3. Connection box for washer/dryer.
 - 4. Connection box for whirlpool.
 - 5. Connectors, piping, flexible.
 - 6. Connectors, pumps.
 - 7. Connectors, pumps, flexible.
 - 8. Drain pan for water heater.
 - 9. Hose bibbs, interior.
 - 10. Hose stations.
 - 11. Hydrants, ground.
 - 12. Hydrants, post.
 - 13. Hydrants, roof.
 - 14. Hydrants, wall.
 - 15. Quick couplers.
 - 16. Shock absorbers.

17. Shut-off valve for washer.
18. Strainers.
19. Tank, expansion, for domestic water system.
20. Tank, expansion, for water heater.
21. Tank, well water hydropneumatics.
22. Thermostatic mixing valves (TMVs).
23. Trap primers.
24. Vacuum breakers.

B. All products of the same type shall be by the same Manufacturer.

1.03 QUALITY ASSURANCE

A. Standards:

1. American National Standards Institute (ANSI).
2. American Society of Heating, Refrigerating and Air Conditioning Engineers (ASHRAE) 188.
3. American Society of Mechanical Engineers (ASME).
4. American Society of Sanitary Engineering (ASSE).
5. Factory Mutual Global (FMG).
6. Food and Drug Administration (FDA).
7. Plumbing and Drainage Institute (PDI).
8. Underwriters Laboratories, Inc. (UL).
9. U.S. Public Health Service.

most of these can probably be deleted. I was never able to find posted BFP requirements and the water dept. did not respond to my email asking about BFP requirements, so unless they mark it up on the permit review should i just leave it at "confirm with AHJ?"

PART 2 PRODUCTS

2.01 BACKFLOW PREVENTERS

- A. Acceptable Manufacturers: Zurn Wilkins or approved equal by Ames Fire & Waterworks, BEECO (MIFAB Inc.), Conbraco Industries Inc., FEBCO (Watts), or Watts.
- B. BFP Manufacturer and Model shall be on the list of accepted equipment published by the AHJ.
1. Note: Before ordering, obtain approval for the specific type and model of BFP to be used from the AHJ.

- C. BFP-XXXX, 3/4 in. to 2 in. Double Check (ASSE Standard 1015 ULI CSA/AW WA C510 Compliant): Zurn Wilkins Model 950XL, Lead free for potable water. Suitable for continuous pressure, inlet and outlet full port resilient seated ball valves, inlet strainer, test cocks, union connections between shut-off valves and device cast bronze and stainless steel construction.
- D. BFP-XXXX, 2-1/2 in. and Above Double Check (ASSE Standard 1015): Zurn Wilkins Model 350OSYFSC. Suitable for continuous pressure, inlet and outlet OS&Y gate valves, inlet strainer, test cocks, (2) independently operated check valves. Entire assembly coated inside and outside with FDA-approved epoxy material.
- E. BFP-XXXX, Dual Check Valve (ASSE Standard 1022): Zurn Wilkins Model 740. 150 PSI SWP, stainless-steel body and stainless-steel parts, 1/2 in. size.
- F. BFP-XXXX, Dual Check Valve (ASSE Standard 1024): Zurn Wilkins Model 700. 150 PSI SWP, cast bronze body and stainless steel parts, 3/4 in. size.
- G. BFP-XXXX, 3/4 in. to 2 in. Reduced Pressure Principle Type (ASSE Standard 1013): Zurn Wilkins Model 975XL2US. Suitable for continuous pressure, lead-free bronze body, inlet and outlet full port resilient seated ball valves, lead-free bronze Y-type inlet strainer, test cocks, differential pressure relief valve with ASME A112.1.2 air gap fitting between (2) independently operated check valves, union connections between shut-off valves and device.
- H. BFP-XXXX, 2-1/2 in. and Above Reduced Pressure Principle Type (ASSE Standard 1013): Zurn Wilkins Model 375OSYFSC. Suitable for continuous pressure, epoxy-coated ductile iron body, inlet and outlet OS&Y gate valves, inlet strainer, test cocks, differential pressure relief valve with ASME A112.1.2 air gap fitting between (2) independently operated check valves. Entire assembly coated inside and outside with FDA-approved epoxy material.
- I. BFP-XXXX for Beverage Dispensing Equipment (ASSE Standard 1022): Zurn Wilkins Model 740. Continuous pressure type with atmospheric vent drain, stainless-steel body and parts, stainless steel springs, double-check valve assembly, plastic replaceable check valves, FDA-approved, 3/8 in. size.
- J. BFP-XXXX for Hose Connection Vacuum Breakers (ASSE Standard 1011): Zurn Wilkins Model BFP-8CH. Chrome finish, ASME B1.20.7, garden-hose threads on outlet. Units may be rough bronze if attached to hose connections that have rough bronze finish.
- K. Refer to Fixture and Equipment Branch Size Schedule on the Drawings for sizes and applications.
- L. Each BFP shall be serviceable without requiring removal from piping.

2.02 CONNECTION BOX FOR ICE MAKER

- A. Acceptable Manufacturers: Guy Gray™ (IPS Corporation) or approved equals by Neptune, Oatey, or Water-Tite (IPS Corporation).
- B. IMB-1: Guy Gray Model BIM875. Recessed wall box with 1/2 in. NPT inlet and 1/4 in. outside diameter outlet compression angle stop.

2.03 CONNECTION BOX FOR WASHER/DRYER

- A. Acceptable Manufacturers: Guy Gray™ (IPS Corporation) or approved equal by Neptune, Oatey, Sioux Chief "OX," or Water-Tite (IPS Corporation).
- B. WB-1: Guy Gray Model (FBED 200 (bottom supply)) (BBED 200TS (top supply)). Recessed galvanized steel wall box with 20-gauge faceplate, 3/4 in. hose end valves, and 2 in. drain pipe with overflow guard. 20 amp 120 volts duplex grounding-type receptacle for washer, and 30 amp 208 volts 1-phase 2-pole 3-wire grounding receptacle for dryer.

2.04 CONNECTION BOX FOR WHIRLPOOL

- A. Acceptable Manufacturers: Guy Gray™ (IPS Corporation) or approved equal by Neptune, Oatey, or Water-Tite (IPS Corporation).
- B. Guy Gray Model T200LV. Recessed wall box, 2 in. drain pipe and overflow guard, white powder coat on cold rolled steel cabinet less valves.
 - 1. Provide Chicago Faucets Model 65-E2-2, polished chrome faucet modified with garden hose threaded outlet, inline vacuum breaker, lever handles.

2.05 CONNECTORS, PIPING, FLEXIBLE

- A. Acceptable Manufacturers: Metraflex or approved equal by Flexicraft, Flexonics Products, Hyspan Precision Products Inc., Minnesota Flexible Corp., Uniroyal General Rubber, or Twin City Hose Inc.
- B. Ethylene propylene diene monomer (EPDM) rubber tube with Teflon seal on threaded swivel, threaded or flanged connections on each end, 300 PSI maximum working pressure.
- C. Metraflex ML/SL. 200 degrees F., 200 PSIG, Type 321 stainless steel corrugated inner tubing and outer wire braid shield, threaded or flanged ends.
- D. Metraflex. Flexible rubber connector, 180 degrees F., 150 PSIG, 150 lb. ASA flanges, steel retaining ring.

2.06 CONNECTORS, PUMPS, FLEXIBLE

- A. Acceptable Manufacturers: Metraflex or approved equal by Flexicraft, Flexonics Products, Hyspan Precision Products Inc., Minnesota Flexible Corp., Uniroyal General Rubber, or Twin City Hose Inc.
- B. 2 in. and Smaller: Metraflex SST. 200 degrees F., 200 PSIG, stainless-steel corrugated inner tubing and outer wire braid shield, threaded ends.
- C. 2-1/2 in. and Larger: Metraflex Metrasphere. 200 degrees F., 150 PSIG, neoprene tube and cover with nylon cord reinforcement, floating flanges with cable bead wire, and tie-rods.

2.07 DRAIN PAN FOR WATER HEATER

- A. Approved Manufacturer: Oatey or approved equal by Water-Tite (IPS Corporation).
- B. Oatey 34000 Series. 2 in. deep aluminum round pan with rolled edges, 1 in. drain outlet fitting.

2.08 HOSE BIBBS, INTERIOR

- A. HB-100: Watts Model SC-8. Exposed domestic cold water supply, bronze body 3/4 in. NPT male threaded inlet, hex shoulder, tee handle, 3/4 in. male hose threaded outlet.
 - 1. Acceptable Manufacturers: Watts or approved equal by Chicago Faucets, MIFAB Inc., T&S Brass and Bronze Works Inc., or Woodford Manufacturing Company.
- B. HB-100: Woodford Model B22CCTK (ASSE Standard 1019). Separate domestic hot and cold water 3/4 in. supply inlets, 3/4 in. male hose thread outlet with vacuum breaker, and flush-mounted chrome-plated brass wall box with lock.
 - 1. Acceptable Manufacturers: Woodford Manufacturing Company or approved equal by Chicago Faucets, MIFAB Inc., T&S Brass and Bronze Works Inc., or Watts.
- C. Hose bibbs shall have integral nonremovable vacuum breakers and shall be furnished as follows:
 - 1. Concealed supply: Nibco #763CL with vacuum breaker on outlet, brass finish.
 - 2. Exposed supply: Nibco #74 with vacuum breaker on outlet, brass finish.
 - 3. Acceptable Manufacturers: Nibco Inc. or approved equal by Chicago Faucets, MIFAB Inc., T&S Brass and Bronze Works Inc., Watts, or Woodford Manufacturing Company.

4. Locker Areas:
 - a. Inside sill fitting: Chicago Faucets #293. Rough chrome-plated, flanged, with lockshield cap, loose key.
 - b. Full-flow hose thread outlet with inline vacuum breaker: Chicago Faucets #E27. Chrome-plated, nonremovable vacuum breaker.
5. Toilet Rooms: Chicago Faucets # [REDACTED]. Hot/cold faucet, polished chrome-plated, flanged, lockshield cap, loose key, and vacuum breaker.

2.09 HYDRANTS, ROOF

- A. Acceptable Manufacturers: Woodford Manufacturing Company or approved equal by Josam, Jay R. Smith Mfg. Co.®, MIFAB Inc., WADE (McWane Inc.), or Zurn Industries LLC.
- B. RH- [REDACTED]: Woodford Model RHY1. Freezeless, with 1 in. NPT female inlet, 3/4 in. brass hose nozzle, 1-1/4 in. galvanized pipe, and 1/8 in. NPT drain hole. Furnish complete with cast iron hydrant support components, well seal, EPDM boot, and deck-mounting system.
 1. Coordinate deck/roofing system with the A/E prior to ordering.
 2. Provide riser-mounted sign stating "NON-POTABLE WATER — DO NOT DRINK."
- C. RH- [REDACTED]: Woodford Model SRH-MS. Freezeless, with 3/4 in. NPT female inlet, 3/4 in. brass hose nozzle, ASSE 1052 double check backflow preventer, integral air vent, 1-1/2 in. galvanized pipe. Furnish complete with cast iron hydrant support components, well seal, EPDM boot, and deck mounting system.
 1. Provide riser-mounted sign stating "NON-POTABLE WATER — DO NOT DRINK."

2.10 HYDRANTS, WALL

- A. Acceptable Manufacturers: Jay R. Smith Mfg. Co.® or approved equal by Josam, MIFAB Inc., WADE (McWane Inc.), Woodford Manufacturing Company, or Zurn Industries LLC.
- B. WH- [REDACTED]: Jay R. Smith Mfg. "Quarterhorse" (Model 5509QT, with bronze faucet face, locking box and cover) (Model 5609QT, with nickel face), (ASSE 1019) bronze freeze-resistant faucet, bronze casing, self-draining, integral vacuum breaker, 3/4 in. size hose connection, length as required. Provide loose key with each hydrant.

2.11 QUICK COUPLERS

- A. Acceptable Manufacturers: Parker or approved equal by Aero, Aeroquip Performance Products, Aro (Parker), Balcrank Corp., Breco (Campbell Fittings), Dynaquip, Foster Hose & Fittings, Hansen (Eaton), Hofmann Fluid Power Inc., Lincoln Industrial, Milton Industries, or Schrader Pacific Advanced Valves.

- B. Parker 20 Series Model 25-F. Steel, 1/2 in. female pipe thread, 3/8 in. body. Verify body size and type with the Owner before ordering. Provide (1) companion male quick coupler per female quick coupler to the Owner.

2.12 SHOCK ABSORBERS

- A. Acceptable Manufacturers: WADE (McWane Inc.) or approved equal by Josam, Jay R. Smith Mfg. Co.®, MIFAB Inc., Precision Plumbing Products, Sioux Chief, Watts, or Zurn Industries LLC.
- B. WADE "Shokstops." (ASSE Standard 1010, PDI-WH201) shock absorbers (water hammer arrestors). Sizes as shown on the Drawings.
 - 1. SA-A: 3/4 in. pipe. ([1] to [11] fixture units.)
 - 2. SA-B: 1 in. pipe. ([12] to [32] fixture units.)
 - 3. SA-C: 1 in. pipe. ([33] to [60] fixture units.)

2.13 SHUT-OFF VALVE FOR WASHER

- A. Acceptable Manufacturers: Watts or approved equal by Apollo Valves, Milwaukee Valve, or Nibco Inc.
- B. WMV-: Watts SeriesI 2M2. Bronze body, single lever ON/OFF, 1/2 in. threaded or solder inlets, 3/4 in. hose end connections.

2.14 STRAINERS

- A. Acceptable Manufacturers: Watts or approved equal by Armstrong International Inc., Hofmann Fluid Power Inc., or Spirax Sarco Ltd.
- B. Y-Pattern: Watts Model 777S. 1/2 in. to 3 in., bronze body, threaded, #20 mesh stainless steel screen, 200 PSI WOG.

2.15 TANK, EXPANSION, FOR DOMESTIC WATER SYSTEM

- A. ET-: Amtrol Therm-X-trol (Model ST-5-C. 2.1 gallon) (Model ST-12-C. 6.4 gallon) (Model ST-42V-C. 17.5 gallon) total volume. Pre-charged welded black steel tank, constructed, labeled, and stamped for 125 lb. working pressure in accordance with ASME. FDA-approved for domestic (potable) water. Vacuum relief cocks, drains, airtrol tank fittings, air charge fitting with protective plastic cap, rigid polypropylene liner, heavy-duty butyl diaphragm, 3/4 in. NPT male system connection, and Manufacturer's standard finish.
 - 1. Acceptable Manufacturers: Amtrol or approved equal by Bell & Gossett (Xylem Inc.), Elbi of America, Goulds Water Technology (Xylem Inc.), State Water Heaters, Watts, or Wessels Company.

- B. ET-XXXX: Amtrol "Therm-X-Trol" Model ST-451C. 158 gallons total and acceptance volume, welded ASME 150 PSI working pressure, steel tank, constructed, labeled, and stamped for 125 lb. minimum working pressure in accordance with ASME. Air charge valve, 1 in. NPT system connection, NSF-approved, replaceable bladder, and air pre-charge at 55 psig.
1. Acceptable Manufacturers: Amtrol or approved equal by ExpanFlex (Calefactio), Goulds Water Technology (Xylem Inc.), State Water Heaters, Watts, or Wessels Company.
- C. ET-XXXX: Wessels Model TXA130. 35 gallons total and acceptance volume, welded ASME 150 PSI working pressure, steel tank, constructed, labeled, and stamped for 125 lb. working pressure in accordance with ASME. Air charge valve, 1 in. NPT system connection, NSF-approved, replaceable bladder, and air pre-charge.
1. Acceptable Manufacturers: Wessels Company or approved equal by Amtrol, ExpanFlex (Calefactio), Goulds Water Technology (Xylem Inc.), State Water Heaters, or Watts.

2.16 TANK, EXPANSION, FOR WATER HEATER

- A. Acceptable Manufacturers: Amtrol or approved equal by Bell & Gossett (Xylem Inc.), ExpanFlex (Calefactio), Goulds Water Technology (Xylem Inc.), State Water Heaters, Watts, or Wessels Company.
- B. ET-XXXX: Amtrol Therm-X-Trol Model ST-447C. Welded ASME 150 PSI working pressure, steel tank, air charge valve, 2 in. NPT cold water connection, NSF-approved, flexible butyl rubber diaphragm, rigid polypropylene liner, and air pre-charge.
- C. ET-XXXX: Amtrol Therm-X-Trol (Model ST-5-C) (Model ST-12C). (0.9) (3.2) gallons acceptance volume, (2.1) (6.4) gallons total volume. (ASME Section VIII Division 1, ASHRAE 188, and ISO 9001.) Welded ASME (125) (150) PSI working pressure, steel tank, air charge valve, 3/4 in. stainless-steel NPT cold water connection, NSF-approved, flexible butyl rubber diaphragm, inlet water agitating device, (rigid polypropylene) (antimicrobial) liner, and 55 PSIG factory air pre-charge.
- D. ET-XXXX: Amtrol Therm-X-Trol Model ST-42V-C. 11.4 gallons acceptance volume, 17.5 gallons total volume. Welded ASME 125 PSI working pressure, steel tank, air charge valve, 3/4 in. NPT cold water connection, NSF-approved, flexible butyl rubber diaphragm, rigid polypropylene liner, and air pre-charge.

2.17 TANK, WELL WATER HYDROPNEUMATIC

- A. Acceptable Manufacturers: Amtrol or approved equal by Myers (Pentair), State Water Heaters, Thrush Co. Inc., or Wessels Company.

- B. Amtrol Well-X-Trol WX-453. 36 in. diameter x 86 in. high, 264 gallons total volume. Welded ASME 125 PSI working pressure steel tank with Schrader air charge valve, flexible heavy-duty butyl rubber bladder, rigid polypropylene liner, air precharge.

2.18 THERMOSTATIC MIXING VALVES

- A. TMV-, for building, (109) (120) degrees F. hot water: Bradley Navigator NRS- assembly (ASSE 1017). Factory-assembled, factory-tested, factory-mounted temperature/pressure water mixing valve, rated for 22 GPM at 15 PSI drop, on enamel-coated strut with GFCI receptacle. Inlet and outlet shut-off valves, temperature/pressure gauges on inlet, outlet set-up connection return line with recirculating pump, and balancing and check valves. Install system as a complete unit.
1. Acceptable Manufacturers: Bradley Corp. or approved equal by Armstrong International, Lawler Manufacturing Company, Leonard Valve Company, Powers (Watts), or Symmons.
 2. RP-: TMV Manufacturer shall provide recirculating pump as an integrated part of the system.

DESIGNER — CHOOSE (a) or (b) or (c):

- a. Inline booster pump: Bell & Gossett Series 100. 3/4 in. size, 10 GPM at 7 ft. hd., 1/12 HP, 120 volts, 1750 RPM.
 - b. Inline booster pump: Xylem Bell & Gossett Series PR. 1 in. size, lead-free copper alloy body, 6 GPM at 17 ft. hd., 1/6 HP, 120 volts, 1,725 RPM.
 - c. Inline booster pump: Bell & Gossett Series PL-36. 1.25 in. size, 20 GPM at 20 ft. hd., 1/6 HP, 120 volts/2.1 full load amperes, 1750 RPM, lead-free, copper alloy body.
 - 1) Acceptable Manufacturers: Bell & Gossett (Xylem Inc.) or approved equal by Armstrong International, Amtrol Inc., Grundfos, PACO Pumps (Grundfos), Taco Comfort Solutions, or Thrush Co. Inc.
 - d. Aquastat controller: Honeywell L6006C. Adjustable temperature range of 60 degrees F. to 200 degrees F. Set at (110) (120) degrees F. 120 volt rating.
 - 1) Acceptable Manufacturers: Honeywell or approved equal by Dwyer Instruments Inc., Johnson Controls, or Schneider Electric.
 - e. Timer: Honeywell Model ST6008C. 7-day time-based switching, battery backup, and NEMA 1 enclosure. UL-listed.
 - 1) Other Acceptable Manufacturers: Honeywell or approved equal by Dwyer Instruments Inc., Johnson Controls, or Schneider Electric.
- B. TMV-, for building, 120 degrees F. hot water: Bradley Model #S59-3080-P-RS assembly (ANSI 1017). Factory-assembled, factory-tested, temperature/pressure water mixing valve

rated for 44 GPM at 15 PSI drop. Inlet check stops, outlet volume/shutoff valve, pressure-regulating valve with pressure gauges.

1. Acceptable Manufacturers: Bradley Corp. or approved equal by Armstrong International, Lawler Manufacturing Company, Leonard Valve Company, Powers (Watts), or Symmons.
 2. Install within recessed 12-gauge stainless-steel cabinet, with keylock.
- C. TMV- , for building, 140 degrees F. hot water: Bradley Model #S59-3260 (HL2x1) high-capacity manifold system (ANSI 1017). Water mixing valve rated for 142 GPM at 15 PSI drop.
1. Acceptable Manufacturers: Bradley Corp. or approved equal by Armstrong International, Lawler Manufacturing Company, Leonard Valve Company, Powers (Watts), or Symmons.
 2. Provide inlet and outlet shut-off valves, temperature/pressure gauges on inlet, outlet set-up connection return line with recirculating pump, balancing and check valves.
 3. RP-1 recirculating pump: Refer to Section 22 11 23.23.
- D. TMV- , outlet temperature 120 degrees F.: Powers Model LFLM 490 (ANSI 1017). Temperature-compensating and pressure-compensating mixing valve, rough chrome finish, union and integral checks with strainers on inlets, vandal-resistant temperature setting and wall-mounting bracket. Maximum expected flow 13 GPM at 15 PSI pressure drop. Minimum flow rate shall be 0.5 GPM without recirculation system. Upon failure, unit shall fail safe.
1. Acceptable Manufacturers and Models: Powers (Watts) or approved equal by Armstrong International, Lawler Manufacturing Company, Leonard Valve Company, or Symmons.
- E. TMV- , outlet temperature 109 degrees F.: Bradley Model #S59 (ANSI 1017). Temperature/pressure-compensating mixing valve, rough chrome finish, union and check stops with strainers on inlets, vandal-resistant temperature setting and wall-mounting bracket. Maximum expected flow GPM at PSI pressure drop. Upon failure, unit shall fail safe.
1. Acceptable Manufacturers: Bradley Corp. or approved equal by Armstrong International, Leonard Valve Company, or Symmons.
- F. TMV- , outlet temperature 120 degrees F.: Bradley Model #S59-4016 (ANSI 1017). Temperature/pressure-compensating mixing valve, union and check stops with strainers on inlets, vandal-resistant temperature setting and wall-mounting bracket. Maximum expected flow 8.5 GPM at 15 PSI pressure drop. Upon failure, unit shall fail safe.
1. Acceptable Manufacturers: Bradley Corp. or approved equal by Armstrong International, Leonard Valve Company, or Symmons.

- G. TMV-, for single fixture, outlet temperature 109 degrees F. hot water: Bradley "Vernatherm Valve," (Model S59-4004XS) (Model S59-4007) (ASSE 1070). Water temperature-limiting device, lead-free body, vandal-resistant adjustment, adjustable temperature setpoint within fixed range, and mounting bracket. Maximum flow (3) (4.5) GPM at 15 PSI pressure drop. Upon failure, unit shall fail safe.
1. Acceptable Manufacturers: Bradley Corp. or approved equal by Lawler Manufacturing Company, Powers (Watts), Symmons, Watts, or Zurn Wilkins.
- H. TMV-, for single fixture, outlet temperature 109 degrees F. hot water: Bradley "Navigator," Model S59-4000A (ASSE 1070). Water temperature-limiting device, lead-free body, vandal-resistant adjustment, adjustable temperature setpoint within fixed range, and mounting bracket. Maximum flow (3) (4.5) GPM at 15 PSI pressure drop. Upon failure, unit shall fail safe.
1. Acceptable Manufacturers: Bradley Corp. or approved equal by Lawler Manufacturing Company, Powers (Watts), Symmons, Watts, or Zurn Wilkins.
- I. TMV-, for multiple fixtures, outlet temperature 105 degrees F. hot water (unless noted otherwise on the Plumbing Drawings): Bradley "Vernatherm Valve," Model S59-4000 (ASSE 1070). Water temperature-limiting device, bronze body, vandal-resistant adjustment, and adjustable temperature setpoint within fixed range. Maximum flow 4.5 GPM at 30 PSI pressure drop. Upon failure, unit shall fail safe.
1. Acceptable Manufacturers: Bradley Corp. or approved equal by Lawler Manufacturing Company, Powers (Watts), Symmons, Watts, or Zurn Wilkins.
- J. TMV-, for single fixture (emergency eye/face wash), outlet temperature 85 degrees F. tepid water: Bradley Model S19-2000-EFX8-R-B-P-0-0 (ASSE 1071). Water temperaturelimiting device, brass body. Minimum-rated flow of 3 GPM at 10 PSI pressure drop. Upon failure, unit shall fail safe, and shall have a cold water bypass in the event of the loss of the hot water supply.
1. Acceptable Manufacturers: Bradley Corp. or approved equal by Lawler Manufacturing Company, Powers (Watts), Symmons, Watts, or Zurn Wilkins.
 2. Provide with wall-mounting bracket.

- K. **TMV's for emergency ey/face wash/full body shower with dedicated water heater, provide to meet ASSE 1085.**
- L. **TMV-**, for single fixture (emergency eye/face wash), outlet temperature 85 degrees F. tepid water: Bradley Model S19-2000 (ASSE 1071). Water temperature-limiting device, bronze body, dial thermometer. Maximum flow 4 GPM at 10 PSI pressure drop. Upon failure, unit shall fail safe, and shall have a cold water bypass in the event of the loss of the hot water supply.
1. Acceptable Manufacturers: Bradley Corp. or approved equal by Lawler Manufacturing Company, Powers (Watts), Symmons, WaterSaver Faucet Co., Watts, or Zurn Wilkins.
 2. Provide in surface-mounted cabinet.
- M. **TMV-**, for single fixture (emergency shower and emergency eye/face wash), outlet temperature 85 degrees F. tepid water: Bradley Model S19-2250-EFX50-R-0-P-SE-0 (ASSE 1071). Water temperature-limiting device, bronze body. Minimum-rated flow of 23 GPM at 10 PSI pressure drop. Upon failure, unit shall fail safe, and shall have a cold water bypass in the event of the loss of the hot water supply.
1. Acceptable Manufacturers: Bradley Corp. or approved equal by Lawler Manufacturing Company, Powers (Watts), Symmons, Watts, or Zurn Wilkins.
 2. Provide in surface-mounted white enamel-painted steel cabinet.
- N. **TMV-**, for multiple fixtures (emergency shower and emergency eye/face wash), outlet temperature 85 degrees F. tepid water: WaterSaver Faucet Model AP3800 (ASSE 1071). Water temperature-limiting device, bronze body. Maximum flow 21 GPM at 5 PSI pressure drop. Upon failure, unit shall fail safe, and shall have a cold water bypass in the event of the loss of the hot water supply.
1. Acceptable Manufacturers: WaterSaver Faucet Co. or approved equal by Bradley Corp., Lawler Manufacturing Company, Powers (Watts), Symmons, Watts, or Zurn Wilkins.
 2. Provide check valves on supply inlets unless integral with TMV.
- O. Provide outlet thermometer. Refer to Section 22 05 19. Do NOT provide the TMV Manufacturer's dial thermometer.
- P. Provide ball valve and check valve on inlet connections, if not integral with TMV.
- Q. Provide 10-year warranty for TMV.

2.19 TRAP PRIMERS

- A. Acceptable Manufacturers: Precision Plumbing Products or approved equal by MIFAB Inc., WADE (McWane Inc.), Watts, or Zurn Industries LLC.

- B. TP-: Precision Plumbing "Pro-Drain," Series PDS (ASSE Standard 1072). Brass housing nonadjustable flow control valve. Provide distribution unit listed below. Units shall adjust automatically to line pressure and flow.
1. Model No. DU-2 for (2) floor drains.
 2. Model No. DU-3 for (3) floor drains.
 3. Model No. DU-4 for (4) floor drains.
- C. TP-: Barrier Type: Precision Plumbing "Pro-Drain," Series PDS (IAPMO-listed, ASSE Standard 1072-listed). Barrier type trap seal device.
- D. TP-: Precision Plumbing "Pro- Rite," Series PR-500 (ASSE Standard 1018). Brass housing nonadjustable flow control valve. Provide distribution unit listed below. Units shall adjust automatically to line pressure and flow.
1. Model No. DU-2 for (2) floor drains.
 2. Model No. DU-3 for (3) floor drains.
 3. Model No. DU-4 for (4) floor drains.
 4. Option: Precision Plumbing "Pro-Drain," Series PDS (IAPMO-listed, ASSE Standard 1072-listed). Barrier type trap seal device.
- E. TP-: Precision Plumbing "Under-Lav Trap Primer Assembly" (ASSE Standard 1018). 35 PSIG to 75 PSIG operating range, includes chrome-plated brass housing, nonadjustable flow control valve, Model No. PR-500, chrome-plated angle stop, compression fittings, copper tubing, and escutcheon plate.
- F. TP-: Precision Plumbing "Prime-Time Electronic Trap Priming Manifold," PT-Series (ASSE Standard 1001). Pre-set 24-hour clock, solenoid valve, manifold, water hammer arrester, pre-assembled inside stainless steel cabinet for flush wall-mounting, stainless steel access door with Allen key latch.

2.20 VACUUM BREAKERS

- A. Watts Model #N36. 200 lb. SWP, bronze body, silicone disc (for use with continuous back pressure).
- B. Atmospheric: Watts (Model #288A (plain brass)) (Model #288A-C (chrome-plated)) (ANSI A112.1). Bronze body, full line size discharge (for use without continuous back pressure).
1. Acceptable Manufacturers: Watts or approved equal by Cash Acme, Cla-Val, Lawler Manufacturing Company, MIFAB Inc., Neptune, Omni Brass Inc., or Zurn Wilkins,.

- C. Watts Model #8A (ANSI A112.1 / ASSE Standard 1011). Bronze body (for use on all items with hose ends).
 - 1. Acceptable Manufacturers: Watts or approved equal by Cash Acme, Conbraco Industries, MIFAB Inc., or Zurn Wilkins.

PART 3 EXECUTION

3.01 INSTALLATION — BACKFLOW PREVENTERS

- A. Install BFPs where indicated or required by Code.
- B. Install each BFP in accordance with the Manufacturer's recommendations.
- C. Install BFPs so as to be easily accessible, and at a maximum of 9 ft. above the floor.
- D. Provide rigid support for BFP.
- E. Locate equipment BFPs below ceilings in area where spillage of water will not cause damage.
- F. Provide shut-off valve ahead of each BFP assembly.
- G. Install BFP on domestic water supply line where shown.
- H. Install BFP on supply line to exterior yard hydrant.
- I. Provide (ball) (gate) valve ahead of strainer on inlet side of BFP assembly.
- J. Install supervisory switch on OS&Y gate valves.
- K. Detector BFP: Install in a horizontal position. Discharge from BFP shall have an open air gap. Indirect waste to nearest drain.
- L. Reduced Pressure BFP: Discharge from BFP shall have an open air gap. Indirect waste full size to nearest floor drain or floor sink using steel or copper piping.
- M. After installation of BFP, flush water supply line to remove debris. Internally clean BFP after flushing.
- N. Each BFP shall have a permanently attached plate indicating type and listing approvals.
- O. Test BFP at time of installation. A person or firm certified by the (State) (Local) Department of Health shall perform the test and provide the testing equipment. Submit test report to the A/E. Provide BFP certification.

3.02 INSTALLATION — CONNECTION BOX FOR ICE MAKER

- A. Install bottom of box at 3 ft.-6 in. above finished floor.

3.03 INSTALLATION — CONNECTION BOX FOR WASHER/DRYER

- A. Provide standpipe and piping.
- B. Install bottom of connection box at 3 ft.-6 in. above finished floor.
- C. Electrical Contractor shall provide power wiring.

3.04 INSTALLATION — CONNECTION BOX FOR WHIRLPOOL

- A. Coordinate exact location with the A/E.
- B. Provide standpipe, 2 in. trap, and drain line.
- C. Install bottom of connection box at 3 ft.-6 in. above finished floor.
- D. Install faucet just above connection box.
- E. Electrical Contractor shall provide power wiring.

3.05 INSTALLATION — CONNECTORS, PIPING, FLEXIBLE

- A. Provide flexible connectors on all piping to equipment 5 HP and larger, and where shown for equipment of lower horsepower.

3.06 INSTALLATION — CONNECTORS, PUMPS, FLEXIBLE

- A. Provide flexible pump connectors for each pump 2 HP and larger.
- B. Connectors shall have limiting tie rods to prevent excessive movement.

3.07 INSTALLATION — DRAIN PAN FOR WATER HEATER

- A. Drain pan shall have a minimum of 2 in. clearance around circumference of water heater.

3.08 INSTALLATION — HOSE BIBBS, INTERIOR

- A. Coordinate installation height of hose bibbs with the A/E.
- B. Install hose bibbs  in. above finished floor.
- C. Provide ball valves on hose bibb.

- D. Provide continuous back pressure type vacuum breakers on all lines to hose bibbs, if not furnished with integral vacuum breaker.

3.09 INSTALLATION — HYDRANTS, ROOF

- A. Coordinate installation height with the A/E.

3.10 INSTALLATION — HYDRANTS, WALL

- A. Coordinate installation height with the A/E. Center hydrants in the middle of block or stone. Installation height is generally 18 in. above finished grade.
- B. Install flush and level with building wall. Careful coordination is required at stone and block walls.
- C. Provide ball valve, with shock absorber between valve and hydrant, on inlet connection to hydrant.

3.11 INSTALLATION — QUICK COUPLERS

- A. Coordinate type and size with the Owner.
- B. Coordinate installation height on water drops with the Owner, Electrical Work, and equipment.
- C. Install where shown on the Drawings.

3.12 INSTALLATION — SHOCK ABSORBERS

- A. Locate where shown on the Drawings, according to PDI Standard WH-201. Size as shown on the Drawings.
- B. Install at the end of each main or branch, unless otherwise noted.

3.13 INSTALLATION — SHUT-OFF VALVE FOR WASHER

- A. Install centerline of valve at 4 ft.-0 in. above finished floor.

3.14 INSTALLATION — STRAINERS

- A. Provide strainer:
 - 1. On upstream line to each water pressure reducing valve.
 - 2. On hot and cold inlets to TMV, past ball valve.
- B. Provide shut-off valve ahead of each strainer.

- C. Provide (drain) (ball) valve on each strainer drain. Extend piping to nearest floor drain.

3.15 INSTALLATION — TANK, EXPANSION, FOR DOMESTIC WATER SYSTEM

- A. Mount and support tank securely from building structure.
- B. Install tank on 4 in. high concrete pad. Refer to Section 22 05 29 for concrete work. Shim to level.
- C. Install dielectric union on inlet.
- D. Connect to cold water supply line piped, with a thermal trap, isolation valve, and drain valve to allow periodic checking and pre-charge pressure adjustment.

3.16 INSTALLATION — TANK, EXPANSION, FOR WATER HEATER

- A. Install expansion tanks for water heaters according to the Manufacturer's instructions.
- B. Support expansion tank from structure.

3.17 INSTALLATION — TANK, WELL WATER HYDROPNEUMATIC

- A. Coordinate location with other Contractors and the Lead Contractor.

3.18 INSTALLATION — THERMOSTATIC MIXING VALVES

- A. Install TMVs so as to be easily accessible, and at a maximum of 9 ft. above the floor.
- B. Provide TMV assembly and piping.
- C. Connect piping according to Manufacturer's requirements.
- D. Provide ball valves on inlet and outlet connection of each TMV.
- E. Provide thermometers and pressure gauges for TMVs. Factory-furnished thermometers and pressure gauges shall be equal to the type specified in Section 22 05 19.
- F. After installation, flush water supply line to remove debris. Clean out shower TMV after flushing.
- G. TMV: Electrical Contractor shall provide:
 - 1. Disconnect switch.
 - 2. Control wiring.
 - 3. Interconnecting wiring.

4. Power wiring.
- H. Domestic Hot Water Recirculating Pump and Aquastat/Timer:
1. Coordinate location and electrical requirements with the Electrical Work.
 2. Electrical Contractor shall provide:
 - a. Disconnect switch.
 - b. Control wiring.
 - c. Interconnecting wiring.
 - d. Power wiring.
 3. Electrical Contractor shall wire through aquastat/timer.
 4. Electrical Contractor shall wire pneumatic-to-electric switch and cycle controller.
- I. Set domestic hot water outlet TMV temperature for (109) (120) (125) degrees F.
- J. Emergency Fixtures: Set TMV temperature for 85 degrees F.

3.19 INSTALLATION — TRAP PRIMERS

- A. Install trap primers according to the Manufacturer's instructions.
- B. TP-: Install below fixture. Provide 3/4 in. gravity drain from trap primer to floor drain within room/area.
- C. TP-: Mount trap primer above ceiling. Provide ball valve ahead of trap primer. Provide access panel where necessary. Provide individual 3/4 in. gravity drains from trap primer to inlet of each floor drain trap where indicated on the Drawings.
- D. TP-: Mount trap primer exposed on wall in accessible location. Provide ball valve ahead of trap primer. Provide individual 3/4 in. gravity drains from trap primer to inlet of each floor drain trap where indicated on the Drawings.
- E. TP-: Mount trap primer recessed in wall in accessible location. Provide individual 3/4 in. gravity drains from trap primer assembly to inlet of each floor drain trap where indicated on the Drawings.
- F. Electronic Trap Priming Manifold:
1. Install manifold and provide piping.
 2. Electrical Contractor shall provide power wiring.

- G. Mechanical trap seals are permitted where approved by the AHJ.
 - 1. Do NOT provide trap seals at kitchen and mechanical room floor sinks; provide trap primers.
 - 2. Spare parts: Furnish (5) of each type of trap seal installed as attic stock. Refer to Section 22 05 00.78.

3.20 INSTALLATION — VACUUM BREAKERS

- A. Install vacuum breakers according to the Manufacturer's instructions.

END OF SECTION 22 11 19

SECTION 22 11 23.23 - DOMESTIC-WATER IN-LINE PUMPS

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 22 05 00.78, "Closeout Submittals for Plumbing."
- B. Section 22 05 48, "Vibration and Seismic Controls for Plumbing Piping and Equipment."

1.02 DESCRIPTION

- A. Provide a hot-water recirculating pump for the building hot water recirculating system where shown on the Drawings.
- B. Provide an aquastat controller for the hot-water recirculating pump.

1.03 RATINGS AND CAPACITIES

- A. Pump Operating Point Requirements: The operating point of each pump at the conditions shown shall be within (5) percentage points of the maximum efficiency on its impeller curve. The impeller furnished shall not exceed 90% of the cut water diameter available for the pump casing furnished. The impeller curve shall be continuously rising throughout the range contained within its efficiency curves. The peak of the impeller curve shall be a minimum of 10% above the total head shown on the Drawings. The pump shall not overload the motor at any point on the impeller curve.
- B. Motors:
 - 1. Provide premium efficiency, NEMA Design B, continuous rated with 1.15 service factor, with Class F insulation and Class B temperature rise, copper windings and leads, 1,750 RPM with the horsepower indicated on the Drawings. All motors shall be equipped with ball bearings.
 - 2. Motors shall be open drip-proof, with thermal overload protection.
- C. Refer to the Drawings for gallons per minute, suction and discharge pressures, feet of head, horsepower, and electrical requirements. Each pump shall be rated for 65% of total design flow.

1.04 QUALITY ASSURANCE

- A. Standards:
 - 1. National Electrical Manufacturers Association (NEMA).

2. Underwriters Laboratories, Inc. (UL), listed and labeled.

1.05 SPARE PARTS

- A. Provide the spare parts listed below. Refer to Section 22 05 00.78.
 1. A spare set of mechanical seals for each pump.
 2. Domestic water pressure gauge with hose end connection.
 3. Set of gaskets — (1) set for each pump.

PART 2 PRODUCTS

2.01 IN-LINE RECIRCULATING PUMP

- A. Acceptable Manufacturers: Bell & Gossett (Xylem Inc.) or approved equal by Amtrol, Armstrong Fluid Technologies, Grundfos, PACO Pumps (Grundfos), Taco Comfort Solutions, or Thrush Co. Inc.
- B. RP-XXXX: Bell & Gossett Series 100. Circulating pump, all-bronze, inline, pipe-mounted, horizontal, 3/4 in. size, 10 GPM at 7 ft. hd., 1/12 HP, 120 volts, 1,750 RPM, 125 PSI working pressure, oil-lubricated, steel shaft, thrust collar, (2) horizontal sleeve bearings, bronze impeller, mechanical seals, flexible motor coupling, resilient motor mounting, and flanged fittings with gaskets.
- C. RP-XXXX: Bell & Gossett "Little Red," Model LR-20B. Recirculating pump, all-bronze, inline, pipe-mounted, horizontal, 3/4 in. size, 5 GPM at 11 ft. hd., 1/20 HP, 120 volts, 2,900 RPM, 125 PSI working pressure, oil-lubricated steel shaft, sleeve bearing, glass-filled Noryl impeller, and mechanical seals.
- D. RP-XXXX: Bell & Gossett "Upstart," Model SLC-25B. Recirculating pump, bronze body, inline, pipe-mounted, horizontal, 3/4 in. size, 4-speed motor, XXXX GPM at XXXX ft. hd., 1/25 HP, 120 volts, 2,480 RPM, 125 PSI working pressure, stainless steel shaft, carbon sleeve bearings, glass-filled polypropylene impeller, and flanged fittings with gaskets.
- E. RP-XXXX: Bell & Gossett Series NBF. Recirculating pump, all-bronze, inline, pipe-mounted, horizontal, 3/4 in. size, 4 GPM at 10 ft. hd., 55 watts, 120 volts, single-phase, 0.48 FLA, 2,800 RPM, 125 PSI working pressure, oil-lubricated steel shaft, sleeve bearing, bronze impeller, and mechanical seals.

2.02 CONTROLS

DESIGNER — EDIT:

- A. Controller: Pulsar. Solid-state, wall-mounted pump controller, with steel cabinet, key lock switch for switching to 24-hour circulation, pump running light, power to panel light, terminal boards, adjustable pause cycle control for pump shut-off time, and adjustable pulse cycle control for pump run time.
 - 1. Acceptable Manufacturer: Pulsar Process Measurement Ltd.
- B. Aquastat: Honeywell Model L6006C. Remotely located strap-on element, adjustable 60 degrees F. to 200 degrees F. temperature range, 120 volt rating, and scale lock assembly.
 - 1. Acceptable Manufacturer: Honeywell.
- C. Timer/Aquastat: Bell & Gossett Model TC-1 Automatic Timer Kit with AQ-3/4 Aquastat.
 - 1. Acceptable Manufacturer: Bell & Gossett (Xylem Inc.).
- D. Timer and Aquastat:
 - 1. Timer: Honeywell Model ST6008C. UL-listed, 7-day time-based switching, battery backup, and NEMA 1 enclosure.
 - 2. Aquastat: Honeywell Model L6006A. UL-listed, ON/OFF circulator controller, strap-on pipe mounting, and horizontal bulb insertion.
 - 3. Acceptable Manufacturers: Honeywell or approved equal by Dwyer Instruments Inc., Johnson Controls, or Schneider Electric.
- E. Timers and Time Delay Relay:
 - 1. Timers: Electric time clock for control of hot-water circulation pump.
 - a. Type: Programmable, 7-day clock with manual override ON/OFF switch.
 - b. Enclosure: Suitable for wall mounting.
 - c. Operation of pump: ON or OFF.
 - d. Transformer: Provide if required.
 - e. Power requirement: (24) (120) volt AC.
 - f. Programmable sequence of operation: Up to (2) ON/OFF cycles each day for (7) days.

2. Time delay relay: Control for hot-water storage tank circulation pump.
 - a. Type: Adjustable time delay relay.
 - b. Range: Up to (5) minutes.
 - c. Setting: (5) minutes.
 - d. Operation of pump: ON or OFF.
 - e. Transformer: Provide if required.
 - f. Power requirement: (24) (120) volt AC.
 - g. Programmable sequence of operation: Limit pump operation to periods of burner operation plus maximum (5) minutes after the burner stops.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Coordinate location and electrical requirements with the Electrical Work.
- B. Provide ball valve on inlet and outlet of recirculating pump for easy removal.
- C. Install recirculating pump aquastat on upstream side of pump.

DESIGNER — CHOOSE (D) or (E):

- D. Set aquastat at 100 degrees F.
- E. Set programmable timer to turn the recirculating pump automatically ON and OFF. Verify times with the Owner. Aquastat will cycle the recirculating pump ON and OFF during timer-controlled intervals, maintaining domestic hot water supply temperature (pump ON at 105 degrees F., OFF at 109 degrees F.).

DESIGNER — CHOOSE (F) or (G):

- F. Adjust system balancing valves and balance waterflow at the end of each project phase in order to provide proper distribution and uniform pressure drop throughout the entire project's duration.
- G. Adjust system balancing valves and balance waterflow at end of each project phase in order to provide proper distribution and uniform pressure drop throughout project duration.

3.02 WIRING

- A. Interlock pump with water heater burner and time delay relay.
- B. Electrical Contractor shall provide:
 - 1. Disconnect switch.
 - 2. Control wiring.
 - 3. Interconnecting wiring.
 - 4. Power wiring for the recirculating pump controller.
- C. Electrical Contractor shall wire through aquastat/timer.

END OF SECTION 22 11 23.23

SECTION 22 13 00 - FACILITY SANITARY SEWERAGE

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 22 05 29, "Hangers and Supports for Plumbing Piping and Equipment."
- B. Section 22 05 76, "Facility Drainage Piping Cleanouts."
- C. Section 22 10 00, "Plumbing Piping."
- D. Section 22 13 23, "Sanitary Waste Interceptors."
- E. Section 22 14 29, "Sump Pumps."
- F. Section 22 40 00, "Plumbing Fixtures."
- G. Refer to Drawings for Fixture and Equipment Branch Size Connection Schedule.

1.02 DESCRIPTION

- A. Provide the site sanitary sewer system.
- B. Connect to site sanitary sewer system 5 ft. outside building foundation wall where shown on the Drawings. Extend sanitary line into building.
- C. Extend building sanitary drain to a point 5 ft. beyond the foundation wall, unless otherwise indicated on the Drawings.
- D. Provide a complete system of interior soil, waste, and vent piping to fixtures and equipment, including the following items:
 - 1. Cleanouts: Refer to Section 22 05 76.
 - 2. Floor drains.
 - 3. Floor drains, inline trap seals.
 - 4. Floor sinks.
 - 5. Food Service equipment drains.
 - 6. Hub drains.
 - 7. Mop service basins: Refer to Section 22 40 00.

8. Sanitary waste interceptors: Refer to Section 22 13 23.
9. Shower drains.
10. Vents.
11. Valves, air admittance.

E. Test complete system.

1.03 TERMINOLOGY

A. Abbreviations:

1. Acrylonitrile Butadiene Styrene (ABS).
2. Drain, waste, and vent (DWV).
3. Polyvinyl chloride (PVC).
4. Volatile organic compounds (VOCs).

1.04 QUALITY ASSURANCE

A. Standards:

1. American National Standards Institute (ANSI).
2. American Society for Testing and Materials (ASTM).
3. American Society of Mechanical Engineers (ASME)
4. American Society of Sanitary Engineers (ASSE).
5. Cast Iron Soil Pipe Institute (CISPI).
6. Factory Mutual Global (FMG).
7. International Code Council, Inc. (ICC): International Plumbing Code (IPC).
8. National Sanitation Foundation (NSF).
9. Code of Federal Regulations (CFR) Title 40, "Protection of Environment," Part 59, "National Volatile Organic Compound Emission Standards for Consumer and Commercial Products," Subpart D, "National Volatile Organic Compound Emission Standards for Architectural Coatings."
10. U.S. Environmental Protection Agency (EPA).
11. Illinois Administrative Code (IAC):
 - 1) IAC Title 35: Environmental Protection.

- 2) IAC Title 41: Fire Protection.
- 3) IAC Title 71: Public Buildings, Facilities, and Real Properties.
- 4) IAC Title 77: Public Health.
- 5) IAC Title 83: Public Utilities.

- B. All pipe and fittings shall conform to the requirements of ASTM A74.
- C. All piping materials and installation shall comply with State and Local Building Code requirements.

1.05 SUBMITTALS

- A. Floor Drains: Each floor drain grate that has a load rating shall have an ANSI load rating indicated on shop drawings. If rating is not indicated on shop drawings, they will be rejected.

PART 2 PRODUCTS

2.01 PIPING AND FITTINGS

- A. Manufacturer and Model Number listed below shall be considered to be the Design Base Manufacturer and Model Number. Other Acceptable Manufacturers will be permitted, unless specifically excluded.
- B. Acceptable Manufacturers:
1. Cast iron: American Cast Iron, Clow, Griffin, MWI, Tyler Pipe, or U.S. Pipe & Foundry.
 2. Copper: American Brass Co., Chase Brass Co., Nibco, or Revere.
 3. ABS or PVC plastic: Charlotte, Crestline, Nibco, or Yardley.
 4. Steel: Armco, Republic, or Wheeling.
- C. Sanitary, waste, and vent piping and fittings under slab:
1. Extra-heavy weight, single hub cast iron (ASTM A74), factory-coated with coal tar enamel.
 2. Hubless cast iron (CISPI 301, ASTM A888), factory-coated with coal tar enamel.
 3. Copper DWV tube, 3 in. and smaller (ASTM B306): Type K drainage tube, drawn temper.

4. Schedule 40 PVC/DWV plastic (ASTM D2665).
 5. Schedule 40 ABS/DWV plastic (ASTM D2661).
- D. Interior above-grade sanitary waste and vent piping and fittings:
1. Service weight, single hub cast iron (ASTM A74), factory-coated with coal tar enamel.
 2. Hubless cast iron (CISPI 301, ASTM A888), factory-coated with coal tar enamel.
 3. Steel pipe, 2-1/2 in. and smaller (ASTM A53): Type E or S, Grade A or B, Schedule 40, galvanized.
 4. Copper DWV tube, 4 in. and smaller (ASTM B306): Type L drainage tube, drawn temper.
 5. PVC Schedule 40 solid wall pipe according to ASTM D 1784, D 1785, and D 2665. PVC DWV fittings conforming to ASTM D 1784 and D 2665.
- E. All fittings shall be compatible with piping for size, material, and joint type.
- F. Joints:
1. Cast iron:
 - a. Tarred jute or oakum with lead poured flush with hub.
 - b. ASTM A 74 / ASTM C 564 Neoprene molded gasket for sealing hub and spigot cast iron fittings; service-weight or extra-heavy pipe fittings, as appropriate.
 - c. Compression-type plastic seal (ASTM C-564).
 - d. Joints in hubless cast iron pipe:
 - 1) Sizes 1-1/2 in. to 10 in.: Coupling approved to FMG 1680 Class 1 and marked with the logo.
 - a) Sizes 1-1/2 in. to 4 in.: Clamp-All Model #80.
 - b) Sizes 5 in. to 10 in.: Clamp-All Model #125.
 - c) Acceptable Manufacturer: Clamp-All Corporation.
 - 2) Sizes 1-1/2 in. to 2 in., where exposure to head pressure cannot exceed 10 ft., coupling approved to CISPI 310.
 - a) Acceptable Manufacturers: Anaco, Charlotte, Jones, or Tyler.
 2. Copper: Cast (ASTM B16.23) or wrought (ASTM B16.29) copper drainage fittings with solder joints.
 3. Galvanized steel: Threaded, galvanized cast iron drainage fittings.

4. ABS or PVC plastic: Solvents and cements as recommended by the Manufacturer.

2.02 SOLVENT CEMENTS AND ADHESIVE PRIMERS

- A. For ABS Pipe and Fittings:
 1. Use ABS solvent cement with a VOC content of 325 grams/liter or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
 2. Use adhesive primer with a VOC content of 550 grams/liter or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
- B. For Joining PVC Piping: ASTM D2564. Include primer according to ASTM F656.
 1. Use PVC solvent cement with a VOC content of 510 grams/liter or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
 2. Use adhesive primer with a VOC content of 550 grams/liter or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

2.03 FLOOR DRAINS

- A. Acceptable Manufacturers: MIFAB Inc. or approved equal by Jay R. Smith Mfg. Co.®, Josam, WADE (McWane Inc.), Watts, or Zurn Industries LLC.
- B. Drains shall be adjustable, coated cast iron double drainage pattern, with bottom outlet, flashing clamp, strainers, sediment buckets, and other accessories and features as noted below. **Coordinate with the Architectural Drawings.**
- C. Each floor drain that is required to have a secured grate shall have at least (3) screws for attachment of grate.
- D. Slot in grates shall be free of flashing (clean cut); tops shall not have sharp edges.
- E. **FD-XXXX**: MIFAB F1000-5-3. Cast iron body, adjustable, 5 in. diameter strainer, secured, stainless steel finish, stainless steel strainer, caulk or push-on outlet, and 3 in. pipe size. **Good for unit R**
 1. In public areas, provide vandal-resistant screws on strainers.
 2. In areas with ceramic tile or quarry tile finish, provide 5 in. x 5 in. square strainer.
- F. **FD-XXXX**: MIFAB F1860-21. Cast iron body, 12 in. square top, 8 in. deep open hopper basin, stainless-steel secondary grate, 4 in. pipe size.
 1. Provide vandal-resistant pin torx screws on strainer.

- G. FD-XXXX: MIFAB F1320-3-11-12-4-7. Cast iron body, adjustable, 9 in. diameter strainer, secured medium-duty load rating, stainless-steel XXXXXX tractor grate, galvanized top assembly, acid-resistant coated strainer, caulk or push-on outlet, 4 in. pipe size, and 3/4 in. trap primer connection.
- H. FD-XXXX: MIFAB F1000-5-3-G. Cast iron body, adjustable, 5 in. diameter strainer, secured, stainless-steel finish, stainless-steel strainer, caulk or push-on outlet, 3 in. pipe size, and 4 in. x 9 in. oval funnel. Good for Mech
- I. TS-XXXX: Floor Drain Trap Seal. MIFAB F1000-5-3-G. Cast iron body, adjustable, 5 in. diameter strainer, secured, stainless-steel finish, stainless-steel strainer, caulk or push-on outlet, 3 in. pipe size, and 4 in. x 9 in. oval funnel.
1. Acceptable Manufacturer: MIFAB Inc.

2.04 FLOOR SINKS

- A. Acceptable Manufacturers: MIFAB Inc. or approved equal by Jay R. Smith Mfg. Co.®, Josam, WADE (McWane Inc., Watts, or Zurn Industries LLC).
- B. FS-XXXX: MIFAB T1200-F-4. Medium-duty grating, 12 in. x 24 in., (2) 1-3/4 in. x 24 in. frame with (2) 20 in. x 24 in. ductile iron grates, 3 in. floor drain FD-XXXX outlet. 24 in. high, 20-gauge Type 304 stainless steel wall guard behind sink.
1. Faucet: Chicago Faucet Model 897-S. Service sink faucet, bucket hook, hose-end rigid spout with top wall brace, vacuum breaker, lever handles, stops in shanks, rough chrome-plated. Mount faucet at (36) (40) in. above finished floor.
2. Hose and hose bracket: Fiat Model 832-AA.
3. Mop hanger: Fiat Model 889-CC. Install mop hanger at 4 ft.-0 in. above finished floor on wall adjacent to faucet and over grating.
4. Provide (2) panels for corner installation.
- C. FS-XXXX: MIFAB FS1720-6-7-150. 12 in. x 12 in. x 6 in. deep cast iron body, with square slotted medium-duty grate, white acid-resistant porcelain interior and top, ABS anti-splash interior bottom dome strainer, vandal-resistant, 1/2 in. trap primer connection, 1/2 grate.
1. Receptor: Sani-Flor. Good for kitchen?
- a. Acceptable Manufacturer: Zurn.

- D. FS-XXXX: Zurn 1900. Sani-Flor receptor, 12 in. x 12 in. x 6 in. deep cast iron body, with square slotted medium-duty grate, white acid-resistant porcelain interior and top, ABS anti-splash interior bottom dome strainer, vandal-resistant, 1/2 in. trap primer connection, Full, 3/4 or 1/2 grate as appropriate for location and fixtures served.

1. Other Acceptable Manufacturers: J. R. Smith, Wade, Josam, or Watts.

2.05 OPTION: FLOOR-DRAINS AND FLOOR SINKS, INLINE TRAP SEALS

- A. Contractor may, in accordance with Local Codes, provide an inline barrier-type trap seal devices in lieu of trap seal primers **except in the following location(s): Commercial kitchen spaces (and XXXXX)**.
- B. Acceptable Manufacturers: Green Drain, Jay R. Smith Mfg. Co.® "Quad Close," Precision Plumbing Products "Pro-Drain Trap Seal," RectorSeal "SureSeal," Sioux Chief "TrapShield," or Zurn Industries LLC "ZShield."
- C. Inline floor drain trap seal device consisting of a frame and sealing flapper designed to allow flow of water and debris, while slowing trap evaporation. The device shall not require tools to install in an open drain, shall be tested in compliance with ASSE 1072, and shall hold an applicable certificate of listing with one or more of the following entities: IAPMO (UPC), ICC (ICC-ES PMG), or ASSE. The Manufacturer shall list product as approved for use as an alternate to trap primer.

2.06 SHOWER DRAINS

- A. Acceptable Manufacturers: MIFAB Inc. or approved equal by Jay R. Smith Mfg. Co.®, Josam, WADE (McWane Inc., Watts, or Zurn Industries LLC).
- B. SHD-XXXX: MIFAB #SD7005-6. Cast iron body, adjustable, 4 in. diameter round strainer, secured, stainless steel finish, strainer, caulk or push-on outlet, 2 in. pipe size, as noted.
1. Provide vandal-resistant screws on strainers.

2.07 VALVES, AIR ADMITTANCE

- A. Acceptable Manufacturers: Oatey or approved equal by Ayrlett, ProSet, or Studor.
- B. AAV-XXXX: Oatey Model 39007. Compliant with NSF Standard 14, ASSE 1051 A and B, and ASSE 1056. PVC, with sliding disk, guidepost, and rubber diaphragm.

2.08 VENTS — CAPS

- A. Acceptable Manufacturers: MIFAB Inc. or approved equal by Jay R. Smith Mfg. Co.®, Josam, Stoneman, WADE (McWane Inc., or Zurn Industries LLC).

- B. MIFAB #R1930. Cast iron with set screws, vandal-resistant.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Examine the areas and the conditions under which interior sanitary piping system work is to be installed. Notify the A/E and Owner of conditions detrimental to the proper and timely completion of the Work.
- B. Do not proceed with the Work until unsatisfactory conditions have been corrected and interior sanitary piping location and depth have been established.

3.02 INSTALLATION — PIPING AND FITTINGS

- A. Coordinate location, size, and depth of site sanitary sewer with the Lead Contractor.
- B. Coordinate location, elevation, and installation of standpipes with the equipment, concrete pads, Lead Contractor, and A/E.
- C. Coordinate location and depth of building sanitary drain installation with the building foundation and the Lead Contractor.
- D. Refer to Section 22 10 00 for general piping installation instructions.
- E. Sanitary lines shall be continuously sloped.
 - 1. Pitch soil and waste piping in direction of flow at not less than 1/8 in./ft. and not more than 1/4 in./ft.
 - 2. Pitch pipes 2 in. and smaller at 1/4 in./ft.
 - 3. Pitch kitchen piping at 1/4 in./ft. to the grease interceptor.
- F. Underground Soil and Waste Piping: Minimum size 2 in.
- G. Do NOT provide “dead-end” sanitary branches longer than 2 ft.-0 in.
- H. Make all connections and changes in direction of the sanitary drainage system with approved drainage fittings.
 - 1. Fittings shall not have ledges or shoulders capable of obstructing piping flow.
 - 2. Install fittings to guide waste in the direction of flow.

3. For lavatory waste drainage fittings, provide sanitary tee branch or combination wye and 1/8 bend for single fixture.
 4. For back-to-back lavatory waste, provide sanitary cross or combination wye and 1/8 bend, double size.
- I. Plastic Piping: Do not install plastic piping:
1. Where it may be vulnerable to physical damage.
 2. Where run exposed.
 3. Where run exposed below joist level.
 4. Where run exposed below 8 ft. above floor.
 5. In ceiling spaces used for return air plenums. Refer to Architectural and HVAC Drawings for the location of these areas.
 6. On waste lines capable of receiving discharge from a boiler, water heater, or other piece of equipment operating above 140 degrees F., for a distance of 25 ft. downstream of the discharge point.
- J. Connect to sanitary drains with "No-Hub" fittings and "No-Hub" stainless steel bands.
- K. Elevator Equipment Rooms: Avoid locating horizontal sanitary piping in elevator equipment rooms. If horizontal sanitary piping shall be located in elevator equipment rooms, obtain written permission from the A/E to do so. Any horizontal sanitary piping located in elevator equipment rooms shall be covered and enclosed by a painted metal enclosure to prevent water dripping onto equipment.
- L. Conceal all piping, unless indicated otherwise on the Drawings.
- M. Cleaning and Testing:
1. After erection, and prior to putting in service, blow or flush lines free of loose materials.
 2. Prior to the completion of work, "power jet" all sanitary mains and branches.
 3. Perform tests in accordance with AHJ requirements. Tests shall be witnessed by the A/E's Representative.
 4. For a minimum test, plug piping at the lowest point and fill with water to highest point on roof to provide a minimum of 10 ft. hd. of water on all parts of the system.
 - a. Maintain for (1) hour with no leakage. If a leak occurs, replace the defective piece or joint with new materials at no cost to the Owner.
 - b. Repair any deficiencies.

- c. Conduct final test with smoke or peppermint at 1 in. W.G. for (15) minutes.
- d. Repeat tests until no leaks occur.
- e. Install gaskets or reset fixtures.

3.03 INSTALLATION — DRAIN RECESSING AND FLOOR SLOPING

A. Building staff only spaces with VCT, tile, or polished concrete floors:

- 1. Recess the top of the drain inlet 1/4 in. below the finish level.
- 2. No slope required to drain.

B. Building staff only spaces with bare or coating finished concrete floors:

- 1. Recess the top of the drain inlet 3/4 in. below the finish level.
- 2. Provide a 1/4 in./ft. slope within a 3-foot radius around the drains.

C. Other spaces with VCT, tile, or polished concrete floors:

- 1. Recess the top of the drain inlet 1/4 in. below the finish level.
- 2. No slope required to drain.

D. Other spaces with bare or coating finished concrete floors:

- 1. Recess the top of the drain inlet 3/8 in. below the finish level.
- 2. Provide a 1/8 in./ft. slope within a 3-foot radius around the drains.

E. Spaces expected to regularly take on water (showers, maintenance garages, hose down areas, etc.):

- 1. Coordinate drain recess and floor sloping with A/E before roughing in..

3.04 INSTALLATION — FLOOR DRAINS

- A. Coordinate location, elevation, and installation of floor drains with equipment, concrete pads, Lead Contractor, and the A/E.
- B. Coordinate size and depth of floor drains.
- C. If not shown on the Architectural Drawings, confirm proper pitch to floor drains with the A/E before roughing in.
- D. Install P-trap below floor for floor drains.

- E. Install top of floor drains flush and level with wall and finished floor.
- F. Where tile is to be installed, adjust top of floor drain to be flush with the finished surface.
- G. Install square floor drains in ceramic tile floors and carefully align edges parallel to room walls.
- H. REMOVE AND REPLACE ALL NEW ITEMS NOT INSTALLED AT PROPER ALIGNMENT, AT PROPER ELEVATION, FLUSH, AND LEVEL, AT NO COST TO THE OWNER.
- I. Cover grates for floor drains to provide protection from scratching and collection of dirt and debris during construction. Remove cover just prior to final inspection.

3.05 INSTALLATION — FLOOR SINKS

- A. Install P-trap below floor for floor sinks.
- B. Where tile is to be installed, adjust top of floor sink to be flush with the finished surface.
- C. Install duct tape over grates for floor sinks to provide protection from scratching and collection of dirt and debris during construction. Remove tape just prior to final inspection.

3.06 INSTALLATION — FOOD SERVICE EQUIPMENT DRAINS

- A. Coordinate location, size, and depth of Food Service equipment drains.

3.07 INSTALLATION — HUB DRAINS

- A. Coordinate location, size, and depth of hub drains.
- B. If not shown on the Architectural Drawings, confirm elevation and proper pitch to drains with the A/E before roughing in.
- C. Install P-trap below floor for hub drains.
- D. Install top of hub drains flush and level with wall and finished floor.
- E. REMOVE AND REPLACE ALL ITEMS NOT INSTALLED AT PROPER ALIGNMENT, AT PROPER ELEVATION, FLUSH, AND LEVEL, AT NO COST TO THE OWNER.

3.08 INSTALLATION — SHOWER DRAINS

- A. Coordinate location, size, and depth of shower drains.

- B. If not shown on the Architectural Drawings, confirm elevation and proper pitch to shower drains with the A/E before roughing in.
- C. Install P-trap below floor for shower drains.
- D. Install top of shower drains flush and level with wall and finished floor.
- E. Where tile is to be installed, adjust top of shower drain to be flush with the finished surface.
- F. **Install all flashing and pans for showers and similar equipment items.**
- G. REMOVE AND REPLACE ALL ITEMS NOT INSTALLED AT PROPER ALIGNMENT, AT PROPER ELEVATION, FLUSH, AND LEVEL, AT NO COST TO THE OWNER.
- H. Install duct tape over grates for shower drains to provide protection from scratching and collection of dirt and debris during construction. Remove tape just prior to final inspection.

3.09 INSTALLATION — TRENCH DRAINS

- A. Coordinate location, size, and depth of trench drains.
- B. Install trench drains and catch basins in strict accordance with Manufacturer's installation procedures.
- C. 4 in. of newly placed concrete shall be provided beneath and around channels. Refer to Section 22 05 29 for concrete requirements.
- D. Provide sealant to interlocking joints between channels and endcaps, and between channels and catch basins to prevent fluid migrating out of trenches.
- E. Plug trench drain outlets temporarily.
- F. Install duct tape over grates for trench drains to provide protection from scratching and collection of dirt and debris during construction. Remove tape just prior to final inspection.

3.10 INSTALLATION — VENTS

- A. Coordinate location of vent penetrations through roof with the HVAC Work, Lead Contractor, and A/E.
- B. Pitch all vents for proper drainage.
- C. All exposed vent piping shall be galvanized steel.

- D. Connect to vents where shown on Drawings.
- E. Install vents through roof as follows:
 - 1. 3 in. minimum size.
 - 2. Locate at least 8 ft. away from outside wall of building and 20 ft. away from outside air intakes or operable windows.
 - 3. Offset vent piping below roof to allow for thermal expansion and contraction of roof. Minimum of 4 ft. offset.
 - 4. Vents shall extend a minimum of 12 in. above roof, except where specifically required to be higher by Code.
 - 5. Roofs: Refer to Section 22 10 00 for additional requirements regarding roofs.
 - a. **Built-up roof:** Vent flashing shall extend at least 12 in. from vent pipe, clamp between built-up roofing, and extend up, over, and turn down inside vent pipe.
 - b. **Single membrane roofs:** Install vent through vent pipe boot.
 - 6. Provide vandal-resistant vent caps on roof vent outlets.

END OF SECTION 22 13 00

SECTION 22 13 23 - SANITARY WASTE INTERCEPTORS

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 22 05 00.78, "Closeout Submittals for Plumbing."
- B. Section 22 05 29, "Hangers and Supports for Plumbing Piping and Equipment."
- C. Section 22 05 76, "Facility Drainage Piping Cleanouts."
- D. Section 22 13 00, "Facility Sanitary Sewerage."
- E. Section 22 40 00, "Plumbing Fixtures."

1.02 DESCRIPTION

- A. Provide the following sanitary waste interceptors:
 - 1. Grease interceptors.
 - 2. Hair interceptors.
 - 3. Lint traps.
- B. For cleanouts, refer to Section 22 05 76.
- C. Connect to sanitary sewer system where shown on the Drawings.
- D. Test complete system.

1.03 QUALITY ASSURANCE

- A. Standards:
 - 1. American Association of State Highway Transportation Officials (AASHTO).
 - 2. American National Standards Institute (ANSI).
 - 3. American Society for Testing and Materials (ASTM).
 - 4. Occupational Safety and Health Administration (OSHA).
 - 5. Environmental Protection Agency (EPA).
 - 6. Illinois Administrative Code (IAC):

- a. IAC Title 35: Environmental Protection.
- b. IAC Title 41: Fire Protection.
- c. IAC Title 71: Public Buildings, Facilities, and Real Properties.
- d. IAC Title 77: Public Health.
- e. IAC Title 83: Public Utilities.

B. Interceptors shall comply with City, County, and State requirements.

1.04 SPARE PARTS

A. Provide (1) set of gaskets for each manhole or handhole. Refer to Section 22 05 00.78.

PART 2 PRODUCTS

2.01 GREASE INTERCEPTORS

Do you have a spec i can
copy for the "Proceptor"
model?

- A. GI-XXXX, Concrete: Babbert Model G5. Rated for 500-gallon capacity. Flow rate and grease capacity, inlet and outlets as shown on the Drawings. Precast concrete, complete with internal baffles, tapplings, vents, manhole extension(s) to finished grade, bolted and gasketed manhole cover(s), reinforced construction, and top suitable for heavy traffic.
- 1. Acceptable Manufacturers: E.C. Babbert Inc. or approved equal by Mack Industries or Oldham.
- B. GI-XXXX: Babbert Model G-20. Rated for 2,000-gallon liquid capacity. Flow rate 100 GPM and 4,000 pounds grease capacity, inlet and outlets as shown on the Drawings. Precast concrete, complete with epoxy-coated internal surfaces, baffles, tapplings, vent, manhole extensions to finished grade, bolted and gasketed manhole covers, reinforced construction, and top suitable for heavy vehicle traffic.
- 1. Option: Polyethylene grease interceptors.
 - a. Acceptable Manufacturer: Schier.
 - 2. Acceptable Manufacturers: E.C. Babbert Inc. or approved equal by National Precast or Oldcastle.

- C. GI-100, Cast Iron: Jay R. Smith Mfg. Model 8120-GT. Low-type, coated steel complete with baffles, access openings, bolted and gasketed cover, attached automatic draw-off line with valve, and 2-minute flow control with vent and line shut-off valve. Above-the-floor installation. Automatic draw-off feature shall permit removal of accumulated grease without cover removal. 20 GPM flow rate and 40 lb. grease capacity.
1. Provide extension to grade.
 2. Obtain specific approval of AHJ before ordering.
 3. Acceptable Manufacturers: Jay R. Smith Mfg. Co.® or approved equal by Zurn Industries LLC "Ejecto-Matic," Josam, or MIFAB Inc.
- D. GI-100, Cast Iron: Zurn Model Z-1170-Y, size 800. Low-type, coated steel complete with baffles, access openings, bolted and gasketed cover, lift-out sediment bucket, vent, enzyme port opening, bronze cleanout plug. Above-the-floor installation. 50 GPM flow rate and 70 lb. grease capacity.
1. Obtain specific approval of AHJ before ordering.
 2. Acceptable Manufacturers: Zurn Industries LLC or approved equal by Jay R. Smith Mfg. Co.®, Josam, or MIFAB Inc.

2.02 HAIR INTERCEPTORS

- A. Acceptable Manufacturers: Zurn Industries LLC or approved equal by Jay R. Smith Mfg. Co.®, Josam, or MIFAB Inc.

- B. HI-100: Zurn Model Z1175. Dura-coated cast iron hair interceptor for installation inside cabinetry. 1-1/2 in. inlet and 1-1/2 in. female NPT connections. Perforated stainless-steel removable basket, bottom access flanged and gasketed cover.

Good for single hair wash sink?

2.03 LINT TRAPS

- A. Acceptable Manufacturer: Zurn Industries LLC.
- B. LI-1: Zurn Model 21185, Size 30. Fabricated steel construction, acid-resistant coated interior, non-skid secured cover, stainless steel lint screen, fixed primary straining baffle, 200 GPM capacity, and 4 in. inlet and outlet.

I used the Z1185 size 3 to serve 2 washers, will revise.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Examine the areas and the conditions under which sanitary waste interceptors are to be installed. Notify the A/E and Owner of conditions detrimental to the proper and timely completion of the Work.
- B. Do not proceed with the Work until unsatisfactory conditions have been corrected and interior sanitary piping location and depth have been established.

3.02 INSTALLATION — GENERAL

- A. Coordinate interceptor and cleanout locations, elevations, and installation with the Lead Contractor and A/E.
- B. Installation of interceptors shall meet State and Local Building Code, Plumbing Code, and Health Code requirements.
- C. Pitch soil and waste piping to interceptor in direction of flow at not less than 1/8 in. per ft. and not more than 1/4 in. per ft.
- D. Testing:
 - 1. Perform tests in accordance with AHJ requirements. Tests shall be witnessed by the A/E's Representative.
 - 2. For a minimum test, plug piping at the lowest point and fill with water to highest point on roof to provide a minimum of 10 ft. hd. of water on all parts of the system.
 - a. Maintain for (1) hour with no leakage. Should leaks occur, remove the defective section of pipe and defective fitting and replace with new materials at no cost to the Owner.
 - b. Repair any deficiencies.
 - c. Conduct final test with smoke or peppermint at 1 in. W.G. for (15) minutes
 - d. Repeat tests until no leaks occur.
 - e. Install gaskets or reset fixtures.

3.03 INSTALLATION — GREASE INTERCEPTORS

- A. Install grease interceptor, in accordance with State Plumbing Code, OEPA, and AHJ requirements, to receive kitchen grease waste from kitchen equipment or fixtures, as shown on the Drawings.

- B. Excavate to the depth required by OSHA and Local Codes and Standards.
- C. Install and level 6 in. bed of pea gravel.
- D. Install and level tank.
- E. Fill tank to pipe invert with water.
- F. Concrete anti-buoyancy backfill shall be poured around tank to depth indicated. Refer to Section 22 05 29 for concrete work.
- G. Backfill up to inlet and outlet pipes level with pea gravel compacted in 6 in. to 8 in. layers.
- H. Install inlet and outlet pipes.
- I. Provide manhole extension collars as recommended by the Interceptor Manufacturer.
- J. Vent interceptor in accordance with AHJ requirements. Vent outlet of interceptor assembly to prevent siphoning. Extend the (2) vents as shown on the Drawings.
- K. Backfill with pea gravel or approved fill equivalent to 12 in. above base of extension collar.
- L. Backfill with coarse sand, gravel, pea gravel, or approved equivalent around the extension collar to the bearing slab.
- M. Install manway frame and cover.
- N. Install all cleanout plugs and covers.

3.04 INSTALLATION — HAIR INTERCEPTORS

- A. Install hair interceptor high enough inside cabinet to allow removal of hair strainer.

3.05 INSTALLATION — LINT TRAPS

- A. Install lint trap according to the Manufacturer's recommendations.

END OF SECTION 22 13 23

SECTION 22 13 29 - SANITARY SEWERAGE PUMPS

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 22 05 00.78, "Closeout Submittals for Plumbing."
- B. Section 22 05 29, "Hangers and Supports for Plumbing Piping and Equipment."
- C. Section 22 05 43, "Underground Plumbing Systems."

1.02 DESCRIPTION

- A. Provide a duplex sewage pump and basin complete with controls.
- B. Test complete system.

1.03 RATINGS AND CAPACITY

- A. Pump Operating Point Requirements: The operating point of each pump at the conditions shown shall be within (5) percentage points of the maximum efficiency on its impeller curve. The impeller furnished shall not exceed 90% of the cut water diameter available for the pump casing furnished. The impeller curve shall be continuously rising throughout the range contained within its efficiency curves. The peak of the impeller curve shall be a minimum of 10% above the total head shown on the Drawings. The pump shall not overload the motor at any point on the impeller curve.

Add? Mike Y says to ask Markus.

- B. Motors: Unless otherwise noted, high-efficiency motors shall be NEMA Design B, continuous rated with 1.15 service factor, with Class F insulation and Class B temperature rise, copper windings and leads, (1,750) (3,450) RPM with the horsepower indicated on the Drawings. All motors shall be equipped with ball bearings.
- C. Refer to Drawings for gallons per minute, suction and discharge pressures, feet of head, horsepower, and electrical requirements. Each pump shall be rated for 65% of total design flow.

1.04 QUALITY ASSURANCE

- A. Standards:
 - 1. National Electrical Manufacturers Association (NEMA).

2. Underwriters Laboratories, Inc. (UL) listed and labeled.
 3. Ohio Environmental Protection Agency (OEPA).
- B. Each pump and drive unit shall be performance-tested at the factory at design conditions. The system shall be hydrostatically tested. The entire assembly shall be painted after shop testing.

1.05 SPARE PARTS

- A. Provide a spare set of mechanical seals for each pump. Refer to Section 22 05 00.78 regarding spare parts.

PART 2 PRODUCTS

2.01 SEWAGE EJECTOR (40 GPM)

- A. Acceptable Manufacturers: Hydromatic (Pentair) or approved equal by Crane/Barnes, Peerless Pump Company, Weinman, or Weil Pump.

is this model still made? Mike Y says to ask Markus

- B. Hydromatic Model Gilx 200 DD. GPM at ft. hd. each pump, HP, volts, -phase, 60 cycles, packaged duplex, nonclog centrifugal type, cast iron casing with integral cast tripod support. Motor shall have inner and outer water seals with moisture-sensing probe in the event of seal failure, and factory-wired controls. Thermally protected motor UL-listed for use in Class 1, Group D hazardous locations. Pump configuration shall permit removal of the motor-shaft-impeller assembly without disconnecting the discharge piping.
- C. Impeller: Enclosed bronze grinder type solids-handling impeller shall pass 3 in. sphere minimum, dynamically balanced, keyed to motor shaft, and secured with a lock screw.
- D. Shaft: Stainless steel motor-impeller type.
- E. Exterior Valve Box: Fiberglass or steel, field-installed, adjacent to basin, containing (2) 4 in. check valves, (2) 4 in. gate valves, (2) 4 in. 90 degree elbows, and (1) 4 in. tee. Lockable valve box steel cover plate suitable for flush installation in concrete slab. Provide drain from valve box to basin, refer to Drawings.
- F. Sump Basin: Reinforced concrete, fiberglass, or steel construction. Provide gasketed, watertight, flexible connections between basin walls and inlet and outlet piping.

- G. Float Controls: Sealed electronic floatless device with electrodes sealed in a chemical-resistant rubber bulb with an electrolytic solution of distilled water and sodium bicarbonate, and NEMA 7 below-grade terminal housing.
 - H. Pump Controls: Provide (4) float level controls as listed below and extend control wiring through cover plate.
 - 1. Lead pump on.
 - 2. Pumps off.
 - 3. Lag pump on.
 - 4. High water alarm with NEMA 4 terminal.
 - I. Pump Control Panel: NEMA (1) (4) (duplex) (simplex) pump control panel complete with nonfused disconnect switches, concealed HAND/OFF/AUTO switches, control transformers, electric automatic alternator, individual pump motor circuit breakers, lightning arrester, pump running lights, relay and indicating lights for seal leak and moisture detection, high water light and alarm with silencing button, extra contacts for remote alarm, selector switch, motor starters with overload protection, high level warning lights, factory-wired controls, and dead front construction panel.
 - 1. Mounting: Provide brackets for mounting on top of basin. Control panel shall be securely mounted to sump basin cover with pedestal mounting stand or structural angles.
 - 2. Wires from sump shall be sealed at basin and control panel to provide vapor-proof seal.
 - 3. Access door: Access door on control panel shall be on opposite side of sump basin and lockable.
 - 4. Provide elapsed time meter for each pump.
 - J. Accessories: Steel basin cover with access opening, vent, and cord grip fittings for motor cable and flexible connector to seal openings for discharge pipe where they pass through the basin cover. Provide hoist above ground to lift pumps from basin as an option for rail type pumps. Provide lockable, bolted, and gasketed basin cover.
 - K. Remote High-Water Alarm Panel: Anchor Scientific Model 1100-1. Wall-mounted, with red light, 4 in. horn, and test/silencing switch. Provide engraved sign for "SEWAGE EJECTOR HIGH WATER ALARM" on enclosure.
 - L. All exposed items shall be rust-proofed.
- 2.02 SEWAGE EJECTOR (100 GPM)
- A. Acceptable Manufacturer: Weil Pump.

- B. Weil Model 2523. [REDACTED] GPM at [REDACTED] ft. hd. each pump, [REDACTED] HP, [REDACTED] volts, [REDACTED]-phase, 60 cycles, packaged duplex, Type 316 stainless steel pump shaft, stainless steel fastening hardware, cast iron motor shell, pump volute, and impeller. Capable of handling raw, unscreened sewage and of passing 2-1/2 in. diameter solids. Each pump shall be equipped with hermetically sealed, Class F insulated motor, installed in a cast iron shell. Pump construction shall include double-lip mechanical seal, with both seals housed in an oil reservoir. Tandem seals with (1) seal operating in the material being pumped are not acceptable. Primer coat and water-resistant enamel finish coat on pump and motor unit.
- C. Pump Impellers: 2-vane, non-clog type, accurately machined to the proper diameter, dynamically balanced prior to installation in the pump, and non-overloading throughout the entire range of the pump curve.
- D. Basin/Valve Box: Fiberglass main basin, 4 in. hubbed inlet, fiberglass valve box, pump mounting plates bolted to main steel plate fastened to main basin bottom, lifting lugs on outside of basin sections, and steel cover, bolted, gasketed, and sealed, with all necessary vent connections, pump access plates, and access doors to valve box.
- E. Piping: 4 in. Schedule 40 galvanized steel pump discharge. Mounted in the main basin, extended through wall and valve box, and joined by flanged connections to piping, valves, and fittings contained within integral valve box. Individual pump discharge lines joined within valve box and cross-connected to common discharge line extending horizontally through valve box wall. Flexible link-type seal with integral sleeve furnished to provide gas-tight, watertight seal. Provide drain with check valve that prevents water from entering the valve box, but allows water to drain back to main basin.
- F. Check and Gate Valves: (2) 4 in. gate valves and (2) 4 in. flanged non-slam check valves, installed either in horizontal position or at maximum 45 degree incline, in valve box with outside levers and adjustable spring tension to ensure valve closing before flow reversal.
- G. Quick Removal System: Weil Series 2163. Each pump shall be able to be removed and returned to its original position without going into the wet well, and shall be furnished with:
1. Removable floor discharge elbow.
 2. Stainless steel lifting cable.
 3. (2) galvanized guide rails.
 4. Upper guide rail bracket.
 5. All necessary hardware.

- H. Electrical Control Panel: NEMA 1 duplex, suitable for wall-mounting, completely factory-wired and bench-tested prior to shipment. Furnish wired and numbered terminal strip, and electrical wiring schematic of system clearly indicating all necessary external field-wiring connections. Panel shall contain in a single enclosure the following items:
1. (2) fusible disconnect switches, sized for motor protection only, with lock-out handles through panel door.
 2. (2) across-the-line magnetic starters with overload and low-voltage protection in all phases. Furnish separate replaceable relay to alternate pumps on each cycle of operation. Furnish additional replaceable relay to cut in standby pump if lead pump fails or cannot keep up with inflow.
 3. (2) HAND/OFF/AUTO selector switches.
 4. (2) pump run lights mounted on door.
 5. Control circuit transformer with primary and secondary fusing.
 6. Alarms: Audible and visual for high water condition and for operating standby pump. Provide auxiliary dry contacts for both alarm conditions for purposes of wiring to remote monitoring or alarm panel. 120 volt alarm circuits, including internal ON/OFF power switch. Audible alarm: 3 in. bell mounted on panel exterior with (2) alarm lights mounted on panel door and (1) alarm-silencing switch.
- I. Basin Level Controls: Weil Series 8230. Mercury float-type liquid level controls shall be furnished for proper pump and alarm operation. Provide (4) mercury switches, each sealed in corrosion-resistant polypropylene float, with PVC-coated UL-listed cable, and attached by corrosive-resistant bracket to galvanized suspension rod. Switch bracket provided with reinforcing sleeve to prevent sharp bending and damage to the cable.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Coordinate location of sewage ejector with the equipment, Electrical Work, and Lead Contractor.
- B. Installation shall comply with all State and Local Building Code requirements.
- C. Sewage ejector discharge piping shall be galvanized steel, to a point of connection to the large gravity line.
- D. Install basin, pump, piping, valves, and control panel where shown on the Drawings. Comply with the Manufacturer's recommendations. Install non-slam check valve in pump discharge piping.
- E. Secure basin cover so that it is absolutely airtight and flush with finished (floor) (grade).

- F. Vent top part of the basin to outdoors.
- G. Refer to Drawings and to Section 22 05 29 regarding the concrete slab for the valve box.
- H. Upon completion, the sewage ejector shall be left clean and free from rubbish until acceptance on the work.
- I. Test sump pumps and controls by filling basin with water and operating through all modes of operation.

3.02 WIRING

- A. Coordinate electrical requirements with the Electrical Work.
- B. Include a mechanical or electrical alternator control panel with starters and circuit breakers or fusible disconnect switches for each motor and factory-wired float control and alarm wiring.
- C. Furnish remote high-water alarm panel to the Electrical Contractor for installation.
- D. Electrical Contractor shall provide:
 - 1. Wiring to control panel and pumps.
 - 2. Control wiring.
 - 3. Power wiring.
- E. Electrical Contractor shall install and wire remote high-water alarm panel.

END OF SECTION 22 13 29

SECTION 22 14 00 - FACILITY STORM DRAINAGE

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 22 05 76, "Facility Drainage Piping Cleanouts."
- B. Section 22 07 00, "Plumbing Insulation."
- C. Section 22 10 00, "Plumbing Piping."
- D. Section 22 14 29, "Sump Pumps."

1.02 DESCRIPTION

- A. Lead Contractor shall provide the site storm sewer system.
- B. Connect to storm sewer system where shown on the Drawings.
- C. Provide a complete system of interior storm-water piping, including the following items:
 - 1. Cleanouts: Refer to Section 22 05 76.
 - 2. Foundation drains.
 - 3. Roof drains.
 - 4. Emergency roof drains.
 - 5. Downspout nozzles.
 - 6. Sump pump. Refer to Section 22 14 29.
- D. Test complete system.

1.03 TERMINOLOGY

- A. Abbreviations:
 - 1. Acrylonitrile Butadiene Styrene (ABS).
 - 2. Drain, waste, and vent (DWV).
 - 3. Polyvinyl chloride (PVC).
 - 4. Volatile organic compounds (VOCs).

1.04 QUALITY ASSURANCE

- A. Standards:
 - 1. American Society for Testing and Materials (ASTM).
 - 2. Cast Iron Soil Pipe Institute (CISPI).
 - 3. Factory Mutual Global (FMG).
 - 4. U.S. Environmental Protection Agency (EPA).
 - 5. Code of Federal Regulations (CFR) Title 40, "Protection of Environment," Part 59, "National Volatile Organic Compound Emission Standards for Consumer and Commercial Products," Subpart D, "National Volatile Organic Compound Emission Standards for Architectural Coatings."
- B. All pipe and fittings shall conform to the requirements of ASTM A74.
- C. All piping materials and installation shall comply with State and Local Building Code requirements.

PART 2 PRODUCTS

2.01 PIPING AND FITTINGS

- A. Acceptable Manufacturers:
 - 1. Cast iron: American Cast Iron, Charlotte, Clow, Griffin, MWI, Tyler Pipe, or U.S. Pipe & Foundry.
 - 2. Steel: Armco, Republic, or Wheeling.
 - 3. Copper: American Brass Co., Chase Brass Co., Nibco, or Revere.
 - 4. ABS or PVC plastic: Charlotte, Crestline, Nibco, or Yardley.
- B. Storm piping and fittings under slab:
 - 1. Extra-heavy weight, single hub cast iron (ASTM A74), factory-coated with coal tar enamel.
 - 2. Hubless cast iron (CISPI 301, ASTM A888), factory-coated with coal tar enamel.
 - 3. Copper DWV tube, 3 in. and smaller (ASTM B306): Type K drainage tube, drawn temper.
 - 4. Schedule 40 PVC/DWV plastic (ASTM D2665).

5. Schedule 40 ABS/DWV plastic (ASTM D2661).
- C. Interior above-grade piping and fittings:
1. Service weight, single hub cast iron (ASTM A74), factory-coated with coal tar enamel.
 2. Hubless cast iron (CISPI 301, ASTM A888), factory-coated with coal tar enamel.
 3. Steel pipe, 2-1/2 in. and smaller (ASTM A53): Type E or S, Grade A or B, Schedule 40, galvanized.
 4. Copper DWV tube, 4 in. and smaller (ASTM B306): Type L drainage tube, drawn temper.
 5. PVC Schedule 40 solid wall pipe according to ASTM D 1784, D 1785, and D 2665. PVC DWV fittings conforming to ASTM D 1784 and D 2665.
- D. Sump Pump Discharge Piping: Standard weight, galvanized steel (ASTM A120), with galvanized cast iron threaded drainage fittings, shall be used to the point of connection to a gravity line.
- E. All fittings shall be compatible with piping for size, material, and joint type.
- F. Joints:
1. Cast iron:
 - a. Tarred jute or oakum with lead poured flush with hub.
 - b. ASTM A 74 / ASTM C 564 Neoprene molded gasket for sealing hub and spigot cast iron fittings; service weight or extra heavy pipe fittings, as appropriate.
 - c. Compression-type plastic seal (ASTM C-564).
 - d. Joints in hubless cast iron pipe:
 - 1) Sizes 1-1/2 in. to 10 in.: Coupling approved to FMG 1680 Class 1 and marked with the logo.
 - a) Sizes 1-1/2 in. to 4 in.: Clamp-All Model #80.
 - b) Sizes 5 in. to 10 in.: Clamp-All Model #125.
 - c) Acceptable Manufacturer: Clamp-All Corporation.
 - 2) Sizes 1-1/2 in. to 2 in., where exposure to head pressure cannot exceed 10 ft., coupling approved to CISPI 310.
 - a) Acceptable Manufacturers: Anaco, Charlotte, Jones, or Tyler.
 2. Copper: Cast (ASTM B16.23) or wrought (ASTM B16.29) copper drainage fittings with solder joints.

3. Galvanized steel: Threaded, galvanized cast iron drainage fittings.
4. ABS or PVC Plastic: Solvents and cements as recommended by the Manufacturer.

2.02 SOLVENT CEMENTS AND ADHESIVE PRIMERS

- A. For ABS Pipe and Fittings:
 1. Use ABS solvent cement with a VOC content of 325 grams/liter or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
 2. Use adhesive primer with a VOC content of 550 grams/liter or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
- B. For Joining PVC Piping: ASTM D2564. Include primer according to ASTM F656.
 1. Use PVC solvent cement with a VOC content of 510 grams/liter or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
 2. Use adhesive primer with a VOC content of 550 grams/liter or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

2.03 ROOF DRAINS

- A. Acceptable Manufacturers: MIFAB Inc. or approved equal by Ancon, Jay R. Smith Mfg. Co.®, Josam, WADE (McWane Inc.), Watts, or Zurn Industries LLC..
- B. RD-: MIFAB #R1200-E-B-M. Adjustable cast iron drain and cast iron dome, bottom outlet, no-hub, membrane clamping collar, roof sump receiver, deck clamp, extension collar where required, and accessories, for roof construction. Size as shown on the Drawings.

2.04 EMERGENCY ROOF DRAINS

- A. Acceptable Manufacturers: MIFAB Inc. or approved equal by Ancon, Jay R. Smith Mfg. Co.®, Josam, WADE (McWane Inc.), Watts, or Zurn Industries LLC...
- B. ERD- (emergency overflow): MIFAB #R1200-EU-M-W4. Adjustable cast iron drain and dome, bottom outlet, no-hub, membrane clamping collar, 4 in. internal water dam, deck clamp, extension collar where required, and accessories, for roof construction. Size as shown on the Drawings.
- C. ERD- (emergency overflow): Mifab #R1200. Same as RD-, except with 4 in. internal water dam. Drain size as shown on the Drawings.

2.05 DOWNSPOUT NOZZLES

- A. Acceptable Manufacturers: MIFAB Inc. or approved equal by Ancon, Jay R. Smith Mfg. Co.®, Josam, WADE (McWane Inc.), Watts, or Zurn Industries LLC.
- B. DN-1: MIFAB #R1940-1-83. Nickel/bronze body, threaded inlet, wall flange, and outlet nozzle with stainless steel screen. Size as shown on the Drawings.

PART 3 EXECUTION

3.01 INSPECTION

- A. Examine the areas and the conditions under which interior storm-water piping system work is to be installed. Notify the A/E and Owner of conditions detrimental to the proper and timely completion of the Work.
- B. Do not proceed with the Work until unsatisfactory conditions have been corrected and interior storm-water piping location and depth have been established.

3.02 INSTALLATION — PIPING AND FITTINGS

- A. Coordinate location, size, and depth of site storm sewer and foundation drain with the Lead Contractor.
- B. Coordinate location and depth of building storm drain installation with the building foundation and the Lead Contractor.
- C. Refer to Section 22 10 00 for general piping installation instructions.
- D. Storm lines shall be continuously sloped. Pitch storm piping in direction of flow at not less than 1/16 in./ft. and not more than 1/4 in./ft., unless otherwise noted.
- E. Do NOT trap storm lines.
- F. Plastic Piping: Do not install plastic piping:
 - 1. Where it may be vulnerable to physical damage.
 - 2. Where run exposed.
 - 3. Where run exposed below joist level.
 - 4. Where run exposed below 8 ft. above floor.
 - 5. In ceiling spaces used for return air plenums. Refer to Architectural and HVAC Drawings for the location of these areas.

- G. **Elevator Equipment Rooms:** Avoid locating horizontal storm piping in elevator equipment rooms. If horizontal storm piping shall be located in elevator equipment rooms, obtain written permission from the A/E to do so. Any horizontal storm piping located in elevator equipment rooms shall be covered and enclosed by a painted metal enclosure to prevent water dripping onto equipment.
- H. Conceal all piping, unless indicated otherwise on the Drawings.
- I. Cleaning and Testing:
 - 1. After erection, and prior to putting in service, blow or flush lines free of loose materials.
 - 2. Perform tests in accordance with AHJ requirements. Tests shall be witnessed by the A/E's Representative.
 - 3. For a minimum test, plug piping at the lowest point and fill with water to highest point on roof to provide a minimum of 10 ft. hd. of water on all parts of the system.
 - a. Maintain for (1) hour with no leakage. If a leak occurs, replace the defective piece or joint with new materials at no cost to the Owner.
 - b. Repair any deficiencies.
 - c. Conduct final test with smoke or peppermint at 1 in. W.G. for (15) minutes.
 - d. Repeat test until no leaks occur.

3.03 INSTALLATION — FOUNDATION DRAINS

- A. **Lead Contractor shall provide** foundation drains.
- B. Connect foundation drain(s) to sump pump basin.

3.04 INSTALLATION — ROOF DRAINS

- A. Coordinate roof drain and cleanout locations, elevations, and installation with the Lead Contractor.
- B. **(Lead Contractor shall install) (Install)** all roof drains at low points in deck and clamp to sump receiver.
- C. Install wall cleanouts and covers at 18 in. above finished floor, and at the base of all downspouts where indicated on the Drawings. Provide exposed cleanout plugs without cover plate in unfinished areas or in above ceiling installations.
- D. Extend waterproofing membrane into roof drain and secure with collar. Install sealing gaskets between collar and drain.

- E. Lead Contractor shall flash into roof construction and make watertight. See detail on the Architectural Drawings.
- F. Offset roof drain piping below roof to allow for thermal expansion and contraction of roof.

3.05 INSTALLATION — EMERGENCY ROOF DRAINS

- A. (Lead Contractor shall install) (Install) emergency roof drains adjacent to each roof drain. Install at a higher elevation than the main roof drain (not in sump pan). See locations on the Drawings. See detail on the Architectural Drawings.

3.06 INSTALLATION — DOWNSPOUT NOZZLES

- A. Lead Contractor shall flash into wall construction and make watertight. See detail on the Architectural Drawings.

END OF SECTION 22 14 00

SECTION 22 14 29 - SUMP PUMPS

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 22 05 29, "Hangers and Supports for Plumbing Piping and Equipment."
- B. Section 22 13 00, "Facility Sanitary Sewerage."
- C. Section 22 14 00, "Facility Storm Drainage."

1.02 DESCRIPTION

- A. Provide a simplex clear water sump pump with controls. Extend sump pump discharge line up on elevator shaft wall, through wall, and indirect above floor drain or mop sink as shown on the Drawings.
- B. Provide sump pit in bottom of elevator shaft.
- C. Provide a simplex clear water sump pump with controls.
- D. Provide a duplex storm water sump pump and basin complete with controls.

1.03 RATINGS AND CAPACITIES

- A. Pump Operating Point Requirements: The operating point of each pump at the conditions shown shall be within (5) percentage points of the maximum efficiency on its impeller curve. The impeller furnished shall not exceed 90% of the cut water diameter available for the pump casing furnished. The impeller curve shall be continuously rising throughout the range contained within its efficiency curves. The peak of the impeller curve shall be a minimum of 10% above the total head shown on the Drawings. The pump shall not overload the motor at any point on the impeller curve.
- B. Motors: Unless otherwise noted, high-efficiency motors shall be NEMA Design B, continuous rated with 1.15 service factor, with Class F insulation and Class B temperature rise, copper windings and leads, 1,750 RPM with the horsepower indicated on the Drawings. All motors shall be equipped with ball bearings.
- C. Refer to Drawings for gallons per minute, suction and discharge pressures, feet of head, horsepower, and electrical requirements. Each pump shall be rated for 65% of total design flow.

1.04 QUALITY ASSURANCE

- A. Standards:
 - 1. National Electrical Manufacturers Association (NEMA).
 - 2. Underwriters Laboratories, Inc., (UL) Label.
 - 3. Installation shall comply with all State and Local Code requirements.
- B. Each pump and drive unit shall be performance-tested at the factory at design conditions. The system shall be hydrostatically tested. The entire assembly shall be painted after shop testing.

1.05 SUBMITTALS

- A. Shop drawings and pump curves shall indicate materials, size, revolutions per minute, gallons per minute, feet of head, efficiency, and brake horsepower at the design conditions. Provide wiring diagrams.

PART 2 PRODUCTS

2.01 SUMP PUMPS

- A. **(Elevator) P-XXXX**: Liberty Pumps Model 283. 50 GPM, 10 ft. hd., permanently sealed ball bearing, 3,450 RPM, 1/2 HP, 120 volts, 1-phase, centrifugal type, cast iron casing, dynamically balanced cast iron impeller, stainless steel shaft, automatic thermal overloads, piggyback float switch controls, length of power cord as required, 1-1/2 in. discharge. Provide (1) pump in basin.
 - 1. Other Acceptable Manufacturers: Approved equal by Liberty, Zoeller, Weinman, Weil Pump, or Crane/Barnes.
- B. **P-XXXX**: Weil Model 1431: 20 GPM at 10 ft. hd., 1,750 RPM, 1/3 HP, 120 volts, 1-phase, centrifugal type, bronze-fitted, cast iron casing, dynamically balanced bronze impeller, strainer inlet, stainless steel shaft keyed to impeller, automatic thermal overloads, piggyback mercury float switch with 10 ft. power cable, power cord shall extend to electrical receptacle, 1-1/4 in. discharge.
 - 1. Provide (1) pump in basin.
 - 2. Acceptable Manufacturers: Weil Pump or approved equal by Armstrong Fluid Technologies, Liberty, Little Giant Pump Co., PACO Pumps (Grundfos), or Zoeller Pump Co.
- C. **P-XXXX**: Peerless Model 3U-4. 15 GPM at 20 ft. hd., 1,725 RPM, 1/3 HP, 120 volts, 1-phase, **duplex**, centrifugal type, bronze-fitted, cast iron casing, dynamically balanced bronze impeller, strainer inlet, stainless steel shaft keyed to impeller, automatic thermal overloads,

(integral) (discharge-mounted) float switch, power cord shall extend to (controls) (electrical receptacles), (1-1/2) (1-1/4) in. discharge.

1. Provide (1) (2) pumps in basin.
 2. Acceptable Manufacturers: Peerless Pump Company or approved equal by Armstrong Fluid Technologies, Crane/Barnes, Weil Pump, or Weinman.
- D. P-XXXX: Hydromatic Model SP40. 55 GPM, 18 ft. hd., ball bearing, 1,750 RPM, 4/10 HP, 120 volts, 1-phase, duplex, centrifugal type, cast iron casing, dynamically balanced cast iron impeller, stainless steel shaft keyed to impeller, automatic thermal overloads, manual controls, shall extend to electrical receptacle, 2 in. discharge.
1. Provide (2) pumps in basin.
 2. Acceptable Manufacturers: Hydromatic or approved equal by Armstrong Fluid Technologies, Crane/Barnes, Weil Pump, or Weinman.

2.02 PUMP CONTROL PANEL

- A. Acceptable Manufacturer: Anchor Scientific.
- B. Anchor Scientific Model 4201-AC. Wall-mounted NEMA 12 enclosure with individual pump circuit breakers, individual double pole, definite purpose contactors rated at 1 HP, terminals for sensors, pumps, incoming power and grounding lugs, individual HAND/OFF/AUTO selector switches, individual pump running lights, lead pump indicating light, alternating pump relay, panel door-mounted red light, and 4 in. horn and test/silencing push-button.
1. Float switches: (4) pump discharge-mounted, normally open mercury float switches, shall extend to electrical receptacle. Functions:
 - a. OFF.
 - b. LEAD PUMP ON.
 - c. LAG PUMP ON.
 - d. HIGH WATER ALARM.
 2. Provide engraved signs on panel door:
 - a. "HIGH WATER ALARM HORN" (top of horn).
 - b. "PUSH TO SILENCE" (next to silence push-button).
- C. Remote High-Water Alarm Panel: Anchor Scientific Model 1100-1: Wall-mounted high-water alarm panel with red light, 4 in. horn, and test/silencing switch. Provide engraved sign for "STORM SUMP HIGH WATER ALARM" on enclosure.

2.03 FLOAT CONTROLS

- A. Basin-cover-mounted, NEMA 4 automatic alternating type with float switch, rod, and float. Provide (4) float level controls:
 - 1. OFF.
 - 2. DUTY PUMP ON.
 - 3. STANDBY PUMP ON.
 - 4. HIGH WATER ALARM, with NEMA 1 terminal.
- B. Extend control wiring through cover plate. Adjust level switches so that there is 1 ft.-0 in. elevation difference before turning on the second pump.

2.04 SUMP BASIN

- A. 30 in. diameter x 6 ft.-0 in. deep fiberglass basin with below grade inlet, below grade outlet, and below grade control wiring openings. Gasketed and bolted nonslip steel cover.
- B. Cover: Zurn Model Z-784-2W. Light-duty grating, 24 in. x 24 in., with slot and grommet for power cord.
 - 1. Acceptable Manufacturer: Zurn Industries LLC.

PART 3 EXECUTION

3.01 INSTALLATION — SUMP PUMPS

- A. Coordinate location of each sump pump with the elevator equipment, HVAC Work, Electrical Work, and Lead Contractor.
- B. Install sump pumps on 8 in. solid concrete blocks, where shown on the Drawings, in accordance with the Manufacturer's recommendations. Refer to Section 22 05 29 regarding concrete work.
- C. Sump pump discharge piping shall be galvanized steel, to a point of connection to the large gravity line.
- D. Arrange piping and provide unions so that the pump can be easily removed.
- E. Install a union, nonslam check valve, and ball valve in each pump discharge piping.
- F. Adjust sump pump level switches.

G. Simplex Clear Water Sump Pump:

1. Extend sump pump discharge line up on elevator shaft wall, through exterior wall, and (indirect above floor drain) (to splash block) as shown on the Drawings.
2. Lead Contractor shall provide sump pit in bottom of elevator shaft.
3. Electrical Contractor shall provide waterproof receptacle for elevator sump pump.

H. Duplex (Storm Water) (Clear Water Waste Sanitary) Sump Pump:

1. Include a mechanical or electrical alternator control panel with starters and circuit breakers or fusible disconnect switches for each motor and factory-wired float control and alarm wiring.
2. Extend storm sump pump discharge line below floor slab, up on wall, through exterior wall, and above exterior grade to splash block.
3. Coordinate size of foundation drain and trench drain line and location with the Lead Contractor.
4. Connect to foundation drain and exterior trench drain.
5. Extend and connect foundation drain and trench drain line to storm sump basin.
6. Provide splash block.
7. Lead Contractor shall provide foundation drain, exterior trench drain, and outlet.
8. Electrical Contractor shall provide all control and alarm wiring, including installation of remote alarm panel.

- I. Test sump pumps and controls by filling basin with water and operating through all modes of operation.

3.02 INSTALLATION — PUMP CONTROL PANEL

- A. Coordinate location of control panel with the equipment, HVAC Work, Electrical Work, and Lead Contractor.
- B. Install control panel where shown on the Drawings in accordance with the Manufacturer's recommendations.
- C. Install top of control panel at 6 ft.-0 in. above finished floor.
- D. Furnish remote high-water alarm panel.
- E. Provide in-panel relay for remote high-water alarm.
- F. Electrical Contractor shall:
 1. Install remote high-water alarm panel.

2. Wire water level switches to control panel and high-water alarm panel.
3. Provide waterproof male and female plugs.

3.03 INSTALLATION — FLOAT CONTROLS

- A. Install float controls in accordance with the Manufacturer's recommendations.

3.04 INSTALLATION — SUMP BASIN

- A. Coordinate location of each sump basin with the **elevator** equipment, HVAC Work, Electrical Work, and Lead Contractor.
- B. Refer to the Drawings for locations of inlet and outlet openings.
- C. Install sump basin where shown on the Drawings in accordance with the Manufacturer's recommendations.
- D. Coordinate the pouring of a minimum of 6 in. of concrete around the sump basin with the Lead Contractor. Refer to Section 22 05 29 regarding concrete work.
- E. Vent the top portion of the basin.
- F. **(Lead Contractor shall install) (Install)** sump basin cover and frame.
- G. Secure sump basin cover so that it is absolutely airtight and flush with finished floor.

3.05 WIRING

- A. Coordinate electrical requirements with the Electrical Work.
- B. Electrical Contractor shall provide:
 1. Wiring to the pumps, control panel, and remote high-water alarm.
 2. Power wiring.
 3. **(Receptacle) (Disconnects)**.

END OF SECTION 22 14 29

SECTION 22 34 36.23 - COMMERCIAL, HIGH-EFFICIENCY, GAS DOMESTIC WATER HEATERS
(CONDENSING)

PART 1 GENERAL

1.01 RELATED DOCUMENTS

- A. Section 22 05 29, "Hangers and Supports for Plumbing Piping and Equipment."
- B. Section 22 11 19, "Domestic Water Piping Specialties."
- C. Section 23 11 23, "Facility Natural-Gas Piping."

1.02 DESCRIPTION

- A. Provide condensing, gas-fired water heater system, tested and ready for service, complete with domestic water heating, domestic water heater circulator pump, interconnecting piping, valves, and controls.
- B. Provide thermostatic mixing valve in accordance with AHJ requirements. Refer to Section 22 11 19.
- C. Provide exhaust and air intake(s) from interior domestic water heater.

1.03 RATINGS AND CAPACITIES

- A. Size, capacity, arrangements, and location as shown on the Drawings.

1.04 QUALITY ASSURANCE

- A. Standards:
 - 1. Air-Conditioning, Heating, and Refrigeration Institute (AHRI).
 - 2. American National Standards Institute (ANSI):
 - a. ANSI Z21.10.3.
 - b. ANSI Z21.13.
 - c. ANSI Z223.1.

3. National Sanitation Foundation (NSF) NSF 61.
 4. American Society of Heating, Refrigerating, and Air Conditioning Engineers (ASHRAE):
 - a. ASHRAE 15, Section 8.13.6.
 - b. ASHRAE 90.1, Current Enforced Edition, "Energy Standard for Buildings Except Low-Rise Residential Buildings."
 5. American Society of Mechanical Engineers (ASME):
 - a. ASME Section IV.
 - b. ASME Section VII.
 6. International Organization for Standardization (ISO) 9001.
 7. National Board of Boiler and Pressure Vessel Inspectors (NB).
 8. National Fire Protection Association (NFPA):
 - a. 2024 NFPA 54, "National Fuel Gas Code."
 - b. 2023 NFPA 70, "National Electrical Code® (NEC)."
 9. 2006 Uniform Mechanical Code (UMC) Section 1107.6.
 10. Ohio Administrative Code (OAC) 4101:1, 2024 Ohio Building Code: OAC 4101:1-13, Energy Efficiency.
 11. National and Local air quality regulations for water heaters with low nitrogen oxides (NO_x) (0 to 20 parts per million NO_x emissions).
- B. Water heater shall be AHRI-certified, ANSI Z21.10.3- or Z21.13-certified, ASME H- or HLW-stamped, ASME V-stamped, and NB-listed.
- C. Water heater shall meet safety standards for exhaust equipment as noted by the 2006 UMC Section 1107.6, and ASHRAE 15, Section 8.13.6.

1.05 SUBMITTALS

- A. Product data sheet (including dimensions, rated capacities, shipping weights, accessories). Documentation that:
1. Units comply with ASHRAE 90.1, Section 7, "Service Water Heating," and "Performance Requirements for Water Heating Equipment" Table..
 2. Field-fabricated piping arrangement for piping-type heat traps complies with ASHRAE 90.1.

- B. Wiring diagram.
- C. Installation and operating instructions.
- D. Warranty information.

1.06 WARRANTY

- A. Furnish a parts-only limited written warranty for commercial use on water heaters, effective for (1) year from date of installation, or a maximum of (18) months from shipment date.
- B. Furnish a 5-year parts-only limited written warranty for water heater heat exchanger, including damage resulting from thermal shock.

PART 2 PRODUCTS

2.01 DOMESTIC WATER HEATER SYSTEMS (400 MBH TO 1,000 MBH)

- A. Acceptable Manufacturers: Lochinvar or approved equal by Advanced Heating and Hot Water Systems (HTP), A. O. Smith, Bradford White, Laars Heating Systems Company, or RBI.
- B. GWH-1 & 2: Condensing water heater. Lochinvar "Armor" Water Heater Model #AWH0650. Minimum 95% efficient system. Input of 650,000 MBH, and a recovery of 645 GPH at 120 degrees F. temperature rise. ASME-stamped. 160 PSI maximum working pressure, and adjustable high-temperature limit set at 140 degrees F. Floor loading of 200 lbs./sq. ft. or less.
 - 1. Cabinet:
 - a. Corrosion-resistant galvanized-steel jacket finished with a baked-on powder coat, applied prior to assembly for complete coverage.
 - b. Water heater, if located on a combustible floor, shall not require a separate combustible floor base.
 - c. Water heater shall connect both the combustion air and flue products through the back of the unit.
 - d. Heater shall have, as standard, an internal combustion air filter rated to MERV 8 (>95% arrestance).

2. Combustion chamber:
 - a. Stainless-steel, sealed combustion type with a high-temperature-rated premix burner.
 - b. Combustion chamber wrapper shall not require insulation to reduce standby radiation losses.
3. Burner:
 - a. Capable of full modulation firing down to 10% or less of rated input with a turndown ratio of 10:1 or greater.
 - b. Able to fire at 100% of rated input when supplied with 4.0 in. water column of inlet gas pressure.
 - c. Shall maintain service under heavy demand conditions.
 - d. Shall use a variable speed combustion air blower.
4. Combustion air blower:
 - a. Shall precisely control fuel/air mixture for maximum efficiency throughout entire modulation range.
 - b. Shall infinitely vary its output by the modulating temperature controller, precisely adjusting the volume of air and gas supplied for combustion.
5. Firing mode: Provide electronic modulating control of the gas input to the water heater.
6. Ignition system:
 - a. Water heater shall be equipped with a 100% safety shutdown.
 - b. Ignition shall be direct spark or hot-surface ignition type with remote flame sensor.
 - c. Ignition system shall use an ignition control module, or a low-voltage ignition control circuit.
 - d. An external viewing port shall be provided, permitting visual observation of burner operation.
7. Gas train:
 - a. Water heater shall have a firing/leak test valve and pressure test valve.
 - b. Water heater shall have dual-seated main gas valve.

- c. Gas control trains shall have a redundant safety shut-off feature, main gas regulation, shut-off cock, and plugged pressure tapping to meet the requirements of ANSI Z21.10.3.
- 8. Dual heat exchangers: 316T (titanium) stainless steel or 316L stainless steel.
 - a. Dual burner, multi-pass design with helically wound 316L stainless steel water tubes that completely enclose (2) combustion chambers for maximum efficiency.
 - b. Designed with a condensate drain at the bottom.
 - c. Water heater shall be able to operate at inlet water temperatures as low as 50 degrees F.
- 9. Water heater pump: Factory-packaged pump system. Stainless-steel or bronze-fitted circulating pump.
- 10. Cold water protection system:
 - a. Automatic proportional bypass system. Ensures heater primary heat exchanger experiences inlet temperatures in excess of 120 degrees F. in less than (7) minutes to avoid damaging condensation.
 - b. Configured with variable-flow injection systems controlled by system that diverts the correct amount of heated outlet water directly into the heater loop to maintain the required minimum inlet temperature. Control temperature sensor shall be located in the inlet header of the primary heat exchanger.
 - c. Exception: A cold water protection system is not required where water heater heat exchange is rated for an inlet water temperature of 40 degrees F. or lower.
- 11. Intake adapter: Water heater shall be shipped with a PVC intake adapter that allows for the use of PVC intake material.
- 12. Exhaust adapter: Factory-mounted to water heater flue outlet.
 - a. Allows use of polypropylene exhaust material when water heater return water temperature does not exceed 190 degrees F.
 - b. Shall include a flue temperature limit mounted to the water heater flue outlet and interlocked into the water heater safety circuit.
 - c. Exhaust termination adapter shall be shipped loose inside of the water heater crating for field installation.
 - d. Water heater shall have a 200 degree F. manual high-limit that is factory-mounted into the water heater outlet header and interlocked into the water heater safety circuit.

- e. Option: Stainless steel exhaust material and adapter.
 - 13. Intake and exhaust piping: Size as recommended by the Manufacturer for longer runs. Pitch all horizontal runs upward away from equipment, at a minimum of 1/4 in. per ft. The first 10 ft. of exhaust piping shall be CPVC or stainless steel.
 - 14. Valves:
 - a. Ball valves (inlet and outlet).
 - b. ASME-rated 125 PSI temperature/pressure relief valve.
 - 15. Multi-stack frame: Lochinvar Part #100341330 for Lochinvar Water Heater Model #AWH1000NPM. If substitute water heaters are used, select compatible frame for alternate heaters
- C. Controls and Accessories:
- 1. Thermostats, switches, relays, safeties, and related control items shall be factory-installed and factory-wired.
 - 2. The Manufacturer shall provide complete piping and wiring diagrams showing control wiring for flow switch, tank thermostat, aquastat, pump, and burner, and location of piping specialties.
 - 3. Provide necessary balanced piping manifold, thermostats, safety flow switch, thermometers, and any other piece of equipment necessary to make a complete and properly functioning domestic water heating system.
 - 4. Controls on heaters and hot water storage tanks shall start/stop circulator pump.
 - 5. Provide flow switch to retard burner flame until water flow has been proven.
 - 6. Water heater operating controls:
 - a. Water heater shall feature an integrated modulating digital controller, mounted and wired.
 - b. Tank sensor shall be shipped loose for field installation. Inlet/outlet sensors are factory-installed.
 - c. Heater shall be able to receive a 0 VDC to 10 VDC signal from the central energy management and direct-digital control system (EMCS) to vary the setpoint control or firing rate. Heater shall have an alarm contact for connection to the central EMCS.
 - d. Heater shall be equipped with Modbus communications compatibility with up to (146) points of data available. B-85 Gateway: BACnet MS/TP, BACnet IP, N2 Metasys or Modbus TCP.

- e. Shall be equipped with a PID modulating temperature controller with liquid crystal display (LCD) that incorporates an adjustable energy-saving pump control relay and freeze protection.
 - f. Control shall have the ability to provide cascade control of up to (4) heaters as a single system via 2-wire communication.
 - g. Safety controls:
 - 1) High-limit control with manual reset.
 - 2) Flow switch, mounted and wired.
 - 3) ASME-rated pressure relief valve, 125 PSIG. Pipe to an approved drain.
 - 4) Temperature and pressure gauge (shipped loose).
7. Water heater diagnostics: Provide external visual indication of:
- a. Water heater status/faults.
 - b. Safeties and internal/external interlocks with fault display.
 - c. Ignition status/faults.

2.02 STORAGE TANKS

- A. ST-1: Lochinvar "Lock-Temp." Vertical storage tank with storage capacity of 650 gallons, constructed in accordance with ASME requirements with the approval for 125/150 lb. PSI working pressure, and glass-lined (optional cement-lined shall be acceptable).
- 1. Inner chamber: Constructed so as to eliminate turbulence, permitting a minimum of 80% drawdown of the tank with no drop in outlet temperature, regardless of the rate of draw.
 - 2. Jacket: Painted metal jacket and 2 in. fiberglass insulation with a minimum "R" factor of 12, covered with a heavy 16 ga. metal jacket fabrication from zinc-coated steel, and protected with a 3-coat enamel finish. Completely encased in high-density fiberglass insulation of sufficient thickness to meet the energy efficiency standards of ASHRAE 90.1b-2007.
 - 3. Install temperature gauge on the front of the cabinet to register the water temperature in the upper portion of the storage tank. Provide brass drain valve and ASME temperature and pressure relief valve.
 - 4. After installation of storage tank, remove all shipping supports, fill any holes in tank jacket with insulation, and install jacket cover plate.

2.03 WATER HEATER ACCESSORIES

- A. Piping-Type Heat Traps: Field-fabricated piping arrangement according to ASHRAE 90.1-2010.
- B. Neutralization Kit: Factory-sized in-line neutralization kit for each water heater condensate drain.
- C. Concentric Vent Kit: Intake and exhaust concentric wall termination kit, factory-matched for each water heater.

PART 3 EXECUTION

3.01 COORDINATION

- A. Coordinate water heater system ingress with the Lead Contractor.
- B. Coordinate location of water heater system with the equipment and Lead Contractor.
- C. Coordinate location and electrical requirements of water heater system with the Electrical Work.

3.02 INSTALLATION

- A. Plumbing Contractor shall be responsible for proper operation of heaters, storage tanks, and circulating pumps and controls.
- B. Installation shall comply with 2017 NEC, 2015 NFPA 54, and ANSI Z223.1.
- C. Install gas water heater system on concrete pad where shown on the Drawings. Shim to level, and bolt water heater to concrete pad. Refer to Section 22 05 29 for concrete requirements.
- D. Install all components according to the Manufacturer's recommendations and installation instructions, including required service clearances and venting guidelines.
- E. Provide clearance to service access panels.
- F. Provide shut-off valve and dielectric union on inlet and outlet of water heater.
- G. Provide thermometer on inlet and outlet to heater, and on hot water storage tank.

- H. Provide exhaust piping and intake air wall termination.
- I. Install exhaust and combustion air intake piping as recommended by the Manufacturer.
- J. Oversize piping as recommended by the Manufacturer for long runs.
- K. Install relief valve, elbow down, and extend a metallic relief valve discharge line to (6 in. above floor) (adjacent to floor drain) (service sink).
- L. Set water heater supply temperature for 140 degrees F.

3.03 START-UP AND TESTING

- A. Water Heater System Manufacturer shall provide a Representative to verify proper and complete installation, and to start up, adjust, and fully prepare system for final operation.
- B. Test the following during operation and adjust if necessary:
 - 1. Safeties.
 - 2. Operating controls.
 - 3. Static and full load gas supply pressure.
 - 4. Gas manifold and blower air pressure.
 - 5. Amp draw of blower.
 - 6. Combustion analysis.
- C. Submit copy of start-up report to the A/E.

3.04 TRAINING

- A. Factory-Authorized Service Representative shall train maintenance personnel on procedures and schedules related to start-up, shut-down, troubleshooting, servicing, and preventive maintenance.
- B. Schedule training at least (7) days in advance.

3.05 WIRING

- A. Electrical Contractor shall provide:
 - 1. Conduit between water heaters, circulating pumps, tank sensor, and flow switches.

2. Disconnect switches.
3. Control wiring.
4. Power wiring.

END OF SECTION 22 34 36.23

SECTION 22 40 00 - PLUMBING FIXTURES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Section 22 07 00, "Plumbing Insulation."
- B. Section 22 11 16, "Domestic Water Piping."
- C. Section 22 11 19, "Domestic Water Piping Specialties."
- D. Section 22 13 00, "Facility Sanitary Sewerage."
- E. Section 22 13 23, "Sanitary Waste Interceptors."

1.2 DESCRIPTION

- A. Provide commercial-quality, defect-free, plumbing fixtures, with vandal-resistant carriers, handles, aerators, nuts, and bolts; supports; water supply and waste piping; trim including escutcheons; fittings; angle stop valves; and all accessories required for a complete installation, as shown on the Drawings.
 - 1. Water closets.
 - 2. Lavatories.
 - 3. Sinks.
 - a. Kitchen hand sinks.
 - b. Scullery sinks.
 - c. Service sinks.
 - d. Mop service basins.
 - 4. Bathtubs.
 - 5. Showers.
 - 6. Disposers (food waste grinders).
 - 7. Shampoo bowls.

8. Laundry trays (laundry tubs).
 9. Plumbing fixture controls:
 - a. Flushometers.
 - b. Automatic flush valve circuit timers.
 - c. Automatic urinal flush valve sensor system.
 - d. Transformers.
 10. Healthcare plumbing fixtures: Clinic sinks.
- B. Color Selections: The A/E shall approve all color selections for plumbing equipment, including fixtures, faucets, and trim. Plumbing fixtures shall be white in color, unless otherwise noted. Refer to the Plumbing Fixture Schedule on the Drawings.
- C. All exposed trim, piping, and metal parts shall be nonferrous and chrome-plated, unless otherwise noted.
- D. Fixture connection sizes are shown on Drawings.
- E. Faucets shall have (renewable seats) (ceramic disc valve cartridges) (cam-activated diaphragm valve cartridges).
- F. Carriers:
1. All carriers shall be specifically chosen to accommodate the particular brand and style of fixture actually installed, the particular type of floor and wall actually present at each fixture location, and the piping arrangement at each fixture. Furnish plastic or metal positioning frames to isolate carrier bolts from wall construction.
 2. Provide a heavy-duty, rectangular vertical support, floor-supported, commercial type fixture carrier for each wall-mounted plumbing fixture, unless noted otherwise.

1.3 QUALITY ASSURANCE

- A. Standards:
1. American National Standards Institute (ANSI).
 2. American Society for Testing and Materials (ASTM).
 3. American Society of Mechanical Engineers (ASME)
 4. American Society of Sanitary Engineering (ASSE).

5. Americans with Disabilities Act (ADA) Accessibility Guidelines.
6. Architectural Barriers Act (ABA) Accessibility Guidelines.
7. Federal Energy Policy Act of 1992.
8. International Association of Plumbing and Mechanical Officials (IAPMO).
9. International Cast Polymer Alliance (ICPA).
10. International Code Council, Inc. (ICC) A117.1, "Accessible and Usable Buildings and Facilities."
11. National Association of Home Builders (NAHB).
12. National Electrical Manufacturers Association (NEMA).
13. National Sanitation Foundation (NSF).
14. Plumbing and Drainage Institute (PDI).
15. Underwriters Laboratories, Inc. (UL).
16. Illinois Administrative Code (IAC):
 - 1) IAC Title 35: Environmental Protection.
 - 2) IAC Title 41: Fire Protection.
 - 3) IAC Title 71: Public Buildings, Facilities, and Real Properties.
 - 4) IAC Title 77: Public Health.
 - 5) IAC Title 83: Public Utilities.
17. Local Plumbing and Energy Codes.

1.4 SUBMITTALS

- A. Submit Manufacturer's product data for all products specified in this Section and shown on the Drawings.
- B. Each shop drawing submittal shall be clearly marked with model number and fixture designation number and shall indicate all required accessories, dimensions, construction, color, and rough-in requirements.
- C. Submit color charts for all fixtures, faucets, and trim (minimum [3] color options).
- D. Product Data: Documentation indicating flow and water consumption requirements.

PART 2 - PRODUCTS

2.1 PLUMBING FIXTURES AND ACCESSORIES

- A. Manufacturer and Model Number listed in the Plumbing Fixture Schedule on the Plumbing Drawings shall be considered to be the Design Base Manufacturer and Model Number. Other Acceptable Manufacturers listed herein will be permitted, unless specifically excluded.
- B. Refer to the Plumbing Fixture Schedule for the Manufacturer, Model, description, and location tags.

2.2 WATER CLOSETS

- A. All water closets shall comply with the Federal Energy Policy Act of 1992 and shall have 1.6 gallons/flush maximum.
- B. Water closet flush (tank trip lever) (valve handle) shall be left-handed operation, unless required to be right-handed operation to be ADA-compliant.
- C. All water closet bowls for public or employee toilet facilities shall be of the elongated type.
- D. Acceptable Manufacturers:
 - 1. Vitreous china fixtures: American Standard, Crane, Gerber, Kohler, Toto, or Zurn Industries LLC.
 - 2. Water closet seats: Bemis, Beneke, Centoco, Church, Kohler, Olsonite, or Sperzel.
 - 3. Flush valves (with vandal-resistant stop): American Standard, Coyne & Delany, Sloan or Zurn Industries LLC.
 - 4. Supplies, escutcheons, and traps: Anso-Flex, Brasscraft, Bridgeport, Consolidated, Dearborn, EBC, Frost, McGuire Mfg. Co., Sanitary-Dash, Speedway, Waterway, or Zurn Industries LLC.

2.3 LAVATORIES

- A. Acceptable Manufacturers:
 - 1. Vitreous china fixtures: American Standard, Crane, Gerber, Kohler, or Zurn Industries LLC.

2. Stainless steel fixtures: Bradley, Metcraft Industries, or Willoughby Industries.
3. Faucets: Delta, American Standard, Chicago Faucet, Elkay, Hyrotek, Moen, Sloan, or Zurn Industries LLC.
4. Supplies, escutcheons, and traps: Anso-Flex, Brasscraft, Bridgeport, Consolidated, Dearborn, EBC, Frost, McGuire Mfg. Co., Sanitary-Dash, Speedway, Waterway, or Zurn Industries LLC.
5. Carriers: Heavy-duty type Zurn Industries LLC Model Z1231 or approved equal by Jay R. Smith Mfg. Co.®, Josam, MIFAB Inc., or WADE (McWane Inc.).

2.4 SINKS

- A. Coordinate size of countertop sink with the Casework Contractor before ordering sink. Install sink in countertop and casework.
- B. Acceptable Manufacturers:
 1. Vitreous china fixtures: American Standard, Crane, Gerber, or Kohler.
 2. Enameled cast iron fixtures: American Standard, Ceco, or Kohler.
 3. Stainless steel sinks: Elkay, Just Mfg. Co., Franke Commercial, Moen "Lancelot," or Republic.
 4. Kitchen stainless-steel sinks: American Standard, Elkay, Franke Commercial, or Just Mfg. Co.
 5. Commercial stainless-steel sinks: Amtekco, Elkay, Just Mfg. Co., Franke Commercial, Metcraft Industries, Moen, or Republic.
 6. Service sinks, mop service basins, and mop receptors: American Standard, Ceco, Creative Industries, Fiat, Florestone, Mustee, Powers, Standard Elsmere, Stern Williams, Swan, or Zurn Industries LLC.
 7. Faucets: Delta, American Standard, Bradley, Chicago Faucet, Crane, Elkay, Gerber, Grohe, Kohler, Moen, Sloan, Speakman, Symmons, T & S Brass Company, or Zurn Industries LLC.
 8. Aerators: American Standard, Bradley, Chicago Faucet, Crane, Delta, Gerber, Grohe, Kohler, Moen, Neoperl, Sloan, Speakman, Symmons, T & S Brass Company, or Zurn Industries LLC.
 9. Supplies, escutcheons, and traps: Anso-Flex, Brasscraft, Bridgeport, Consolidated, Dearborn, EBC, Frost, McGuire Mfg. Co., Sanitary-Dash, Speedway, Waterway, or Zurn Industries LLC.

2.5 BATHTUBS

A. Acceptable Manufacturers:

1. Fiberglass reinforced or acrylic plastic fixtures: Penner, AquaBath, Aqua Glass, Aquarius, American Standard, Ceco, Clarion, or Kohler.
2. Faucets: American Standard, Bradley, Chicago Faucet, Crane, Delta, Elkay, Gerber, Grohe, Kohler, Moen, Sloan, Speakman, Symmons, T & S Brass Company, or Zurn Industries LLC.
3. Supplies, escutcheons, and traps: Anso-Flex, Brasscraft, Bridgeport, Consolidated, Dearborn, EBC, Frost, McGuire Mfg. Co., Sanitary-Dash, Speedway, Waterway, or Zurn Industries LLC.

2.6 SHOWERS

A. All shower heads shall comply with the Federal Energy Policy Act of 1992 and shall have 2.5 gallon/minute maximum at 80 PSIG.

B. Acceptable Manufacturers:

1. Fiberglass reinforced or acrylic plastic fixtures: Freedom, American Standard, Aqua Bath, Aqua Glass, Aquarius, Dura Glass, Fiat, Fibersheen, Kohler, or National.
2. Shower receptors: Freedom, American Standard, Ceco, Creative Industries, Fiat, Florestone, Mustee, Powers, Standard Elsmere, Stern Williams, Swan, or Zurn Industries LLC.
3. Shower heads and controls: Acorn, American Standard, Bradley, Chicago Faucet, Crane, Elkay, Grohe, Haws, Kohler, Lawler, Leonard, Moen, Powers, Sloan, Speakman, Symmons, or T & S Brass Company.
4. Supplies, escutcheons, and traps: Anso-Flex, Brasscraft, Bridgeport, Consolidated, Dearborn, EBC, Frost, McGuire Mfg. Co., Sanitary-Dash, Speedway, Waterway, or Zurn Industries LLC.

2.7 DISPOSERS (FOOD WASTE GRINDERS)

A. Acceptable Manufacturers:

1. Food waste grinders: Hobart, InSinkErator, Kitchen-Aid, Master, or Waste King.
2. Supplies, escutcheons, and traps: Anso-Flex, Brasscraft, Bridgeport, Consolidated, Dearborn, EBC, Frost, McGuire Mfg. Co., Sanitary-Dash, Speedway, Waterway, or Zurn Industries LLC.

Can probably take out as
it should be provided by
the kitchen guy

2.8 SHAMPOO BOWLS

A. Acceptable Manufacturers:

1. Shampoo bowls: Kaemark, Belvedere or Willoughby Industries.
2. Supplies, escutcheons, and traps: Anso-Flex, Brasscraft, Bridgeport, Consolidated, Dearborn, EBC, Frost, McGuire Mfg. Co., Sanitary-Dash, Speedway, Waterway, or Zurn Industries LLC.

2.9 LAUNDRY TRAYS (LAUNDRY TUBS)

A. Acceptable Manufacturers:

1. Laundry tubs: American Standard, Ceco, Creative Industries, Fiat, Florestone, Mustee, Powers, Standard Elsmere, Stern Williams, Swan, Franke Commercial, or Zurn Industries LLC.
2. Faucets: American Standard, Bradley, Chicago Faucet, Crane, Delta, Elkay, Gerber, Grohe, Kohler, Moen, Sloan, Speakman, Symmons, T & S Brass Company, or Zurn Industries LLC.
3. Supplies, escutcheons, and traps: Anso-Flex, Brasscraft, Bridgeport, Consolidated, Dearborn, EBC, Frost, McGuire Mfg. Co., Sanitary-Dash, Speedway, Waterway, or Zurn Industries LLC.

2.10 HEALTHCARE PLUMBING FIXTURES

A. Acceptable Manufacturers:

1. Clinic sinks: American Standard, Crane, or Kohler.
2. Flush valves (with vandal-resistant stop): Coyne & Delany, Sloan "Regal XL," or Zurn Industries LLC "Aqua Flush."
3. Supplies, escutcheons, and traps: Anso-Flex, Brasscraft, Bridgeport, Consolidated, Dearborn, EBC, Frost, McGuire Mfg. Co., Sanitary-Dash, Speedway, Waterway, or Zurn Industries LLC.

PART 3 - EXECUTION

3.1 DELIVERY, STORAGE, AND HANDLING

- A. Fixtures and trim shall remain crated and stored until installation to prevent physical damage and moisture and dirt contamination.
- B. Adequately protect installed fixtures from damage with cardboard, plywood, or other rigid material.

3.2 INSTALLATION — GENERAL

- A. Coordinate exact location and installed height of plumbing fixtures with the Lead Contractor and A/E.
- B. Coordinate mounting height of all fixtures and controls to be used by the physically handicapped. Refer to ADA Accessibility Guidelines.
- C. Verify that colors of fixtures, faucets, and trim match approved shop drawings.
- D. Install fixtures and trim according to the Manufacturer's recommendations.
- E. Install carriers and accessories on wall-mounted fixtures. Anchor carriers securely to floor.
- F. Carefully assemble and connect fixtures to the required inlets and outlets.
- G. Provide stops on all hot- and cold-water supplies to fixtures.
- H. Provide all required seals, gaskets, nuts, bolts, and washers.
- I. Insulate domestic water supply and drain piping that could come in contact with wheelchair occupants. Refer to Section 22 07 00.
- J. Install chrome-plated brass escutcheons on waste and supply piping at walls, including piping located inside cabinets.

K. **Food Service** Equipment:

1. Provide angle stop valves and chrome supply piping to all **food service** equipment requiring domestic water.
2. Provide chrome cast brass tailpiece, trap, and chrome 17-gauge drain to wall on all **food service** equipment.
3. Tailpieces provided with all **food service** equipment.

L. **(Lead Contractor shall caulk) (Caulk)** between fixture, floor, and wall, with caulking compatible with the wall finish, at all new **or relocated** floor-mounted and wall-mounted fixtures. **Coordinate with the A/E.**

M. After the installation of the fixtures and trim is completed, flush out all connecting pipes through the fixtures to eliminate scale. Clean **faucet** strainers. Refer to Section 22 11 16 regarding sterilization of water lines.

N. Test to ensure that the fixtures shall function correctly when the Work is completed.

O. Thoroughly clean all fixtures of paper and dirt before final acceptance.

3.3 INSTALLATION — WATER CLOSETS

A. Mount wall-mounted water closets at the following heights or according to Manufacturer's recommendations, unless noted or directed otherwise:

Fixture	Height
Water Closet	15 in. to top (not including seat).
ADA-Compliant Water Closet	17 to 19 in. to top of seat. Flush valve at 44 in. maximum.

B. Field-adjust self-sustaining water closet seats to self-sustain in any position.

C. Provide anchors and supports for flush valve supply piping behind walls and chases.

D. Mount **tank trip lever** on the wide side of the toilet stall. Verify correct side.

E. Conform to ADA Accessibility Guidelines concerning location of flush valve trip lever. Mount flush valve trip lever on the wide side of the toilet stall. Verify correct side.

3.4 INSTALLATION — LAVATORIES

- A. Coordinate location of prefabricated counter/lavatory and rough-in with the Lead Contractor before roughing in.
- A. Before ordering countertop lavatories, coordinate countertop fixture selection with cabinet sizes and the Lead Contractor.
- B. Coordinate countertop openings with the Lead Contractor and the A/E. Furnish templates and locate. **Lead Contractor shall cut** countertop openings.
- C. Mount wall-mounted lavatories at the following heights or according to Manufacturer's recommendations, unless noted or directed otherwise:

Fixture	Height
Lavatory	31 in. to rim, unless noted otherwise.
ADA-Compliant Lavatory	29 in. to bottom of apron. Maximum height to rim 34 in.
Juvenile Lavatory	21 in. to bottom of apron. Maximum height to rim 28 in.

- D. Adjust self-closing lavatory faucets for closing time of (8) seconds.
- E. Provide chrome cast brass tailpiece, trap, and chrome 17-gauge drain to wall.

3.5 INSTALLATION — SINKS

- A. Before ordering countertop sinks, coordinate countertop fixture selection with cabinet sizes and the Lead Contractor.
- B. Coordinate countertop openings with the Lead Contractor and the A/E. Furnish templates and locate. **Lead Contractor shall cut** countertop openings.
- C. Before ordering equipment, coordinate **hair or plaster** trap placement and size with sink and under-counter cabinet space available.
- D. Brace floor-mounted sinks to walls. Provide structural support to secure sink.

- E. Coordinate location of food service equipment and rough-in requirements with the Food Service Equipment Contractor before roughing in.
- F. Sink Sanitizing Heater: Electrical Contractor shall provide power wiring.
- G. Mop Service Basins:
 - 1. Install P-trap below floor for mop service basins.
 - 2. Install duct tape over grates for mop service basins to provide protection from scratching and collection of dirt and debris during construction. Remove tape just prior to final inspection.
 - 3. Mount faucet at 36 in. above finished floor.
 - 4. Install mop hanger at 4 ft.-0 in. above finished floor on abutting wall from faucet and centered on basin.
- H. Provide chrome cast brass tailpiece, trap, and chrome 17-gauge drain to wall.

3.6 INSTALLATION — BATHTUBS

- A. Coordinate pit locations for bathtubs on slab with the Lead Contractor. Install all bathtub traps so they are accessible.
- B. Mount bathtub accessories at the following heights or according to Manufacturer's recommendations, unless noted or directed otherwise:

Accessory	Height
Bathtub Valves	14 in. from top tub to centerline of valve.
Bathtub Spout	4 in. from top tub to centerline of spout.

- C. Install check valves on thermostatic mixing valve inlets.
- D. Adjust thermostatic mixing valves to provide a leaving water temperature of 105 degrees F. MAXIMUM.
- E. Hand-held shower heads shall be provided with backflow prevention or anti-siphon device.

F. Whirlpool Bathtub:

1. Coordinate access to pump with the Lead Contractor.
2. Lead Contractor shall provide raised platform.

3.7 INSTALLATION — SHOWERS

- A. Mount shower accessories at the following heights or according to Manufacturer's recommendations, unless noted or directed otherwise:

Accessory	Height
Shower Controls	48 in. maximum to centerline.
Shower Heads	6 ft.-6 in. to floor.
ADA-Compliant Shower Slide Bar	6 ft.-6 in. from floor to top.

- B. Install check valves on thermostatic mixing valve inlets.
- C. Adjust mixing valves to provide a leaving water temperature of 105 degrees F. MAXIMUM.

3.8 INSTALLATION — DISPOSERS (FOOD WASTE GRINDERS)

- A. Electrical Contractor shall provide:
1. (Pilot light switch) (Wall switch) (Rocker switch in a style to match light switches).
 2. Power wiring.

3.9 INSTALLATION — SHAMPOO BOWLS

- A. Before ordering countertop shampoo bowls, coordinate countertop fixture selection with cabinet sizes and the Lead Contractor.
- B. Coordinate countertop openings with the Lead Contractor and the A/E. Furnish templates and locate. Lead Contractor shall cut countertop openings.
- C. Before ordering equipment, coordinate hair trap placement and size with sink and under-counter cabinet space available at salon.

3.10 INSTALLATION — LAUNDRY TRAYS (LAUNDRY TUBS)

- A. Install in accordance with the Manufacturer's recommendations.

3.11 INSTALLATION — HEALTHCARE PLUMBING FIXTURES

- A. Clinic Sinks: Install in accordance with the Manufacturer's recommendations.

END OF SECTION 22 40 00